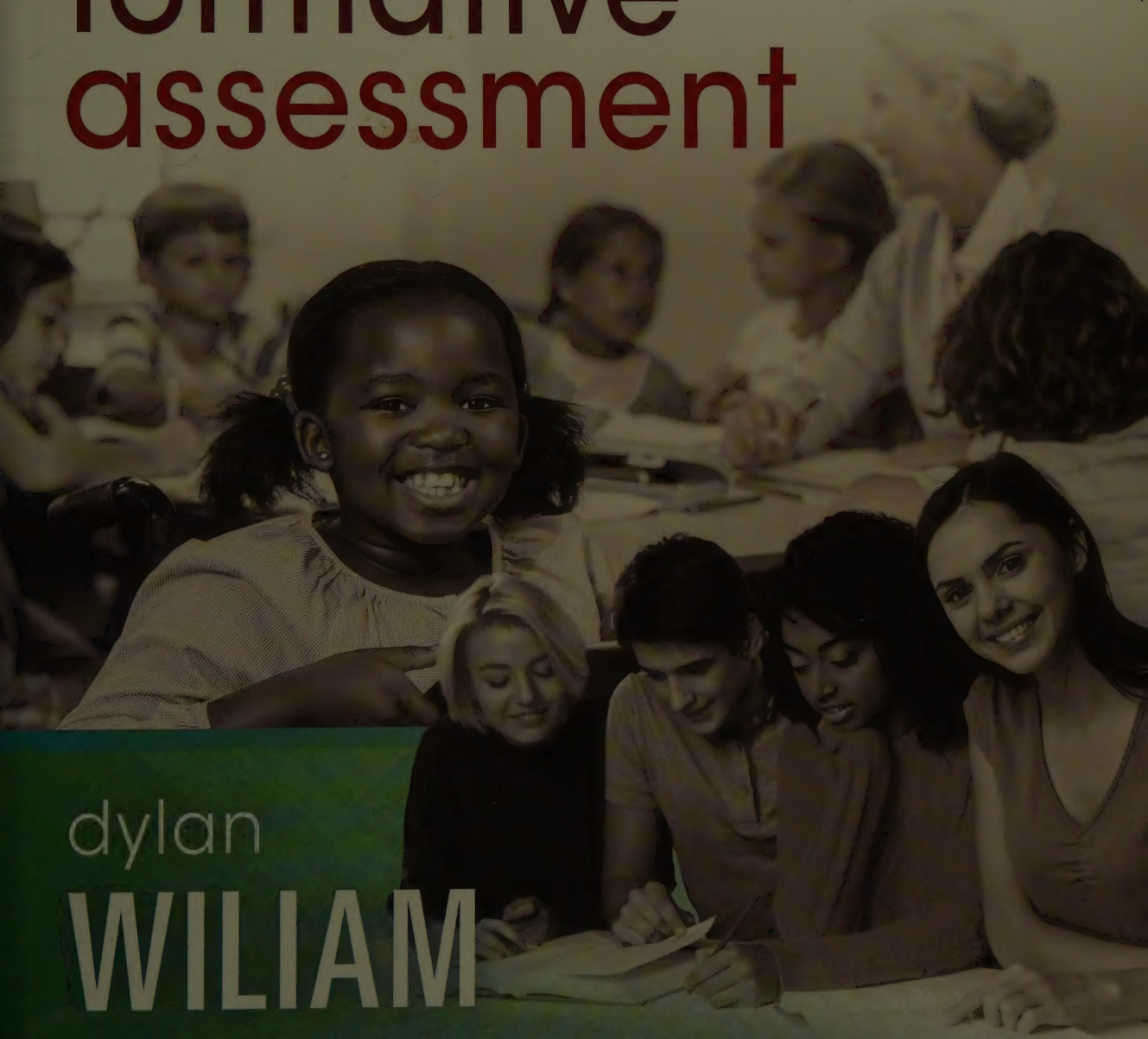
[second edition]

# EMBEDDED

## formative assessment



dylan WILIAM

# EMBEDDED formative assessment

second edition

DYLAN WILIAM

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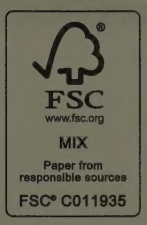
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Printed in the United States of America

21 20 19 18 2 3 4 5




Library of Congress Cataloging-in-Publication Data

Names: Wiliam, Dylan, author.
Title: Embedded formative assessment / author, Dylan Wiliam.
Description: Second Edition. | Bloomington, Indiana : Solution Tree Press, [2018] | Includes bibliographical references and index.
Identifiers: LCCN 2017020334 | ISBN 9781945349225 (perfect bound)
Subjects: LCSH: Effective teaching. | Educational tests and measurements. | Academic achievement.
Classification: LCC LB1025.3 .W47 2018 | DDC 371.102--dc23 LC record available at https://lccn.loc.gov/2017020334

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*Proofreader:* Jessi Finn
*Editorial Assistants:* Jessi Finn and Kendra Slayton

# Acknowledgments

The first edition of this book was the result of a thirty-five-year journey that took me from public school teacher to teacher educator to researcher and university administrator. Given the length of this journey, you will not be surprised that I have a lot of people to thank. First, the students at Christopher Wren School and North Westminster Community School in London taught me most of what I know about learning. The research I have read and carried out since then has done little more than help me make sense of what I learned there. Second are the teachers I have worked with in those two schools and in the extended networks of teachers of which I have been a part. As someone who went straight from a bachelor's degree to the classroom, with no training or preparation, I was fortunate to start teaching in one of London's toughest schools—fortunate because every teacher in that school struggled and was willing to talk about his or her struggles. It is only subsequently that I have realized how lucky I was, because that openness is not found everywhere.

Mentioning individuals is always invidious, but I owe particular debts of gratitude to Sarah Sharkey and Jerry Hardy (Christopher Wren School) and Dick Gmiterek (North Westminster Community School). When I joined the University of London, I lucked out again, being mentored by two great thinkers about education—first, Margaret Brown, and later, Paul Black. As well as being extraordinary academics, they are two of the nicest people one could ever hope to meet, and their generosity to younger academics is a model I have tried to emulate.

Special thanks must go to Greg Ashman (@Greg_Ashman) for permission to use the graphic in chapter 3 from his blog *Filling the Pail.* Thanks also are due to Maureen Wellbery, at the American International School of Vilnius in Lithuania, for the story about all-student response systems in chapter 4. I am also grateful to the editorial staff at Solution Tree, and particularly Lesley Bolton (and Alissa Voss for the second edition), for all the work they have done in getting my manuscript to the

finished book you have before you. Finally, and most important, my greatest thanks go to my partner, Siobhán Leahy. We first met at a mathematics teachers' conference almost forty years ago, and for thirty-seven years, she has been my greatest inspiration and greatest support and has kept me grounded in the realities of daily school life. That is why the book is dedicated to her.

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# About the Author



**Dylan Wiliam, PhD,** is a consultant who works with educators in North America, the United Kingdom, and many other countries to develop effective, research-based formative assessment practices. He is former deputy director of the Institute of Education at the University of London. From teaching in urban public schools to directing a large-scale testing program to serving in university administration, his professional path has led to a variety of positions at the forefront of education. Dr. Wiliam was also, from 2003 to 2006, senior research director at the Educational Testing Service in Princeton, New Jersey.

During his early years of teaching in inner-city classrooms, Dr. Wiliam focused on physics and mathematics. He later joined the faculty of Chelsea College, University of London, which later became part of King's College London. Here, he worked on developing innovative assessment schemes in mathematics before accepting leadership of the King's College Mathematics Education Program.

For three years, Dr. Wiliam served as the academic coordinator of the Consortium for Assessment and Testing in Schools, which developed a variety of assessments for the national curriculum of England and Wales. He then returned to King's College to serve as dean of the School of Education before being promoted to assistant principal of the College.

In 1998, he coauthored a major review of research evidence on formative assessment with Paul Black, and he has worked with many teachers in the United Kingdom and the United States on developing formative assessment practices to support learning.

In addition to holding a doctor of education degree, Dr. Wiliam has numerous degrees in mathematics and mathematics education. You can keep up with what he is writing at www.dylanwiliam.net and at www.dylanwiliamcenter.com, and you can follow him on Twitter @dylanwiliam.

# Introduction

In 1984, I left my job teaching mathematics and science in an urban public school in London to join a research project at the University of London that was exploring the potential of formative assessment to improve student learning. Over a quarter of a century later, this book is the result of that journey.

The book has two main purposes. The first is to provide simple, practical ideas about changes that every teacher can make in the classroom to develop his or her practice of teaching. The second is to provide the evidence that these changes will result in improved outcomes for learners.

In chapter 1, I show why educational achievement is so important and why raising educational achievement needs to be a priority. With higher levels of educational achievement, people are healthier, live longer, contribute more to society, and earn more money. For society, the benefits include reduced criminal justice costs, reduced health care costs, and increased economic growth.

I also outline briefly how previous attempts at reform—including changes to the structure of schooling, to the governance of schools, and to the curriculum, and an increased role for digital technology—have been largely ineffective. These reform efforts have failed to take into account three crucial points.

1. The quality of teachers is the single most important factor in the education system.
2. Teacher quality is highly variable.
3. Teacher quality has a greater impact on some students than others.

In chapter 1, I also show that attempts at improving the quality of entrants into teaching and removing the least effective teachers will result in small effects that will take a generation to materialize. The fundamental argument of chapter 1, therefore,

is that to secure our future economic prosperity, we need to help the teachers who are already serving in our schools improve.

In chapter 2, I explore some of the ways that teachers might develop, and estimate how big an impact these changes would have on student outcomes. I show that some popular initiatives, such as learning styles, have no discernible impact on student achievement at all, while others, such as increasing teacher content knowledge, *do* improve student learning but by much less than people generally assume.

In the second part of chapter 2, I summarize the research on classroom formative assessment practices and show that these practices appear to have a much greater impact on educational achievement than most other reforms. Chapter 2 concludes by outlining what formative assessment is, and what it is not, and presents the five key strategies of formative assessment.

Chapters 3, 4, 5, 6, and 7 deal in turn with each of the five key strategies of formative assessment.

1. Clarifying, sharing, and understanding learning intentions and success criteria
2. Eliciting evidence of learning
3. Providing feedback that moves learning forward
4. Activating learners as instructional resources for one another
5. Activating learners as owners of their own learning

In each of these five chapters, I present a summary of the research evidence that shows the impact of the strategy, and I offer a number of practical techniques that teachers have used—some in the literature, and others in classrooms that I have observed—to incorporate the strategy into their regular classroom practice. Although there is a definite progression throughout the chapters, I have made each chapter as self-contained as possible.

In all, these five chapters describe over seventy practical techniques for classroom formative assessment. Most of these techniques are not new. What *is* new is the framework for formative assessment presented in chapter 2, which shows how these disparate techniques fit together, and the research evidence that shows that these techniques are powerful ways to increase student engagement and help teachers become more responsive to their students' needs.

As I have encountered many hundreds of teachers over the course of my career investigating formative assessment, I have not been able to recall the specific names

of teachers and the dates on which I observed their classrooms in every instance in this book. Additionally, other observations occurred as part of research studies in which we granted the teachers anonymity for their contributions. As a result, many of the examples and techniques throughout this book are not cited directly, but rather presented through the lens of my personal observations.

In preparing this second edition, I show how the strategies and techniques presented in this book are just as relevant in colleges and universities as they are in K–12 settings. While the five strategies are equally applicable to college-aged students (and, indeed, to adult learning), the ways in which they are implemented *do* need to be adapted, and in this second edition, I provide a number of examples of practical techniques for doing formative assessment in higher education, and especially in lecture settings.

Additionally in this second edition, I have obviously updated many of the research studies, particularly those related to the changing nature of the working environment. I have also updated the research evidence on the various ideas that have been proposed for improving schools—and while the evidence has changed, the conclusions have not. First, this updated evidence continues to indicate that most current methods of school improvement won't help improve schools much. Second, it indicates that the power of classroom formative assessment to improve students' learning is incredibly strong, with evidence from all over the world showing that not only does classroom formative assessment work, but it is manageable in ordinary classrooms, without extra resources. I hope that this book convinces every reader about the impact that formative assessment can have on student achievement and provides some guidance about how best to begin the difficult, challenging, but worthwhile task of applying research to practice.

chapter 1

# Why Educational Achievement Matters

Educational achievement matters—more now than at any time in the past. It matters for individuals, and it matters for society. For individuals, higher levels of education mean higher earnings, better health, and increased life span. For society, higher levels of education mean lower health care costs, lower criminal justice costs, and increased economic growth. In this chapter, we will explore why education and educational achievement are vital to the prosperity of every nation and why the vast majority of attempts by policymakers to improve the achievement of school students have failed. We will then discuss three generations of school effectiveness research, the impact of teacher quality, and research-proven ways to increase teacher quality.

## The Increasing Importance of Educational Achievement

Education has always been important, but it has never been as important as it is now. In 1979, the median salary of those with bachelor's degrees was $30,000 higher than the median salary of those with a high school diploma or GED (in constant 2012 dollars). By 2012, the annual *earnings gap* had widened to over $58,000 (Autor, 2014).

Higher levels of education are also associated with better health; people with more education are less susceptible to a whole range of diseases, including cancer, and are less likely to have a significant period of disability toward the end of their lives (Jagger et al., 2007). No doubt this is partly due to lifestyle choices, such as

smoking, but it is also due in part to the kinds of work that are available to those with limited education. According to the Organisation for Economic Co-operation and Development (OECD, 2010), approximately 75 percent of American adults who did not complete high school say they are in good health, compared with 95 percent of those with college degrees.

Perhaps more surprisingly, people with more education live longer. Between 1915 and 1939, at least thirty states changed their child labor laws and periods of compulsory schooling. As a result, a number of students were required to attend school for one more year than children in other states. By looking at the life spans of those who had been required to attend an extra year of school, Adriana Lleras-Muney (2005) estimates that each additional year of schooling adds 1.7 years to one's life. These are high stakes indeed.

Educational achievement also matters for society. Henry Levin and his colleagues at Columbia University estimate that preventing one high school dropout produces a net benefit to society of $209,000 (Levin, Belfield, Muennig, & Rouse, 2007). The main components of this total are:

- $139,000 in extra taxes the individual would pay because he or she would be earning more money
- $40,500 in reduced health care costs, partly because the individual would be healthier, as noted previously, but also partly because he or she would be more likely to get health benefits from an employer and, therefore, be less dependent on public assistance
- $26,600 in reduced criminal justice costs (largely because the individual would be less likely to be incarcerated)

With higher levels of education, the U.S. economy will also grow faster. Using data from OECD's Programme for International Student Assessment (PISA) and a variety of other sources, Eric Hanushek and Ludger Woessmann (2015) examine the impact of increased achievement on economic growth. They estimate that if educators could raise the scores of American fifteen-year-olds on the triennial PISA tests by twenty-five points—the improvement that Poland made over a period of ten years—then by 2095, the U.S. economy would be 30 percent larger than it would otherwise be. And if American fifteen-year-olds could achieve a level of reading and mathematics that allowed them to participate effectively in modern society (defined by a score of 420 on PISA as compared with the international average of 500), the U.S. economy would grow by an extra $30 trillion (Hanushek & Woessmann, 2015).

The reason that higher levels of education are so valuable is because employers' educational demands are increasing steadily, nowhere more so than in manufacturing. According to the U.S. Bureau of Labor Statistics (2016), seventeen million Americans worked in manufacturing in 2000. Ten years later, the figure was less than twelve million. This means that in the first decade of the 21st century, the U.S. economy lost 2,700 manufacturing jobs *every day*. It is common to hear people say, "We don't make stuff in America anymore," but the sentiment is wrong. It turns out that more goods were manufactured in the United States in 2016 than at any other point in its history, surpassing the previous peak reached in 2008 (Federal Reserve Bank of St. Louis, n.d.). The United States makes more stuff than it has ever made before. It just doesn't use so many humans to do it—and that's a good thing. Across all U.S. manufacturing, between 2002 and 2015, manufacturing output per hour of labor went up by 47 percent (Levinson, 2016). Compared to the heyday of American manufacturing, the average American worker employed in manufacturing in 2016 is more than *six times* as productive per hour as a worker in 1950. The reason that American workers are so much more productive is because they can work with more sophisticated technology, but this means that modern workers need higher levels of skill. Almost half of the manufacturing jobs that did not require a high school diploma in 2000 were gone by 2015, while the number of manufacturing jobs that required at least a master's degree rose by 32 percent (Levinson, 2016).

One common reaction to such changes in the working world is the fear that we are going to run out of jobs. Many people believe that there are only a certain number of jobs to go around, and if some of these jobs are destroyed, then there will not be enough work for everyone. This is an incorrect, albeit common, belief. Despite the millions of jobs that were lost in manufacturing, according to the Bureau of Labor Statistics, there are more people working in the United States than at any time in its history—160 million as of May 2017 (U.S. Bureau of Labor Statistics, 2017).

Interestingly, many of the new jobs being created do not demand much in the way of educational qualifications. In 2013, the U.S. Bureau of Labor Statistics projected that between 2012 and 2022, the U.S. economy will create just over four million new jobs for those with college degrees, but it will also create three million that require some education beyond high school but not a college degree, four million that require only a high school diploma, and another four million jobs that do not even require a high school diploma (U.S. Bureau of Labor Statistics, 2013). As far as can be seen, there will be jobs for people whatever their level of education in the United States. It is therefore probably not accurate to say to young people that they need to get a good education to get a job, but it does seem that education will be important to getting a *good* job.

We have already seen this in the changes that have occurred since the late 1990s. The greatest job destruction has not been for the lowest-skilled workers; rather, it has been for those doing routine jobs, whatever the skill level. And because computers are simpler and less expensive than robots, things like routine office work—what economists call *routine cognitive jobs*—have been easier to automate than manual work (Dvorkin, 2016). We think playing chess is an amazing human achievement, but for a few dollars, you can now buy a smartphone app that will beat most humans on the planet. What employers haven't been able to do yet is to use robots to stack shelves in Walmart in a cost-effective way, which is why humans do the job right now. But the message since the Industrial Revolution has been that as soon as a job can be done cost-effectively by a machine, it will be.

If having a valued skill no longer guarantees employment, then the only way to be sure of being employable is to be able to develop new skills, as Seymour Papert (1998) observes:

> So the model that says learn while you're at school, while you're young, the skills that you will apply during your lifetime is no longer tenable. The skills that you can learn when you're at school will not be applicable. They will be obsolete by the time you get into the workplace and need them, except for one skill. The one really competitive skill is the skill of being able to learn. It is the skill of being able not to give the right answer to questions about what you were taught in school, but to make the right response to situations that are outside the scope of what you were taught in school. We need to produce people who know how to act when they're faced with situations for which they were not specifically prepared.

This is why education—as opposed to training—is so important. Not only does education confer skills, but it also produces the ability to develop new skills.

The fundamental idea that education is the engine of future economic prosperity has been understood for many years, but studies have shown just how much education increases economic growth (or, conversely, just how much economic growth is limited by low educational achievement).

## The Difficulties of Raising Student Achievement

Successive governments have understood the importance of educational achievement and have sought to raise standards through a bewildering number of policy initiatives. Although most of these seemed like sensible measures at the time, the

depressing reality is that the net effect of the vast majority of these measures on student achievement has been close to, if not actually, zero.

A number of reform efforts have focused on the *structures* of schooling. In the United States, particular attention was given to reducing school size. The logic was simple: many high schools are very large and impersonal, and so the creation of smaller high schools should create more inclusive learning communities, which should then result in better learning. Advocates for smaller high schools also pointed to evidence that in many states, the highest test scores were in small high schools. However, they forgot to look at the other end of the distribution. The lowest test scores were also in small high schools (Wainer & Zwerling, 2006). The evidence suggests that small schools aren't any better, on average. They are just more likely to have extreme results—whether high or low—because they are small (Kahneman, 2011). The fewer students there are in a class, the greater the chance that, in a particular year, the students happen to be either very strong or very weak academically. In fact, the evidence suggests that smaller high schools are actually less effective than larger ones because teachers have to teach a range of courses, and therefore have fewer opportunities to specialize. As one high school student in Seattle summarizes, "There's just one English teacher and one mathematics teacher. They end up teaching things they don't really know" (Geballe, 2005).

The creation of smaller high schools can also be rather inefficient. In many cases, large high schools of around three thousand students are divided into five or six smaller high schools, each with five hundred or six hundred students but housed in the same building. Often, in such cases that I have seen, the only change is increased administrative costs, as a result of appointing six new principals for each of the small high schools and increasing the compensation of the existing principal for looking after six newly appointed junior principals.

In other cases, students have not experienced all the potential benefits of small high schools because leaders assumed the creation of small high schools was an end in itself, rather than a change in structure that would make other needed reforms easier to achieve. One benefit leaders hoped for was that smaller high schools would improve staff-student relationships, and with improved relationships, students would become more engaged in their learning. Students would interact with a smaller number of teachers, thus fostering the development of better staff-student relationships. This may well be effective, although it should be said that getting students engaged so that they can be taught something seems much less efficient than getting them engaged *by* teaching them something that engages them. But every student would still have a language arts teacher, a mathematics teacher, a science teacher, a social

studies teacher, and so on. The size of a high school does not affect the number of teachers a student meets in a day. Staff-student relationships can grow stronger if teachers loop through with their students so that the same teacher teaches a class for more than a single semester or year; however, this requires amendments to schedules and depends on having teachers who can teach multiple grades. Large high schools could easily incorporate this system, if they considered it a priority.

Other countries are going in the opposite direction. In England, for example, high-performing schools are asking their principals to assume responsibility for less successful schools by forming *federations* of schools—groups of schools with a single principal—but as yet, there is no evidence that this has led to improvement.

Other reforms have involved changes to the governance of schools. The most widespread such reform in the United States has been the introduction of charter schools. According to the Education Commission of the States (2017), forty-three states and the District of Columbia now have charter laws, but the evaluations of their impact on student achievement do not allow for any easy conclusions.

There is no doubt that some charter schools are achieving notable success, but others are not, and it appears that, at least to begin with, there were more of the latter than the former. In 2009, the Center for Research on Education Outcomes (CREDO) at Stanford University reported that across fifteen states and the District of Columbia, approximately one-half of charter schools obtain similar results to traditional public schools, one-third get worse results, and one-sixth get better results (CREDO, 2009). For the first twenty years of their operation, the net effect of charter schools was to *lower* student achievement rather than increase it, although this may well have been partly because most charters get less money per student (Miron & Urschel, 2010). Some charter schools, such as those that the Knowledge Is Power Program (KIPP) operate, are undoubtedly much more effective than comparable traditional public schools. Students at KIPP schools typically make an extra three to four months more progress each year, but they achieve this by having longer school days, some Saturday classes, and a longer school year (Tuttle et al., 2013). Each year KIPP school students spend 45 percent more time in school and make about 30 percent more progress—a clear example of diminishing returns. Moreover, while some charter schools are highly effective, most are not. An evaluation of the charter school system in Chicago (Hoxby & Rockoff, 2004) finds that students attending charter schools score higher on the Iowa Test of Basic Skills, but the effects are small: an increase of 4 percentile points for reading and just 2 percentile points for mathematics. As states have become better at closing less effective charter schools, the performance of charter schools has improved relative to traditional public schools, but a report from the CREDO team at Stanford—now covering twenty-two states

and Washington, DC—finds that the differences are small (CREDO, 2013). In mathematics, performance was higher in 29 percent of charter schools, about the same in 40 percent, and lower in 31 percent of schools. For reading, the figures were 56 percent, 25 percent, and 19 percent (CREDO, 2013). To put this in perspective, on average, a student attending a charter school in the United States would make 4 percent more progress (equivalent to eight days) than if he or she attended a traditional public school—an improvement worth having, but much less improvement than we need.

As the characteristics of successful charter schools become better understood, it will, no doubt, be possible to ensure that charter schools are more successful, but it is worth noting that the organizations that run the best charter schools are not keen to expand quickly, so any impact on the whole education system will be slow. For example, if we assume that the North American school population increases, as it has in the past, at a rate of around 0.7 percent per year, and the number of charter school places increases by 250,000 each year (the average rate over the last few years), then even if these new charter schools are as good as KIPP schools, it will be 2058 before students achieve an extra three weeks' learning per year (Wiliam, in press a). Of course, we could expand charter schools more aggressively, but this would be likely to result in lower quality, thus weakening the impact. Whatever their benefits, the creation of charter schools is not likely to have a substantial and immediate impact on student achievement (Carnoy, Jacobsen, Mishel, & Rothstein, 2005).

In England, the government has reconstituted many low-performing schools as "academies" that are run by philanthropic bodies but receive public funds equivalent to public schools, in addition to a large capital grant for school rebuilding. The principals of these academies have far greater freedom to hire and fire staff and are not required to follow national agreements on teacher compensation and benefits, nor to follow the national curriculum. Student test scores in these academies have risen faster than those in regular public schools, but this is to be expected, since such schools start from a lower baseline, and therefore have more room for improvement. A comparison with similarly low-performing schools *not* reconstituted as academies shows that they improve at the same rate (Machin & Wilson, 2009).

One of the most radical experiments in the organization of schooling has been taking place in Sweden. In 1992, the Swedish government invited for-profit providers to run public schools. Although many evaluations of this initiative found some successes, each of these studies contained significant methodological weaknesses. An evaluation from the Institute for the Study of Labour (Böhlmark & Lindahl, 2008), which corrected the flaws of earlier studies, found that the introduction of

for-profit education providers did produce moderate improvements in short-term outcomes such as ninth-grade GPA and in the proportion of students who chose an academic high school track. However, these improvements appeared to be concentrated in more affluent students and were transient. There was no impact on longer-term outcomes such as high school GPA, university attainment, or years of schooling (Böhlmark & Lindahl, 2008), nor on employment, earnings, or engagement with the criminal justice system (Wondratschek, Edmark, & Frölich, 2014).

In England, since 1986, secondary schools have applied for *specialist school* status, along the lines of magnet schools in the United States. Specialist schools do get higher test scores than traditional secondary schools in England, but they also get more money—around $200 more per student per year. The improvement in results achieved by specialist schools turns out to be just what you would expect if you gave traditional public schools an extra $200 per student per year (Mangan, Pugh, & Gray, 2007). Moreover, specialist schools do not get better results in the subjects in which they specialize than they do in other subjects (Smithers & Robinson, 2009).

Other reform efforts have focused on curriculum. Almost every country aspires to have a 21st century curriculum. For instance, the Scottish government has adopted a Curriculum for Excellence, but whether anything changes in Scottish classrooms remains to be seen, and the short-term results are not encouraging (OECD, 2016). Trying to change students' classroom experiences through changes in curriculum is very difficult. A bad curriculum well taught is usually a better experience for students than a good curriculum badly taught; pedagogy trumps curriculum. Or more precisely, pedagogy *is* curriculum, because what matters is *how* things are taught, rather than *what* is taught.

There is no standard definition of the term *curriculum*. The word originally (in English at least) described the selection of courses in Scottish universities in the 17th century, but over the years, it has come to mean *activities that educational organizations arrange to help their students learn the intended material.* However, there are at least three levels at which the word might be applied: (1) intended, (2) implemented, and (3) achieved. The *intended* curriculum includes the things that states or national governments determine students should learn in school. The *implemented* curriculum includes the textbooks and other materials that schools and districts adopt, and the *achieved* curriculum is what actually happens in classrooms. Needless to say, there is often some slippage between these three kinds of curricula. Often the textbooks that schools adopt do not align well with the intended curriculum (even though the publishers often claim that they do), and the way that teachers use those materials often does not accord with the intentions of those who wrote the textbooks (Wiliam, 2013).

While the way that teachers use textbooks may not always accord with the intentions of textbook publishers, textbooks do influence how teachers teach, and so there has been a great deal of interest in whether some textbooks are more effective than others. Some textbooks may align better with one state's standards than others. However, as more and more states have either adopted common standards, such as the Common Core State Standards or the Next Generation Science Standards, or other similar local standards, alignment has become less of an issue. What has become apparent, however, and particularly in mathematics, is that some textbooks are much more effective than others at teaching the same content.

Initially, most studies of textbook adoption have found little evidence that changes in textbooks alone have much impact on student achievement. However, particularly in elementary schools, just changing the textbooks can have a marked impact on student achievement, increasing the rate of student learning by up to 25 percent (Agodini & Harris, 2016). When researchers consider all grade levels, it is only when the programs change teaching practices and student interactions that a significant impact on achievement occurs (Slavin & Lake, 2008; Slavin, Lake, Chambers, Cheung, & Davis, 2009; Slavin, Lake, & Groff, 2009). However, at present, it does not seem to be possible to predict which textbooks are likely to be the most effective. We know that textbooks make a difference, but we don't know what makes the difference in textbooks. Thus, while textbook choice is important, it does not seem to be, at present, a reliable way of raising student achievement.

Many reforms look promising at the pilot stage but, when rolled out at scale, fail to achieve the same effects. In 1998, after less than a year in office, Tony Blair's Labour Party launched the National Literacy Strategy and, a year later, the National Numeracy Strategy for elementary schools in England and Wales. Although these programs showed promising results in the early stages, their effectiveness when rolled out to all elementary schools was equivalent to only one extra eleven-year-old in each elementary school reaching proficiency per year (Machin & McNally, 2009). Bizarrely, the fastest improvement in the achievement of English eleven-year-olds was in science, which had not been subject to any government reform efforts.

Other reform efforts have emphasized the potential impact of educational technology, such as computers, on classrooms. While there is no shortage of boosters for the potential of computers to transform education, reliable evidence of their impact on student achievement is rather hard to find. The history of computers in education is perhaps best summarized by the title *Oversold and Underused* (Cuban, 2002). This is not to say that computers have no place in education; some computer programs can be effective at teaching challenging content matter. One good example is Carnegie

Learning's Cognitive Tutor: Algebra I, developed over a period of twenty years at Carnegie Mellon University (Ritter, Anderson, Koedinger, & Corbett, 2007). The program has a very specific focus—teaching procedural aspects of ninth-grade algebra—and therefore should be used only for two or three hours per week, but it is more effective than many teachers at teaching this particular content (Pane, Griffin, McCaffrey, & Karam, 2014; Ritter et al., 2007). However, such examples are rare, and computers have failed to revolutionize our classrooms in the way leaders predicted (Bulman & Fairlie, 2016). As Heinz Wolff once said, "The future is further away than you think" (as cited in Wolff & Jackson, 1983).

Attention has become focused on the potential of interactive whiteboards. In the hands of expert practitioners, these are stunning pieces of educational technology, but as tools for improving educational outcomes at scale, they appear to be very limited. We know this from an experiment that took place in London. The English secretary for education, Charles Clarke, was so taken with the interactive whiteboard that he established a fund that doubled the number of interactive whiteboards in London schools. The net impact on student achievement was zero (Moss et al., 2007). But, say the technology boosters, you should provide professional development to go with the technology. This may be so, but if interactive whiteboards are only effective when teachers are given a certain number of hours of professional development, then surely it is right to ask whether the same number of hours of professional development could be more usefully, and less expensively, used in another way.

As a final example of an effort to produce substantial improvement in student achievement at scale, it is instructive to consider the impact of teachers' aides in England. One large-scale evaluation of the impact of support staff on student achievement found that teachers' aides actually lowered the performance of the students they intended to help (Blatchford et al., 2009), largely because in many schools, aides were routinely tasked to help the students with the most profound learning needs—a task for which they were not well suited. Of course, this does not mean that the use of teachers' aides cannot increase student achievement. Evidence from North Carolina suggests that teachers' aides can be cost-effective—particularly for minority students—if they are well managed and assigned suitable classroom roles (Clotfelter, Hemelt, & Ladd, 2016). However, all this means is that two or three teachers' aides can be as effective as a regular teacher. Where qualified teachers are in short supply, the deployment of teachers' aides may be a useful short-term measure. However, it is unlikely to have much impact on overall student achievement.

The reform efforts discussed here, and the history of a host of other reform efforts, show that improving education at scale is clearly more difficult than we often

imagine. Why have we pursued such ineffective policies for so long? Much of the answer lies in the fact that we have been looking in the wrong place for answers.

## Three Generations of School Effectiveness Research

Economists have known about the importance of education for economic growth for years, and this knowledge has led to surges of interest in studies of school effectiveness. Some schools appeared to get consistently good test results, while others seemed to get consistently poor results. The main thrust of the first generation of school effectiveness research, which began in the 1970s, was to understand the characteristics of the most effective schools. Perhaps if we understood that, we could reproduce the same effect in other schools.

Unfortunately, things are rarely that simple. Trying to emulate the characteristics of today's most effective schools would lead to the following three measures.

1. First, get rid of the boys. All over the developed world, girls are outperforming boys, even in traditionally male-dominated subjects such as mathematics and science (OECD, 2016). The more girls you have in your school, the better you are going to look.
2. Second, become a parochial school. Again, all over the world, parochial schools tend to get better results than other schools, although this appears to be more due to parochial schools tending to be more socially selective than public schools (see, for example, Cullinane, Hillary, Andrade, & McNamara, 2017).
3. Third, and most important, move your school into a nice, leafy, suburban area. This will produce three immediate benefits. First, it will bring you much higher-achieving students. Second, parents will better support their students, whether this is in terms of supporting the school and its mission or paying for private tuition. Third, the school will have more money—potentially lots more. Some American schools receive more than $40,000 per student per year, compared with others that receive less than $5,000 per student per year (National Public Radio, 2016).

In case it wasn't obvious, these are not, of course, serious suggestions. Girls' schools, parochial schools, and schools in affluent areas get better test scores primarily because of who goes there, rather than how good the schools are, as researchers pointed out in the second generation of school effectiveness studies (see, for

example, Thrupp, 1999). These researchers pointed out that most of the differences between school scores are due to the differences in students attending those schools rather than any differences in the quality of the schools themselves. The OECD data (Programme for International Student Assessment, 2010) are helpful in quantifying this. The PISA data show that 74 percent of the variation in the achievement of fifteen-year-olds in the United States is within schools, which means that 26 percent of the variation in student achievement is between schools (such as some schools getting better test scores than others). However, around two-thirds of the between-school variation is caused by differences in the students attending that school. This, in turn, means that only 8 percent of the variability in student achievement is attributable to the school—or, in reverse, that 92 percent of the variability in achievement is *not* attributable to the school (PISA, 2010). What this means in practice is that if fifteen out of a class of thirty students achieve proficiency in an average school, then seventeen out of thirty would do so in a "good" school (one standard deviation above the mean, or one of the best one-third of all schools) and thirteen out of thirty would do so in a "bad" school (one standard deviation below the mean). While these differences are no doubt important to the four students in the middle who are affected, they are, in my experience, much smaller than people imagine. It seems that Basil Bernstein (1970) was right, therefore, when he said that "education cannot compensate for society" and that we should be realistic about what schools can, and cannot, do (as cited in Thrupp, 1999).

However, as higher-quality data sets have become available, we have been able—in the third generation of school effectiveness studies—to dig a little deeper. In particular, where a data set allows us to compare a student's achievement at the beginning of the year and at the end of the year, we can estimate the school's *value added* (the difference between what a student knew when she arrived at a school and what she knew when she left). It turns out that as long as you go to school (and that's important), then it doesn't matter very much which school you go to, but it matters very much which classrooms you're in.

In the United States, the classroom effect appears to be at least four times the size of the school effect (PISA, 2007), which, predictably, has generated a lot of interest in what might be causing these differences. It turns out that these substantial differences between how much students learn in different classes have little to do with class size, how the teacher groups the students for instruction, or even the presence of between-class grouping practices (for example, tracking). The most critical difference is simply the quality of the teacher.

Parents have always understood how important having a good teacher is to their children's progress, but only since the mid-1990s have we been able to quantify exactly how much of a difference teacher quality makes.

## The Impact of Teacher Quality

For a long time, it seems, many people involved in education assumed that the correlation between teacher quality and student progress was effectively zero. In other words, if properly qualified, all teachers were equally good, and so on average, students should progress at the same rate in all classrooms. Of course, different students would progress at different rates according to their talents and aptitudes, but the assumption was that all teachers were comparable and therefore able to function like a *commodity.*

To an economist, a commodity is a good for which there is a demand, and it is *fungible*—a person can substitute one unit for another, since all units are assumed to be of equal quality. It is convenient for policymakers to treat teachers as a commodity, because then they can determine teacher compensation on a supply-and-demand basis. Teacher compensation could—like that for traders on the financial markets—be set based on the value they contribute, but this would mean that the best teachers would cost way too much—over $300,000 per year according to one study (Chetty et al., 2010). It is convenient for politicians to set a standard for "the qualified teacher," so that everyone who meets that standard gets in. Teacher compensation can then be determined by supply and demand—how much needs to be paid to get a qualified teacher in every classroom (although in this context, it is worth noting that this is not the basis that politicians tend to use to determine their own compensation!).

The desire of teacher unions to treat all teachers as equally good is understandable, because it generates solidarity among their members, but more important because performance-related pay is *in principle* impossible to determine fairly. Consider a district that tests its students every year from third through eighth grade and then uses the test score data to work out which teachers have added the most value each year. This looks straightforward, but there is a fatal flaw: no test can capture all that is important for future progress. A fourth-grade teacher who spends a great deal of time developing skills of independent and collaborative learning, who ensures that her students become adept at solving problems, and who develops her students' abilities at speaking, listening, and writing in addition to teaching reading may find that her students' scores on the fourth-grade mathematics and reading tests are not as

high as those of other teachers in her school who have been emphasizing only what is on the test. And yet, the teacher who inherits this class in fifth grade will look very good when the results of the fifth-grade tests are in, not because of what the fifth-grade teacher has done, but because of the firm foundations that the fourth-grade teacher laid.

In addition, evidence suggests that paying teachers bonuses for the achievement of their students does not raise test scores. Between 2006 and 2009, researchers selected teachers in Nashville, Tennessee, at random and offered bonuses of $15,000 for getting their students' achievement to match the highest-performing 5 percent of students, with lesser bonuses of $10,000 and $5,000 for matching the highest-performing 10 percent and 20 percent, respectively. An evaluation of the incentives found that the scores of the students taught by teachers offered bonuses were no higher than the scores of those taught by other teachers (Springer et al., 2010).

Such results seem to surprise many economists. They tend to assume that people's primary motivation is economic reward, and so offering cash incentives for people to try harder must surely increase results. They forget that such incentives work only when people are not already trying as hard as they can. There are, no doubt, some teachers who do not care about how well their students do, and for this small minority of teachers, incentives may work. But the vast majority of teachers are trying everything they can to increase their students' achievement. There is certainly no evidence that there are teachers who are holding on to a secret proven method for teaching fractions until someone pays them more money. So, performance-related pay is impossible to implement fairly, does not seem to work, and even if it can be made to work, will make a difference only for that small minority of teachers who are not already trying their best.

As noted previously, for many years, researchers and politicians assumed that one teacher was as good as another, providing each was adequately qualified for the job. However, in 1996, William Sanders and June Rivers published a paper based on an analysis of three million records on the achievement of Tennessee's entire student population from second to eighth grade. Because of the way they collected the data, it was possible to track the progress each individual student made and match that to the teacher who had been teaching them each year. They found that there were differences in how much students learned with different teachers, and that these differences were large. To show how large the differences were, they classified the teachers into five equally sized groups based on how much progress their students had made (low, below average, average, above average, and high). They then examined how an average eight-year-old student would fare, depending on what teachers he or she

got. What they found was rather surprising. A student who started second grade at the 50th percentile of achievement would end up at the 90th percentile of achievement after three years with a high-performing teacher but, if assigned to the classes of low-performing teachers for three years, would end up at the 37th percentile of achievement—a difference of over 50 percentile points. They found that increases in teacher quality were most beneficial for lower-achieving students, and the general effects were the same for students from different ethnic backgrounds (Sanders & Rivers, 1996).

Subsequent studies (for example, Rivkin, Hanushek, & Kain, 2005; Rockoff, 2004) confirmed the link between teacher quality and student progress on standardized tests. While different studies have found slightly different results, there is now a clear consensus among researchers that the correlation between teacher quality and student progress is at least 0.1, and may be over 0.2, especially for mathematics, as the data in table 1.1 clearly indicate. A correlation of 0.1 would mean that if a student was taught by an above-average teacher (for example, a teacher who is one standard deviation above the mean), then that student would make 0.1 standard deviations' more progress in a year than a student taught by an average teacher. Since for most students in these studies, one year's progress is around 0.4 standard deviations, this would equate to a 25 percent increase in the rate of learning.

**Table 1.1: Correlation Between Teacher Quality and Student Progress in Reading and Mathematics**

| Study | Location | Reading | Mathematics |
|---|---|---|---|
| Rockoff (2004) | New Jersey | 0.1 | 0.11 |
| Nye, Konstantopoulos, and Hedges (2004) | Tennessee | 0.26 | 0.36 |
| Rivkin, Hanushek, and Kain (2005) | Texas | 0.15 | 0.11 |
| Aaronson, Barrow, and Sander (2007) | Chicago | — | 0.13 |
| Kane, Rockoff, and Staiger (2008) | New York City | 0.08 | 0.11 |
| Jacob and Lefgren (2008) | — | 0.12 | 0.26 |
| Kane and Staiger (2008) | — | 0.18 | 0.22 |
| Koedel and Betts (2011) | San Diego | — | 0.23 |
| Rothstein (2010) | North Carolina | 0.11 | 0.15 |
| Hanushek and Rivkin (2010) | — | — | 0.11 |
| Chetty, Friedman, and Rockoff (2014) | — | 0.12 | 0.16 |

The estimates in table 1.1 are based on the progress made from one year to the next on standardized tests, and these numbers could be very different if we looked at other measures of student achievement. However, there is no evidence that having a

good teacher is more important for performance on standardized tests than it is for other measures of performance. As far as we know, having a good teacher makes a difference no matter what the subject, or the age of the student.

One objection to this argument is that teachers may appear to make more progress with their students not because they are better teachers, but because they teach higher-achieving students. There is no doubt that in many school districts, teachers with seniority have influence over the classes they are assigned to teach. To test this explanation, the Measures of Effective Teaching (MET) project, funded by the Bill and Melinda Gates Foundation, identified teachers who had been successful in one school, and reassigned them to teach in another school, often teaching students of different socioeconomic backgrounds. The project found that the teachers who were more successful in one school were more successful in a very different school (Kane, McCaffrey, Miller, & Staiger, 2013). While the circumstances in which teachers work—the time they have to plan instruction, the quality of curriculum with which they are working, the size of classes they teach—undoubtedly have an influence, there is something that successful teachers carry around with them in their heads that makes them more effective wherever they work.

The average correlations for reading and mathematics in table 1.1 are 0.14 and 0.18 respectively, so 0.15 is a reasonable average value for the correlation between teacher quality and student progress. What this means is that an increase of one standard deviation of teacher quality would result in an increase of 0.15 standard deviations in student achievement, equivalent to moving a student from the 50th to the 56th percentile of achievement. Averaged across K–12, one year's progress is equivalent to moving a student from the 50th to the 65th percentile of achievement, so having an above-average teacher (one standard deviation above the mean, or capable of increasing a student's achievement by 6 percentile points *in addition to* the 15 percentile points of one year's progress) improves student learning by an extra five months' progress per year.

At the extremes, these effects are even more pronounced. Take a group of fifty teachers. The students of the most effective teacher in that group will learn in six months what the students of the average teachers will learn in a year. And those students of the least effective teacher in that group of fifty teachers are likely to take two years to learn the same material. In other words, the very best teachers generate learning in their students at four times the rate of the least effective teachers.

Just as important, teacher quality appears to play a significant role in promoting equality of outcomes. In the United States, many policymakers seem to have assumed that excellence and equity are somehow in tension—that we can have one

or the other, but not both. However, evidence from international comparisons has shown that the countries with the high average scores also tend to have a narrow range of achievement (Bursten, 1992; Mullis, Martin, & Foy, 2008; OECD, 2016).

As noted previously, Sanders and Rivers (1996) find that increases in teacher quality confer greater benefits on low achievers than high achievers, and a particularly well-designed study of fifteen- and sixteen-year-olds (Slater, Davies, & Burgess, 2008) also finds that the benefits of having a high-performing teacher are greatest for low achievers (although, interestingly, it also finds that high achievers benefit more than those of average achievement). In their work in kindergarten and first-grade classrooms, Bridget Hamre and Robert Pianta (2005) find that in the classrooms where students make the most progress in reading and mathematics (as measured by scores on the revised Woodcock–Johnson Psychoeducational Battery, a standardized test of cognitive ability and academic achievement), students from socioeconomically disadvantaged backgrounds make as much progress as those from wealthy homes, and those with behavioral difficulties (such as aggressive or defiant behaviors) progress as well as those without. In other words, when teachers are more effective on average, they are disproportionately *more* effective for at-risk students.

This last finding is particularly important because it shows that Basil Bernstein was wrong—education *can* compensate for society, *provided it is of high quality.* Ideally, in the short term, we would concentrate on how to get the highest-quality teachers to the students who need them most—schools will only secure equitable outcomes by ensuring that the lowest-achieving students get the best teachers. In the short term, this means taking the best teachers away from the high-achieving students, which is politically challenging, to say the least.

In the longer term, a focus on improving teacher quality will mean that teacher allocation will no longer be a zero-sum game. Our focus on the achievement gap draws attention to the gap between the high achievers and the low achievers. The problem with thinking about this as an issue of gaps is that one can reduce the gap either by improving the performance of the lowest achievers or by reducing the achievement of the highest achievers. This leads back to the traditional, and now discredited, thinking that equity is the enemy of excellence. Rather than thinking about narrowing the gap, we should set a goal of *proficiency for all, excellence for many,* with all student groups fairly represented in the excellent. And the way to achieve this is simply to increase teacher quality. As Michael Barber says, "The quality of an education system cannot exceed the quality of its teachers" (Barber & Mourshed, 2007, p. 19).

## Ways to Increase Teacher Quality

The realization that teacher quality is the single most important variable in an education system has led to an exploration of how teacher quality can be improved, and there are only two options. The first is to attempt to replace existing teachers with better ones, which includes both deselecting serving teachers and improving the quality of entrants to the profession. The second is to improve the quality of teachers already in the profession.

Because past attempts to improve the performance of serving teachers have achieved so little success, some authors suggest that the only way to improve the profession is through replacement, including both rigorous deselection and increasing the threshold for entry into the profession (see, for example, Hanushek, 2010).

Teacher deselection may be politically attractive—after all, who could possibly be against getting rid of ineffective teachers? But it is hard to do, it may not work anyway, and even when it does work, it is a slow process (see, for example, Winters & Cowen, 2013). First, it is hard to do because, although we know that teachers make a difference, it is very difficult, if not impossible, to work out who really are the least effective teachers. Second, it may not work because, to be effective, you must be able to replace the teachers you deselect with better ones, and that depends on whether there are better potential teachers not currently teaching. Jack Welch famously believes in getting rid of the lowest-performing 10 percent of employees each year (Welch & Welch, 2005). Such an approach sounds a little like the joke that "firings will continue until morale improves," but even if it does not have a negative impact on those who remain employed, it is only effective if one can replace the deselected 10 percent with better employees. When those recruited to fill the vacancies are worse than those fired, the 10 percent rule is guaranteed to *lower* average employee quality.

The third problem with deselection is that it is very slow. Replacing the lowest 10 percent with teachers who are only slightly better will take many years to have any noticeable impact on average teacher quality. Given the difficulty with deselecting the right teachers, it is natural that much attention has focused instead on improving the quality of entrants into the profession.

When looking for ways to improve educational achievement, many people look at high-performing countries like Finland and Singapore and note that these countries hold teaching in high regard (Tucker, 2011). Therefore, not surprisingly, many people want to be teachers, so competition for places on preservice teacher preparation programs is intense. For example, at the end of twelfth grade, students at Finnish

high schools take a series of end-of-course tests. In addition, all students who want to go on to university in Finland must take a national written test. Those who perform well then take a second test set by the university to which they have applied. In 2014, the University of Helsinki had 1,650 applicants for the 120 places on its preservice teacher program (Sahlberg, 2015). The university selected approximately seventy students based on a combination of their school end-of-course test scores and their scores on the university tests, and fifty students based on the university tests alone. While it is not true to say that Finland recruits only the highest-achieving students, it certainly draws from the highest-achieving one-third of students (Ingersoll, 2007).

In contrast, the United States tends to draw teachers from the lower end of the college achievement range. According to Marigee Bacolod (2007), only around 10 percent of teachers who began their careers in the mid-1980s were academic *high achievers* in high school (in the top 20 percent), while for other professions, the figure is over 60 percent. This has, predictably, led to calls to recruit more academic high achievers into teaching. The problem with this attractive solution is that there is little evidence that academic achievement has much of a relationship with teacher quality (Harris & Sass, 2009).

Now this does not, of course, mean that academic ability is unimportant in teaching—clearly a certain amount of academic ability is necessary to be a teacher—and so it probably makes sense to require teachers to have college degrees. But what many people find surprising is that, beyond this, qualifications do not seem to matter much (Harris & Sass, 2009). Teachers with higher college grade point averages do not seem to be any more effective than other teachers. Perhaps even more surprisingly, teachers with master's degrees in education are no more effective than those with just a bachelor's degree (Harris & Sass, 2009). Some researchers have gone so far as to claim that the only teacher variable that consistently predicts how much students will learn is teacher IQ (Hanushek & Rivkin, 2006), although other studies (like Harris & Sass, 2009) find no consistent relationship between teachers' intellectual abilities and the progress of their students.

In the 21st century, researchers have made some progress in determining what kinds of teacher knowledge do contribute to student progress. For example, elementary school teachers' scores on a test of mathematical knowledge for teaching (MKT) correlated significantly with their students' progress in mathematics (Hill, Rowan, & Ball, 2005). Although the effect was greater than the impact of socioeconomic status or race, it was, in real terms, small; a one standard deviation increase in a teacher's mathematical knowledge for teaching resulted in a 4 percent increase in a student's rate of learning. In other words, students whose teacher scored highly

on the MKT (that is, one standard deviation above the mean) would learn in fifty weeks what a student whose teacher scored the average would learn in fifty-two weeks—a difference, but not a big one. Or, to put it another way, we saw earlier that one standard deviation of teacher quality increases the rate of student learning by around 40 percent, and we have just seen that one standard deviation of mathematical knowledge for teaching increases the rate of student learning by 4 percent. This suggests that subject knowledge accounts for only around 10 percent of the variability in teacher quality.

While the impact of teacher knowledge on student progress in Hill et al.'s (2005) research may be disappointingly small, this is, in fact, one of the strongest results in the research literature. A study of over thirteen thousand teachers, involving almost one million items of data on over three hundred thousand students in the Los Angeles Unified School District (LAUSD), shows that student progress is unrelated to teachers' scores on licensure examinations, nor are teachers with advanced degrees more effective (Buddin & Zamarro, 2009). Most surprising, in the study, there is no relationship between LAUSD teachers' scores on the Reading Instruction Competence Assessment (which all elementary school teachers must pass) and their students' scores in reading. As the researchers themselves note, since this test is a requirement for all elementary school teachers, those who fail the test are not permitted to teach, and so we cannot conclude that the test is not effective in screening out weaker teachers, but the results do suggest that the relationship between teachers' knowledge of reading pedagogy and student progress in reading is, at best, weak and perhaps nonexistent (Buddin & Zamarro, 2009).

In an article in the *New Yorker*, Malcolm Gladwell (2008a) likens this situation to the difficulties of finding a good quarterback for the National Football League (NFL). Apparently, for most positions, how well a player plays in college predicts how well he will fare in the NFL, but at quarterback, how well a player plays in college seems to be useless at predicting how well he will play in the pros.

One theory about why good—and often even outstanding—college quarterbacks fail in the NFL is that the professional game is so complex (Gladwell, 2008a). To try to mitigate this discrepancy, all players drafted in the NFL now take the Wonderlic Personnel Test—a fifty-item test that assesses arithmetic, geometric, logical, and verbal reasoning. Unfortunately, as several studies have shown (for example, Mirabile, 2005), there does not appear to be any clear relationship between scores on the Wonderlic and how good a quarterback will be in the NFL. In 1999, for example, of the five quarterbacks taken in the first round of the draft, only one—Donovan McNabb—is likely to end up in the Hall of Fame, and yet his score was the lowest of the five. Other quarterbacks scoring in the same range as McNabb include Dan

Marino and Terry Bradshaw—widely agreed to be two of the greatest quarterbacks ever (Mirabile, 2005). Although efforts continue to try to predict who will do well and who will not within the NFL, Gladwell (2008a) suggests that there is increasing acceptance that the only way to find out whether someone will do well in the NFL is to try him out in the NFL.

The same appears to be true for teaching. It may be that the only way to find out whether someone has what it takes to be a teacher is to try him or her out in the classroom, even though Thomas Kane and Douglas Staiger have estimated that we might need to try out four prospective teachers to get one good one (as cited in Gladwell, 2008a).

Even if we could identify in advance who would make the best teachers, doing anything useful with that information would take a long time. Suppose, for example, we could predict exactly how good each teacher was going to be. Suppose also that we had the luxury of so many people wanting to be teachers that we could raise the bar to a level whereby only two-thirds of those who are currently entering the profession would get in. Over time, this would certainly raise the quality of teachers. However, if we "raise the bar" for entry into teaching today, it would still be forty years before those who had begun teaching before the raising of the bar finally retired.

We can't wait that long. While deselecting the least effective teachers and trying to raise the quality of those entering the profession will have some effects, they are likely to be small, and nothing like the kinds of improvements in teacher quality our students need. If we are serious about securing our economic future, we have to help improve the quality of those teachers already working in our schools—what Marnie Thompson, my former colleague at the Educational Testing Service, calls the "love the one you're with" strategy.

## Conclusion

Improving educational outcomes is a vital economic necessity, and the only way that we can achieve this is by increasing the quality of the teaching force. Identifying the least effective teachers and deselecting them has a role to play, as does trying to increase the quality of those entering the profession, but as the data and the research studies examined in this chapter have shown, the effects of these measures will be small and will take a long time to materialize. In short, if we rely on these measures to raise student achievement, the benefits will be too small and will arrive too late to maintain the United States' status as one of the world's leading economies. Our future economic prosperity, therefore, depends on investing in those teachers already working in our schools.

chapter 2

# The Case for Formative Assessment

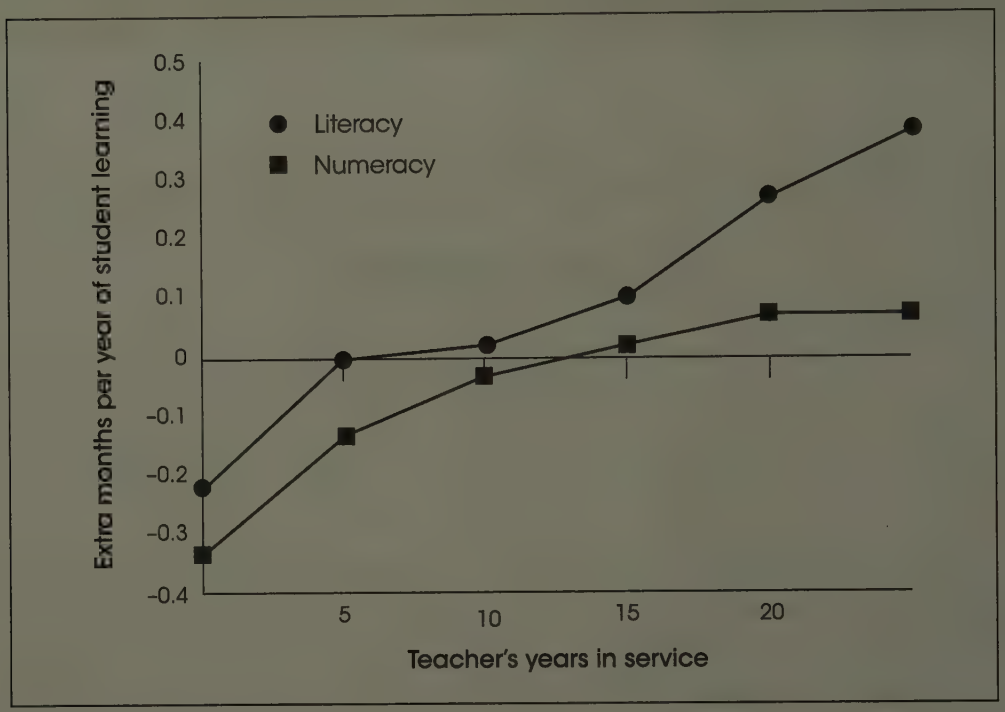
We've discussed how increasing the educational achievement of students is a national economic priority, and the only way to do that is to improve teacher quality. We also saw that deselecting existing teachers and improving the quality of entrants into the profession will have, at best, marginal effects, and so securing our economic future boils down to helping current teachers become more effective.

This chapter reviews the research on teacher professional development—specifically focusing on learning styles, educational neuroscience, and content-area knowledge—and shows that while there are many possible ways in which we could seek to develop the practice of serving teachers, attention to minute-by-minute and day-to-day formative assessment is likely to have the biggest impact on student outcomes. It continues by discussing the origins of formative assessment and by defining what, exactly, formative assessment is. The chapter concludes by presenting the strategies of formative assessment, which will be the subjects of each subsequent chapter in this book, and by discussing assessment as the bridge between teaching and learning.

## The Importance of Professional Development

Andrew Leigh (2010) analyzed a data set that includes test scores on ninety thousand Australian elementary school students and found that, as in the American research, whether the teacher has a master's degree or not makes no difference. He

did, however, find a statistically significant relationship between how much a student learns and the experience of the teacher, as seen in figure 2.1.




*Source: Adapted from Leigh, 2010.*

**Figure 2.1: Increases in teacher productivity with experience.**

The value added by a teacher increases particularly quickly in the first five years of teaching, but what is most sobering about figure 2.1 is the vertical axis. If a student's literacy teacher is a twenty-year veteran, the student will learn more than he will if his teacher is a novice, but not much more. In a year with a twenty-year veteran, a student will make an extra half-month's progress—in other words, a twenty-year veteran teacher achieves in thirty-four weeks what a novice teacher will take thirty-six weeks to achieve. Because of the size of the study, this result is statistically significant, and the improvement is worth having, but it is not a large difference. Therefore, it's not surprising that many have argued that the answer is more, and better, professional development for teachers.

Indeed, it would be hard to find anyone who would say that teacher professional development is unnecessary. Professional development for serving teachers is a statutory requirement in most states. However, most of these requirements are so loosely worded as to be almost meaningless. Pennsylvania's Act 48 (Act of Nov. 23, 1999, P.L. 529, No. 48) requires teachers to complete 180 hours of professional development that relates to an educator's certificate type or area of assignment every five years. Note

that there is no requirement for teachers to improve their practice or even to learn anything. The only requirement is to endure 180 hours of professional development.

Many states justify these requirements with the need for teachers to "keep up to date" with the latest developments in the field, but such a justification merely encourages teachers to chase the latest fad. One year, it's language across the curriculum; the next year, it's differentiated instruction. Because teachers are bombarded with innovations, none of these innovations has time to take root, so nothing really changes. And worse, not only is there little or no real improvement in what happens in classrooms, but teachers get justifiably cynical about the constant barrage of innovations to which they are subjected. The reason that teachers need professional development has nothing to do with professional updating. Teachers need professional development because the job of teaching is so difficult, so complex, that one lifetime is not enough to master it.

The fact that teaching is so complex is what makes it such a great job. At one time, André Previn was the highest-paid film-score composer in Hollywood, and yet one day, he quit. People asked him why he had given up this amazing job, and he replied, "I wasn't scared anymore." Every day, he was going in to his office knowing that his job held no challenges for him. This is not something that any teacher is ever going to have to worry about.

Even the best teachers fail. Talk to these teachers, and no matter how well the lesson went, they always can think of things that didn't go as well as they would have liked, things that they will do differently next time. But things get much, much worse when we collect the students' notebooks and look at what they thought we said. That's why Doug Lemov (2010) says that, for teachers, no amount of success is enough. The only teachers who think they are successful are those who have low expectations of their students. They are the sort of teachers who say, "What can you expect from these kids?" The answer is, of course, a lot more than the students are achieving with those teachers. The best teachers fail all the time because they have such high aspirations for what their students can achieve (generally much higher aspirations than the students themselves have).

People often contact me and ask whether I have any research instruments for evaluating the quality of teaching. I don't, because working out which teachers are good and which teachers are not so good is of far less interest to me than helping teachers improve. No teacher is so good—or so bad—that he or she cannot improve. That is why we need professional development.

Although there is widespread agreement that professional development is valuable, there is much less agreement about what form it should take, and there is

little research about what should be the focus of teacher professional development. However, there does seem to be a consensus that *one-shot deals*—sessions ranging from one to five days held during the summer—are of limited effectiveness, even though they are the most common model (Muijs, Kyriakides, van der Werf, Creemers, Timperley, & Earl, 2014). The following sections highlight some of the more popular areas of focus for professional development.

## Learning Styles

Many teachers are attracted to developments such as theories pertaining to students' learning styles. The idea that each learner has a particular preferred style of learning is attractive—intuitive even. It marries up with every teacher's experience that students really are different; it just feels right. However, there is little agreement among psychologists about what learning styles are, let alone how to define them. One review of the research in this area finds seventy-one different models of learning styles (Coffield, Moseley, Hall, & Ecclestone, 2004). Indeed, it is difficult not to get the impression that the proposers of new classifications of learning styles have followed Annette Karmiloff-Smith's advice: "If you want to get ahead, get a theory" (Karmiloff-Smith & Inhelder, 1974/1975). Some of the definitions, and the questionnaires used to measure them, are so flaky that one may classify an individual as having one learning style one day and a different one the next (Boyle, 1995). Others do seem to tap into deep and stable differences between individuals in how they think and learn, but there does not appear to be any way to use this in teaching.

Although many studies have tried to show that taking students' individual learning styles into account improves learning, evidence remains elusive (Coffield et al., 2004). The Association for Psychological Science asked a blue-ribbon panel of America's leading psychologists of education to review the available research evidence to see whether there was evidence that teaching students in their preferred learning style would have an impact on student achievement. They realized that any experiment that showed the benefit of teaching students in their preferred learning style (what they called the *meshing hypothesis*) would have to satisfy three conditions.

1. Following some assessment of their presumed learning style, teachers would divide learners into two or more groups (for example, visual, auditory, and kinesthetic learners).
2. Teachers would randomly allocate learners within each of the learning-style groups to at least two different methods of instruction (for example, visual- and auditory-based approaches).
3. Teachers would give all students in the study the same final test of achievement.

In such an experiment, the meshing hypothesis would be supported if the results showed that the learning method that optimized test performance of one learning-style group (for example, visual learners) was different from the learning method that optimized the test performance of a second learning-style group (for example, auditory learners). In their review, Harold Pashler and colleagues found only one study that gave even partial support to the meshing hypothesis, and two that clearly contradicted it. Their conclusion was stark: "If classification of students' learning styles has practical utility, it remains to be demonstrated" (Pashler, McDaniel, Rohrer, & Bjork, 2008, p. 117).

Now, of course, the fact that there is currently no evidence in favor of the meshing hypothesis does not mean that such evidence will not be forthcoming in the future; absence of evidence is not evidence of absence. However, it could be that the whole idea of learning styles research is misguided because its basic assumption—that the purpose of instructional design is to make learning easy—may just be incorrect.

Since the pioneering work of Hugh Carlton Blodgett in the 1920s, psychologists have found that performance on a learning task is a poor predictor of long-term retention (for a summary of this research, see Soderstrom & Bjork, 2015). More precisely, when learners do well on a learning task, they are likely to forget things more quickly than if they do badly on the learning task; good instruction creates "desirable difficulties" (Bjork, 1994, p. 193) for the learner. As Daniel Willingham (2009) says, "memory is the residue of thought" (p. 41). By trying to match our instruction to our students' preferred learning style, we may, in fact, be reducing learning. If students do not have to work hard to make sense of what they are learning, then they are less likely to remember it in six weeks' time. Perhaps the most important takeaway from the research on learning styles is that teachers need to know about learning styles if only to avoid the trap of teaching in the style they believe works best for them. A review of the literature on learning styles and learning strategies (Adey, Fairbrother, Wiliam, Johnson, & Jones, 1999) concludes that:

> The only feasible "solution" is that teachers should NOT try to fit their teaching to each child's style, but rather that they should become aware of different styles (and help students also to become aware of different styles) and then encourage all students to use as wide a variety of styles as possible. Students need to learn both how to make the best of their own learning style and also how to use a variety of styles, and to understand the dangers of taking a limited view of their own capabilities. (p. 36)

As long as teachers vary their teaching style, then it is likely that all students will get some experience of being in their comfort zone and some experience of being pushed

beyond it. Ultimately, we should remember that teaching is interesting because our students are so different, but only possible because they are so similar.

## Educational Neuroscience

Another potential area for teacher professional development—and one that has received a lot of publicity—is applying what we are learning about the brain to the design of effective teaching. Cognitive psychologists work to understand *what* the brain does and *how* it does what it does, while neuroscientists try to connect what the brain does to its physiology.

Some of the earliest attempts to relate brain physiology to educational matters relate to the respective roles of the left and right sides of the brain in various kinds of tasks in education and training, despite clear evidence that the conclusions being drawn were unwarranted (see, for example, Hines, 1987). Schools have been inundated with suggestions for how they can use the latest findings from cognitive neuroscience to develop *brain-based education*, and despite the wealth of evidence that these claims are at best premature and at worst simply disingenuous (for example, Bruer, 1997, 1999; Goswami, 2006; Howard-Jones, 2009), many *neuromyths* still abound.

- Approximately 50 percent of teachers in China, Greece, the Netherlands, Turkey, and the United Kingdom believe that we only use about 10 percent of our brains, and more than 90 percent of teachers in these countries believe that instruction in students' preferred learning styles is more effective (Howard-Jones, 2014). Neither of these claims is actually true.
- People are more likely to believe a psychological report if the explanation claims to be based in neuroscience, even if the explanation is nonsense (Weisberg, Keil, Goodstein, Rawson, & Gray, 2008).
- Over 50 percent of teachers in the Netherlands and the United Kingdom believe that children are less attentive after consuming drinks or snacks that contain a lot of sugar (they're not), and 90 percent believe that differences in whether the left or the right brain is dominant can help explain individual differences among learners (they can't; Dekker, Lee, Howard-Jones, & Jolles, 2012).
- Many believe that people remember 10 percent of what they read, 20 percent of what they hear, 30 percent of what they see, 50 percent of what they hear and see, 70 percent of what they see and write, and

> 90 percent of what they do, despite the fact that there is absolutely no evidence to support these suspiciously neat percentages (De Bruyckere, Kirschner, & Hulshof, 2015).

Other neuromyths include the idea that the left side of our brain is analytical and the right side is creative, that you can train your brain with activities like Brain Gym (www.braingym.org), that male and female brains are different, that listening to classical music can improve a child's cognitive development (the so-called *Mozart effect*), or that we can learn when we are asleep. None of these is true as far as we know right now (De Bruyckere, Kirschner, & Hulshof, 2015). In fact, we know a great deal about how the brain works and what kinds of activities help students learn, but these findings come from cognitive science rather than neuroscience. Neuroscience, rather, provides plausible explanatory mechanisms for things we already knew from cognitive science. Two leading experts in the field of neuroscience and education, Sergio Della Sala and Mike Anderson (2011), sum it up thus in their "opinionated introduction" to their book, *Neuroscience in Education*:

> While the use of the term "neuroscience" is attractive for education it seems to us that it is cognitive psychology that does all the useful work or "heavy lifting." The reason for this is straightforward. We believe that for educators, research indicating that one form of learning is more efficient than another is more relevant than knowing where in the brain that learning happens. There is indeed a gap between neuroscience and education. But that gap is not filled by the "interaction" of neuroscientists and teachers (nearly always constituted by the former patronizing the latter) or "bridging" the two fields by training teachers in basic neuroscience and having neuroscientists as active participators in educating children. Rather what will ultimately fill the gap is the development of evidence-based education where that base is cognitive psychology. (p. 3)

## Content-Area Knowledge

If training teachers in cognitive neuroscience isn't going to help, what about increasing teachers' knowledge of their subjects? After all, surely the more teachers know about their subjects, the more their students will learn.

There is evidence that teachers in countries that are more successful in international comparisons than the United States appear to have stronger knowledge of the subjects they are teaching (Babcock et al., 2010; Ma, 1999), and this, at least in part, appears to be responsible for a widespread belief that teacher professional development needs to be focused on teachers' knowledge of the subject matter they are teaching.

It is important to note that not all kinds of subject-matter knowledge have the same impact on student progress. A study of German high school mathematics teachers found that students did not make more progress when their teachers had advanced mathematics knowledge (such as knowledge of mathematics studied at university). However, when teachers had a profound *understanding* of the school-level mathematics they were teaching, then, echoing Heather Hill, Brian Rowan, and Deborah Ball's (2005) study, students did make more progress (Baumert et al., 2010). Thus, it appears that an in-depth understanding of the curriculum may be more beneficial to student progress than advanced study of a subject on the part of the teacher.

Most studies of the relationship between teacher subject-matter knowledge and student progress, including those by Hill et al. (2005) and Jurgen Baumert et al. (2010) discussed previously, are cross-sectional in nature; researchers look to see whether the teachers whose classes make more progress have higher levels of subject knowledge. However, even if a link is found, it is not clear what this means. It could be that what really matters is general intellectual ability—that those with higher intellectual ability find learning their subject easier, and also make more effective teachers. To rule this out, we need experimental studies, where some teachers work on improving their subject knowledge while others work on something else, and then we compare their students' progress. Here, the results are rather disappointing.

Summer professional development workshops do increase teachers' knowledge of their subjects (Hill & Ball, 2004), but most studies that have increased teachers' subject knowledge find little or no knock-on effects on student achievement. For example, an evaluation of professional development designed to improve second-grade teachers' reading instruction found that an eight-day content-focused workshop increased teachers' knowledge of scientifically-based reading instruction and also improved the teachers' classroom practices on one out of three instructional practices that had been emphasized in the professional development (Garet et al., 2008). However, at the end of the following school year, there was no impact on the students' reading test scores. More surprising, even when supplementing the workshop with in-school coaching, the effects were the same.

A similar story emerges from an evaluation of professional development for middle school mathematics teachers in seventy-seven schools in twelve districts (Garet et al., 2010). The districts implemented the program as intended, which resulted in an average of fifty-five hours of additional professional development for participants (who had been selected by lottery). Although the professional development had been specifically designed to be relevant to the curricula that teachers were using in their classrooms and did have some impact on teachers' classroom practice (specifically the

extent to which they engaged in activities that elicited student thinking), there was no impact on student achievement, even in the specific areas on which the intervention focused (ratio, proportion, fractions, percentages, and decimals). A study that attempted to improve mathematics and science learning in early years teaching found that increasing teachers' subject knowledge had no impact on student achievement (Piasta, Logan, Pelatti, Capps, & Petrill, 2015).

These findings are clearly counterintuitive. It seems obvious that teachers need to know about the subjects they are teaching, yet the relationship between teachers' knowledge of the subjects and their students' progress is weak. Attempts to improve student outcomes by increasing teachers' subject knowledge appear to be almost entirely failures.

Of course, these failures could be due to our inability to capture the kinds of subject knowledge that are necessary for good teaching, but they suggest that there is much more to good teaching than just knowing the subject. We know that teachers make a difference, but we know much less about what makes the difference in teachers. However, there is a body of literature that shows a large impact on student achievement across different subjects, across different age groups, and across different countries, and that is the research on formative assessment.

## The Origins of Formative Assessment

Polymath and academic philosopher Michael Scriven coined the term *formative evaluation* in 1967 to describe the role that evaluation could play "in the on-going improvement of the curriculum" (p. 41). He contrasted this with summative evaluation. Summative evaluation's job is:

> To enable administrators to decide whether the entire finished curriculum, refined by use of the evaluation process in its first role, represents a sufficiently significant advance on the available alternatives to justify the expense of adoption by a school system. (Scriven, 1967, pp. 41–42)

Two years later, Benjamin Bloom (1969) applied the same distinction to classroom tests:

> Quite in contrast is the use of "formative evaluation" to provide feedback and correctives at each stage in the teaching-learning process. By formative evaluation we mean evaluation by brief tests used by teachers and students as aids in the learning process. While such tests may be graded and used as part of the judging and classificatory function of evaluation, we see much more effective use of formative evaluation if it is separated from the grading process and used primarily as an aid to teaching. (p. 48)

Bloom (1969) went on to say, "Evaluation which is directly related to the teaching-learning process as it unfolds can have highly beneficial effects on the learning of students, the instructional process of teachers, and the use of instructional materials by teachers and learners" (p. 50).

Although educators used the term *formative* infrequently in the twenty years after Bloom's (1969) research, a number of research reviews began to highlight the importance of using assessment to inform instruction, the best known of which is *cognitively guided instruction* (CGI).

In the original CGI project, a group of twenty-one elementary school teachers participated, over a period of four years, in a series of workshops that showed teachers extracts of videotapes selected to illustrate critical aspects of children's thinking. The researchers then prompted teachers to reflect on what they had seen, by, for example, challenging them to relate the way a child had solved one problem to how he or she had solved or might solve other problems (for a summary of the whole project, see Fennema et al., 1996). Throughout the project, researchers encouraged the teachers to make use of the evidence they had collected about the achievement of their students to adjust their instruction to better meet their students' learning needs. Students taught by CGI teachers did better in number fact knowledge, understanding, problem solving, and confidence (Carpenter, Fennema, Peterson, Chiang, & Loef, 1989), and four years after the end of the program, the participating teachers were still implementing the principles of the program (Franke, Carpenter, Levi, & Fennema, 2001).

The power of using assessment to adapt instruction was vividly illustrated in a 1991 study of the implementation of the measurement and planning system (MAPS), in which 29 teachers, each with an aide and a site manager, assessed the readiness for learning of 428 kindergarten students (Bergan, Sladeczek, Schwarz, & Smith, 1991). The researchers tested students in mathematics and reading in the fall and again in the spring. Their teachers learned to interpret the test results and to use the *classroom activity library*—a series of activities typical of early grades instruction but tied specifically to empirically validated developmental progressions—to individualize instruction. The researchers then compared these students' performances with the performances of 410 other students taught by 27 different teachers. At the end of the year, 27 percent of the students in the control group were referred for placement and 20 percent were actually placed into special education programs for the following year. In the MAPS group, only 6 percent were referred, and fewer than 2 percent were placed in special education programs.

In addition to these specific studies, in the late 1980s, a number of research reviews began to highlight the importance of using assessment to inform instruction. A review by Lynn Fuchs and Douglas Fuchs (1986) synthesized findings from twenty-one different research studies on the use of assessment to inform the instruction of students with special needs. They found that regular assessment (two to five times per week) with follow-up action produced a substantial increase in student learning. A couple of other findings were notable. First, some studies required teachers, before assessing their students, to make up systematic evaluation rules that would tell the teachers when or how they were to make changes to the instructional plans they had made for their students. In other studies, teachers made judgments about what instructional changes might be needed only after seeing their students' results. Both strategies increased student achievement, but the benefit was twice as great when the teachers used rules, rather than judgments, to determine what to do next. Second, when teachers tracked their students' progress with graphs of individual students' achievements as a guide and stimulus to action, the effect was almost three times as great as when they didn't track progress. These findings suggest that when teachers rely on evidence to make decisions about what to do next, students learn more.

Over the next two years, two further research reviews, one by Gary Natriello (1987) and the other by Terence Crooks (1988), provided clear evidence that while classroom assessment can improve learning, it can also have a substantial negative impact on student achievement. Natriello (1987) concludes that much of the research he reviewed was difficult to interpret because of a failure to make key distinctions in the design of the research (for example, between the quality and quantity of feedback). As a result, although some studies showed that assessment could be harmful, it was not clear why. He also points out that assessments serve a number of different purposes in schools, and many of the studies showed that assessment that is designed for selecting students (for example, by giving a grade) is not likely to improve student achievement as much as assessment that is specifically designed to support learning. Crooks's (1988) paper focuses specifically on the impact of assessment practices on students and concludes that although classroom assessments do have the power to influence learning, too often the use of assessments for summative purposes—grading, sorting, and ranking students—gets in the way.

In 1998, Paul Black and I sought to update Natriello's and Crooks's reviews. One of the immediate difficulties we encountered was how to define the field of study. Their reviews cited 91 and 241 references respectively, and yet only 9 references were common to both papers. Neither paper cited Fuchs and Fuchs's (1986) review. While many reviews of research use electronic searches to identify relevant studies, we found that the keywords used by authors were just too inconsistent to be of much

help. Where one researcher might use the term *formative assessment*, another might use *formative evaluation*, and a third might use *responsive teaching*. In the end, we decided that there was no alternative to actually going into the library and physically examining each issue, from 1987 to 1997, of seventy-six education and psychology journals that we thought most likely to contain relevant research studies. We read the abstracts, and where those looked relevant, read the studies. Through this process, we found just over 600 studies related to the field of classroom assessment and learning, of which approximately 250 were directly relevant.

We did consider at this point conducting a formal meta-analysis of the studies we had identified, but we quickly realized that with such a diverse range of studies, meta-analysis would simply not be appropriate (see Wiliam, 2016, for an extended analysis of the problems with meta-analysis in education). Instead we conducted what we might call a *configurative review* (Gough, 2015) because our main purpose was to make sense of the field rather than to quantify the effects of classroom assessment processes on students. However, many of the studies that we reviewed provided considerable evidence that attention to classroom assessment processes could substantially increase the rate of student learning, in some cases effectively doubling the speed of student learning. We realized that because of the diversity of the studies, there was no simple recipe to easily apply in every classroom, but we were confident we had identified some fruitful avenues for further exploration:

> Despite the existence of some marginal and even negative results, the range of conditions and contexts under which studies have shown that gains can be achieved must indicate that the principles that underlie achievement of substantial improvements in learning are robust. Significant gains can be achieved by many different routes, and initiatives here are not likely to fail through neglect of delicate and subtle features. (Black & Wiliam, 1998a, pp. 61–62)

While we did not conduct a formal meta-analysis, in a subsequent publication (Black & Wiliam, 1998b), we did try to provide some indication for practitioners and policymakers of the likely potential benefits of formative assessment. We suggested that the effective use of formative assessment would increase achievement by between 0.4 and 0.7 standard deviations, which would be equivalent to a 50 to 70 percent increase in the rate of student learning (see Wiliam, 2006, for details).

While we were confident that the research evidence that we had compiled made a compelling case for making classroom formative assessment a priority, we were not sure that these ideas could be implemented in real classrooms, especially where students were regularly subjected to external standardized tests, and where teachers were held accountable for their students' achievement.

We therefore recruited twenty-four (later expanded to thirty-six) secondary school mathematics and science teachers in six schools in two districts in England to help us explore what classroom formative assessment might look like in classrooms (Black, Harrison, Lee, Marshall, & Wiliam, 2003). The work with teachers had two main components. The first was a series of eight workshops over an eighteen-month period, which introduced teachers to the research base underlying how assessment can support learning, allowed them the opportunity to develop their own plans for implementing formative assessment practices, and, at later meetings, provided them the time to discuss with colleagues the changes they had attempted to make in their practice. Most of the teachers' plans contained reference to two or three important areas in their teaching in which they were seeking to increase their use of formative assessment, generally followed by details of techniques that they could use to make this happen. The second component was a series of visits to the teachers' classrooms, so that the researchers could observe teachers implementing some of the ideas they had discussed in the workshops and could discuss how their ideas could be put into practice more effectively.

Because each teacher had made his or her own decisions about what aspect of formative assessment to emphasize and which classes to try it with, it was impossible to use a traditional experimental design to evaluate the effects of our intervention. Therefore, we designed a *poly-experiment*. For each class with which a teacher was trying out formative assessment techniques, we looked for the most similar comparison class and set up a mini-experiment in which we compared the test scores of the class that was using formative assessment with the test scores of the comparison class. In some cases, this was a parallel class taught by the same teacher; in some cases, it was a similar class to one the teacher had taught in previous years; and in other cases, it was a similar class taught by a different teacher. This experimental design is not as good as a random-allocation trial, because the teachers participating in the experiment might have been better teachers to begin with, and so the results need to be interpreted with some caution. Nevertheless, in this study, using scores on externally scored standardized tests, the students with which the teachers had used formative assessment techniques made twice as much progress over the year (Wiliam, Lee, Harrison, & Blade, 2004).

## Definitions of Formative Assessment

As the evidence that formative assessment can have a significant impact on student learning accumulates, many researchers have proposed a variety of definitions of *formative assessment*. In our original review, Paul Black and I (1998a) define formative

assessment "as encompassing all those activities undertaken by teachers, and/or by their students, which provide information to be used as feedback to modify the teaching and learning activities in which they are engaged" (p. 7). Writing around the same time, Bronwen Cowie and Beverley Bell (1999) qualify this slightly by requiring that teachers and students act upon the information from the assessment while learning takes place. They define formative assessment as "the process used by teachers and students to recognise and respond to student learning in order to enhance that learning, *during the learning* [emphasis added]" (Cowie & Bell, 1999, p. 32, emphasis added). Others also emphasize the need for action during instruction and define formative assessment as "assessment carried out during the instructional process for the purpose of improving teaching or learning" (Shepard et al., 2005, p. 275). Reviewing practice across eight countries, OECD defines formative assessment as "frequent, interactive assessments of students' progress and understanding to identify learning needs and adjust teaching appropriately" (as cited in Looney, 2005, p. 21).

What is notable about these definitions is that, however implicitly, they regard formative assessment as a *process*. Others tend to regard formative assessment as a tool. For example, Stuart Kahl (2005), cofounder of Measured Progress, defines formative assessment as "a tool that teachers use to measure student grasp of specific topics and skills they are teaching. It's a 'midstream' tool to identify specific student misconceptions and mistakes while the material is being taught" (p. 11). Indeed, it appears educators more often use *formative assessment* to refer to a particular kind of assessment instrument than a process to improve instruction.

The difficulty with trying to make the term *formative assessment* apply to a thing (the assessment itself) is that it just does not work. Consider an advanced placement (AP) calculus teacher who is getting her students ready to take their examination. Like many teachers, she has her students take a practice examination under formal test conditions. Most teachers would then collect the papers, score them, write comments for the students, and return the papers to the students so that they could see where they went wrong. However, this calculus teacher does something slightly different. She collects the papers at the end of the examination, but she does not score them. Instead, during her next period with the class, each group of four students receives its unscored papers and one blank examination paper, and has to compile the best composite examination paper response that it can. Within each group, the students review their responses, comparing their answers to each question and discussing what the best answer would be. Toward the end of the period, the teacher reviews the activity with the whole class, asking each group to share its agreed-on answers with the rest of the class.

The AP calculus assessment that the teacher uses was designed entirely for summative purposes. The College Board designs AP exams to confer college-level credit so that students passing the exams at a suitable level are exempt from introductory courses in college. However, this teacher uses the assessment instrument formatively—what Black and I have called "formative use of summative tests" (Black et al., 2003, p. 53). Describing an assessment as formative is, in fact, what philosopher Gilbert Ryle (1949) calls a *category error*: the error of ascribing to something a property that it cannot have, like describing a rock as happy. Because the teacher can use the same assessment both formatively and summatively, the terms *formative* and *summative* make much more sense as descriptions of the function that assessment data serve, rather than of the assessments themselves (Wiliam & Black, 1996).

Some people (for example, Popham, 2006; Shepard, 2008) call for the term *formative assessment* not to be used at all, unless instruction is improved. In the United Kingdom, the Assessment Reform Group argues that using assessment to improve learning requires five elements to be in place (as cited in Broadfoot et al., 1999).

1. Providing effective feedback to students
2. Actively involving students in their own learning
3. Adjusting teaching to take into account the assessment results
4. Recognizing the profound influence assessment has on students' motivation and self-esteem, both of which are crucial influences on learning
5. Needing students to be able to assess themselves and understand how to improve

The group suggests that *formative assessment*—at least in the way many people use it—is not a helpful term for describing such uses of assessment because, as it says, "the term 'formative' itself is open to a variety of interpretations and often means no more than that assessment is carried out frequently and is planned at the same time as teaching" (Broadfoot et al., 1999, p. 7). Instead, it suggests that it would be better to use the phrase *assessment for learning*.

The earliest use of the term *assessment for learning* appears to be in the book *Assessment for Learning in the Mentally Handicapped* (Mittler, 1973). Harry Black (1986) used the term as the title of a chapter in the book *Assessing Educational Achievement*, and Mary James brought it to a wider audience as the title of a paper presented at the 1992 annual conference of the Association for Supervision and Curriculum Development in New Orleans (James, 1992), but the term has become common in North

America as a result of the work of Rick Stiggins, who adopted the term *assessment for learning* as being very different from *formative assessment.*

In the United States, for many years, educators used the term *formative assessment* to describe a process for monitoring student achievement. Students took assessments at regular periods (typically four to ten weeks), and teachers then looked at the resulting data to determine which students were making sufficient progress and which were not. Where students were not making sufficient progress, teachers would investigate what might be done to improve progress (such assessments are also called *benchmark assessments* or *interim assessments*).

Now it is important to realize that monitoring student progress is a good thing to do. Any well-run organization should be able to monitor its progress toward its goals. As W. Edwards Deming is reputed to have said, "In God we trust. All others bring data" (Hastie, Tibshirani, & Friedman, 2009, p. vii). However, if formative assessment merely identifies which students are falling behind, then it limits the impact on student achievement. It is in response to this limited view of formative assessment that Rick Stiggins (2005), founder of the Assessment Training Institute, writes:

> If formative assessment is about more frequent, assessment FOR learning is about continuous. If formative assessment is about providing teachers with evidence, assessment FOR learning is about informing the students themselves. If formative assessment tells users who is and who is not meeting state standards, assessment FOR learning tells them what progress each student is making toward meeting each standard while the learning is happening—when there's still time to be helpful. (pp. 1–2)

However, just replacing the term *formative assessment* with the term *assessment for learning* merely clouds the definitional issue (Bennett, 2011). What really matters is what kind of processes we value, not what we call them. The problem, as researcher Randy Bennett (2011) points out, is that it is an oversimplification to say that formative assessment is *only* a matter of process or *only* a matter of instrumentation. Good processes require good instruments, and instruments are useless unless teachers use them intelligently.

The original, literal meaning of the word *formative*, according to *Merriam-Webster's Online Dictionary*, is "capable of alteration by growth and development" ("formative," 2017). This suggests that formative assessments should shape instruction—our formative experiences are those that have shaped our current selves—and so we need a definition that can accommodate all the ways in which assessment can shape instruction. And there are many. Consider the following eight scenarios.

1. In spring 2016, a science curriculum supervisor needs to plan the summer workshops that the district will offer to its middle school science teachers. She analyzes the scores the district's middle school students obtained on the 2015 state tests and notes that while the science scores are generally comparable to those of the rest of the state, the students in her district appear to be scoring rather poorly on items involving physical sciences when compared with those testing life sciences. She decides to make physical science the focus of the professional development activities offered in summer 2016, which are well attended by the district's middle school science teachers. Teachers return to school in fall 2016 and use the revised instructional methods they have developed over the summer. As a result, when students take the state test in spring 2017, the achievement of middle school students in the district on items involving physical sciences increases, and so the district's performance on the state tests, reported in summer 2017, improves.
2. Each year, a group of algebra 1 high school teachers reviews students' performance on a statewide algebra 1 test and, in particular, looks at the facility (proportion correct) for each item on the test. When item facilities are lower than the group expects, the group looks at how teachers prepared and delivered instruction on that aspect of the curriculum and considers ways in which teachers can strengthen the instruction in the following year.
3. A school district administers a series of interim tests, tied to the curriculum, at intervals of six to ten weeks to check on student progress. The district uses past experience to determine a threshold that gives students an 80 percent chance of passing the state test, and requires students whose interim test scores fall below the threshold to attend additional instruction on Saturday mornings.
4. Since 2003, the School District of Philadelphia has mandated a core curriculum that includes a tightly sequenced planning and scheduling timeline, in which the school year is divided up into a number of six-week cycles. In each six-week cycle, the district expects teachers to use the first five weeks for instruction, at the end of which students take a multiple-choice test, which the teachers can use to determine how to spend the final week of the cycle. If students have done well, teachers typically schedule enrichment and enhancement activities, but if there

are significant weaknesses in students' understanding, the final week becomes a "re-teaching week" (Oláh, Lawrence, & Riggan, 2010).

5. A middle school science teacher is designing a unit on pulleys and levers. She allocates fourteen periods to the unit, but plans to cover all the content in the first eleven periods. Building on ideas common in Japan (see, for example, Lewis, 2002), in period twelve, the teacher gives the students a quiz and collects the papers. Instead of grading the papers, she reads through them carefully, and based on what she discovers about what the class has and has not learned, she plans appropriate remedial activity for periods thirteen and fourteen.
6. A history teacher has been teaching about the issue of bias in historical sources. Three minutes before the end of the lesson, students pack away their books and receive an index card on which the teacher asks them to respond to the question "Why are historians concerned about bias in historical sources?" The students turn in these exit passes as they leave the class at the end of the period. After all the students leave, the teacher reads through the cards and then discards them, concluding that the students' answers indicate a good-enough understanding for the teacher to move on to a new chapter.
7. A language arts teacher has been teaching his students about different kinds of figurative language. Before moving on, he wants to check his students' understanding of the terms he has been teaching, so he uses a real-time test. The teacher gives each student a set of six cards bearing the letters A, B, C, D, E, and F; and on the board, he displays the following.

   A. Alliteration

   B. Onomatopoeia

   C. Hyperbole

   D. Personification

   E. Simile

   F. Metaphor

   He then reads a series of statements.

   - This backpack weighs a ton.
   - He was as tall as a house.
   - The sweetly smiling sunshine melted all the snow.

- He honked his horn at the cyclist.
- He was a bull in a china shop.

After the teacher reads each statement, he asks the class to hold up a letter card (or cards) to indicate which kind or kinds of figurative language features in each of the six statements. All students respond correctly to the first question, but in responding to the second, each student holds up a single card (some hold up *E*, and some hold up *C*). The teacher reminds the class that some statements might be more than a single type of figurative language. Once the students realize that there can be more than one answer, the class responds correctly to statements two, three, and four. About half the students, however, indicate that they think statement five is a simile. The teacher then leads a whole-class discussion during which students give their reasons for why they think statement five is a simile or a metaphor, and after a few minutes, all the students agree that it is a metaphor, because the statement does not include *like* or *as*.

8. An AP calculus teacher has been teaching students about graph sketching and wants to check quickly that the students have grasped the main principles. She asks the students, "Please sketch the graph of $y = 1$ over $1 + x^2$." Each student sketches the graph on a whiteboard and holds it up for the teacher to see. The teacher sees that the class understands and moves on.

In each of these eight examples, the teachers use evidence to elicit and interpret student achievement and make a decision about what to do next, but whether this is enough to make each of these an example of formative assessment is a matter of some debate.

I often ask teachers which of these eight cases they would regard as formative, and there is rarely any consensus. In example 1, assessment modifies instruction, especially if you regard the supervisor as the teacher and the teachers as her students, but many people are unhappy that it is two years before the changes occur. Similar concerns are raised about example 2, especially since the students on whom the data are collected do not benefit from the process, and, moreover, it is not clear that next year's algebra 1 students will have the same problems. Example 3 raises concerns for many teachers that assessment is being used in a punitive way, but as Harvard economist Roland Fryer (2014) points out, some students need more instructional time to reach proficiency on their state's standards. He calls this the *basic physics* of

education: "If your students are falling behind, you have two choices: spend more time in school or convince the high-performing schools to give their kids four-day weekends. The key is to change the ratio."

Requiring students to attend additional classes on Saturdays may not be ideal, especially in rural areas where transportation requirements create additional difficulties. However, a school that provides additional instruction on Saturdays for students who need it has at least found a solution to the problem of how it is going to get more instructional input to the students who need it. Any school that hopes to "close the gap" without having a way of getting more instructional input to the students who need it is just paying lip service to the idea of equality.

Example 5 is interesting because it is a districtwide policy in which the formative assessment process is hardwired into the school year. To create the slack needed for the system to work, teachers have to prioritize content, which is difficult because teachers and administrators are generally told that all the state standards are essential. The problem is that most state standards contain so much material that only the fastest-learning students have any chance of mastering the required material in the time available. Robert Marzano and his colleagues asked teachers how much time they would need to cover all the content in their state standards for each year, and the average figure was twenty months (Marzano, Kendall, & Gaddy, 1999).

While that figure may have been reduced somewhat in states that have adopted variants of the Common Core State Standards, the fact is that most states specify considerably more content for their students to learn than can possibly be achieved by most students in the time available. The teacher could, of course, cover the required material at a rate that guarantees all the standards are covered, but that would mean that most students would be floundering. In Philadelphia's school system, as discussed previously, teachers have to make choices about which of the standards are essential and which are desirable, teach the prioritized standards, and then assess. If students have made enough progress, then the teacher can spend some of the "re-teaching week" on new material, but if a substantial number of students have not made enough progress on the essential standards, they remain the priority. The defining feature of this system is that the teacher does not know what he or she will be teaching in the re-teaching week until she sees how the students have done on the assessment.

Most teachers, in my experience, are happy to regard example 5 as an example of formative assessment, although some teachers suggest that twelve periods is a long time to wait to find out whether students are learning anything. On the other hand, many teachers are disturbed by example 6 because the teacher discarded the

students' responses, rather than giving students individual feedback. However, this rather misses the point, because the reason that the teacher used exit passes in this situation was because she did not want to give students individual feedback. Her aim was to decide how to start the next lesson, as I discovered when I asked her why she had discarded the cards.

> ***Me:*** *Why did you discard the exit passes?*
>
> ***Teacher:*** *Because I know where to start tomorrow's lesson.*
>
> ***Me:*** *What did you decide?*
>
> ***Teacher:*** *They mostly got it right. I'm moving on.*
>
> ***Me:*** *What would you have done if they weren't ready to move on?*
>
> ***Teacher:*** *I would have taught it again, but slower and louder. I'm joking of course. I would have taught it again, but in a different way.*
>
> ***Me:*** *What would you have done if half the students had answered correctly and half of the students had answered incorrectly?*
>
> ***Teacher:*** *I would have kept two of the cards—one with a good answer, and one with a less good answer—and put them both on the document camera at the beginning of the lesson, and ask the students to vote by holding up one or two fingers which one they thought was the best, and take it from there.*
>
> ***Me:*** *Why didn't you give students individual feedback?*
>
> ***Teacher:*** *I couldn't. The students didn't write their names on the cards.*
>
> ***Me:*** *Why didn't you get the students to write their names on the cards?*
>
> ***Teacher:*** *That would be really stupid. If I wanted to give the students individual feedback, I would have got them to write their answers in their history notebooks, which already have their names written on them.*

What this teacher did was really smart. She didn't want to generate a whole lot of marking. She wanted to get a quick read on the achievement of the whole class in order to make a decision about what to do next. She wasn't doing *data-driven decision making*. She was doing what I call *decision-driven data collection*. She was collecting the minimal amount of information she needed to make the decision she needed to make, and this illustrates a really important principle about assessment. People who espouse data-driven decision making tend to focus on the data. They collect data hoping that they might come in useful at some point in the future. Those who focus on decision-driven data collection decide what they want to do with the data *before* they collect the data, and collect only the data they need. That way, they always know what to do with the data.

Most teachers are happy to regard the last two examples (7 and 8) as formative assessment, although it is worth noting that in example 8, the assessment did not change the instruction. The teacher was planning to move on, but the evidence she collected told her that moving on was the right thing to do. She did not make a better decision because she had the data; rather, the decision she made was exactly the same decision she would have made if she had not collected the data. It wasn't a better decision, but it was a better-founded decision, because she now had evidence, rather than just a hunch, that moving on was the right thing to do.

Given the diversity of opinions about the definition of formative assessment, it is unlikely we will ever be able to get agreement about which of these is and is not formative assessment. Some people believe that the students on whom data were collected have to be direct beneficiaries of the assessment, while others are happy that the assessment functions formatively for other students. Some believe that the assessment must affect instruction almost immediately, while others are happy for longer assessment–interpretation–action cycles to be counted as formative assessment. Some believe that students must get individual feedback for the assessment to qualify as formative, while others are happy that the assessment provides the teachers with guidance about what to do next with a whole group of students. Moreover, since no one can trademark or copyright the term *formative assessment*, it seems relatively futile to try to restrict the usage of the term. People will use the term in the way they want to use it. That is why I believe that we should define formative assessment inclusively, rather than trying to get others to accept a more restricted definition. The trouble with using a wider definition, of course, is that it becomes too vague, and is therefore not helpful in guiding action. Having spent a great deal of time thinking about this, I believe the following definition provides a reasonable compromise between precision (so that it is clear what is and is not formative assessment) and comprehensiveness (so that all the things we might want to include as formative assessment are included):

> An assessment functions formatively to the extent that evidence about student achievement is elicited, interpreted, and used by teachers, learners, or their peers to make decisions about the next steps in instruction that are likely to be better, or better founded, than the decisions they would have made in the absence of that evidence.

The first point to make about this definition is that it uses the term *formative* to describe the function that evidence from the assessment actually serves, rather than the assessment itself.

The second point concerns who is actually doing the assessment. While in many cases the teachers make the decisions, the definition also includes individual learners or their peers as agents in making such decisions.

The third point is that the focus is on decisions instead of on the intentions of those involved, as is the case with some definitions of *assessment for learning*. Evidence that teachers collect with the intent of using, but never actually use, is unhelpful.

The fourth point continues the third, in that the focus is on the resulting *action* instead of the intent. As stated earlier, some definitions require that teachers use the evidence to make adjustments that actually improve learning beyond what would have happened without those adjustments. This, however, would make the definition of formative assessment too stringent. Learning is just too unpredictable for us ever to guarantee that learning will take place on a particular occasion. Moreover, if we require the assessment to result in better learning than will occur without the assessment, it will be impossible to establish that any assessment was ever formative, since we would need to establish a counterclaim: that what actually happened is different (and better) than what would have otherwise happened (but did not). The probabilistic formulation we use in our definition—that the decisions are *likely* to be better—reflects the fact that even the best-designed interventions will not *always* result in better learning for *all* students.

The fifth point is that the focus is on decisions about the next steps in *instruction*. In much of the English-speaking world, the word *instruction* connotes training or *transmission* approaches to teaching. Here, the term *instruction* refers to the combination of teaching and learning, or to any activity that intends to create learning (defined as an increase, brought about by experience, in the capacities of an individual to act in valued ways).

The sixth point is that decisions are *either* better *or* better founded than decisions that would have been made without the evidence that the assessment process elicits. The second possibility is included because the formative assessment might, as we saw in the fifth and seventh examples, indicate to the teacher that the best course of action is what the teacher had intended to do all along. The formative assessment might not change the course of action but instead simply show that the proposed course of action was right.

The emphasis on decisions as being at the heart of formative assessment also assists with the assessment design process. With many so-called formative assessments, assessment data are generated and then communicated to teachers with the expectation that teachers will be able to use the information in some way. However, if the formative assessment is designed without any clear decision in mind, then there is

a good chance that the information from the assessment will be useless. For example, many vendors now offer schools regular student testing (typically every four to ten weeks), and then feed the results back to the teachers. Sometimes these vendors report the results simply in terms of which students are on target to reach proficiency on the state tests, but even when the results are more detailed, they are often of little use to the teachers for two reasons. First, the results are usually at the level of state or provincial standards, which are generally too coarse to guide teachers' instructional decision making. Second, the results sometimes arrive weeks after the teacher has moved on to a different topic of study. Caroline Wylie and I describe this kind of formative assessment as *data-push* (Wylie & Wiliam, 2006). Data are pushed at teachers, and although those designing the assessments aren't really clear about what the teacher should do with the information, they expect the teacher to be able to use these data.

The alternative, and the ideal method of formative assessment design, is to design the assessments backward from the decisions. When the focus is on the decision that he or she needs to make, the teacher can then look at relevant sources of evidence that would contribute to making that decision in a smarter way. With such a *decision-pull* approach, the teacher always knows what to do with these data once he or she collects them because that has been thought through *before* the data are collected.

The definition of formative assessment on page 48 allows us to describe all the preceding examples as examples of formative assessment, since in each case assessment was elicited and used to make decisions that were better, or better founded, than the decisions that would have been made in the absence of the evidence. That said, I think it is worth distinguishing between different kinds of formative assessment, because there are differences in what is actually formed by formative assessment.

Examples 1, 2, 3, and 4 are what might be called *long-cycle formative assessment*, since the length of the formative assessment cycle is at least four weeks (including one case, example 1, that lasted over a year). The primary purpose of such long-cycle formative assessment processes is monitoring student progress, and, in the case of example 3, securing better alignment among curriculum, standards, and assessment. These kinds of assessments are often called *benchmark* or *interim assessments*.

Example 5 is an example of a *medium-cycle formative assessment*, because the cycle occurs within a teaching unit. Other examples of medium-cycle formative assessment would be any formative assessment approaches that seek to ensure that students understand the standards against which they will be graded, so that rather than saying, "She gave me a C," students say, "I got a C," because they understand the standards being applied to their work, and they understand how their work falls

short. This is a particular feature of the work of Rick Stiggins and his colleagues on *Classroom Assessment for Student Learning* (Chappuis, Stiggins, Chappuis, & Arter, 2012; Stiggins, 2001).

Examples 6, 7, and 8 are what we might call *short-cycle formative assessment*, because the cycle length is minute-to-minute and day-by-day, and the impact is less on generating data and more on helping teachers respond in real time to their students' learning needs. Moreover, because these processes collect evidence from all students, rather than just those who are happy to volunteer, they increase student engagement.

The relationship between these different kinds of formative assessment is shown in table 2.1.

**Table 2.1: Perspectives on Formative Assessment**

| | Long Cycle | Medium Cycle | Short Cycle |
|---|---|---|---|
| **Span** | Across terms or teaching units | Within and between teaching units | Within and between lessons |
| **Length** | Four weeks to a year (or more) | One to four weeks | Minute-by-minute and day-by-day |
| **Impact** | Monitoring, curriculum alignment | Student-involved classroom assessment, teacher cognition improvement | Student engagement, teacher responsiveness improvement |

As noted several times already, it is reasonable to describe each of the processes described in the eight preceding examples as formative assessment. What is not reasonable is to claim that all kinds of formative assessment are equally effective, because the evidence is clear that the shorter the assessment–interpretation–action cycle becomes, the greater the impact on student achievement (Wiliam, 2016). Long-cycle, medium-cycle, and short-cycle formative assessment all have their roles to play in ensuring effective instruction. But short-cycle formative assessment has to be the priority for schools and teachers, because the impact on students is greater.

## Strategies of Formative Assessment

The discussion thus far establishes that any assessment can be formative, and that assessment functions formatively when it improves the instructional decisions that teachers, learners, or learners' peers make. These decisions can be immediate, on-the-fly decisions, or longer term. However, if we want to really see what formative assessment looks like on the ground, we must dig a little deeper.

All teaching really boils down to three key processes and three kinds of individual roles involved. The processes are (1) finding out where learners are in their learning, (2) establishing where they are going, and (3) determining how to get there. The roles are (1) teacher, (2) peer, and (3) learner. Crossing the roles with the processes gives us a three-by-three grid of nine cells. However, for the sake of convenience, we can group these nine cells into five key strategies of formative assessment with one big idea (Leahy, Lyon, Thompson, & Wiliam, 2005), as shown in figure 2.2. The five key strategies are:

1. Clarifying, sharing, and understanding learning intentions and success criteria
2. Eliciting evidence of learning
3. Providing feedback that moves learning forward
4. Activating learners as instructional resources for one another
5. Activating learners as owners of their own learning

<table>
<tr><th></th><th>Where the learner is going</th><th>Where the learner is right now</th><th>How to get there</th></tr>
<tr><td>Teacher</td><td rowspan="3">Clarifying, sharing, and understanding learning intentions and success criteria</td><td>Eliciting evidence of learning</td><td>Providing feedback that moves learning forward</td></tr>
<tr><td>Peer</td><td colspan="2">Activating learners as instructional resources for one another</td></tr>
<tr><td>Learner</td><td colspan="2">Activating learners as owners of their own learning</td></tr>
</table>

*Source: Adapted from Leahy et al., 2005.*

**Figure 2.2: The five key strategies of formative assessment.**

The big idea is that evidence about learning is used to adjust instruction to better meet student needs—in other words, teaching is *adaptive* to the learner's needs. Over the next five chapters, we discuss each of these strategies in greater detail. Before moving on, however, it is worth considering why assessment should occupy such a central position in teaching.

## Assessment: The Bridge Between Teaching and Learning

Assessment occupies such a central position in good teaching because we cannot predict what students will learn, no matter how we design our teaching. In a very

carefully designed experiment, Brenda Denvir developed a detailed taxonomy of children's early number skills and showed that some were prerequisites for others (Denvir & Brown, 1986a). For example, she found that before students could subtract one single-digit number from another, they needed to be able to count backward by one (in other words, be able to name the number immediately preceding a given number).

One student—called Jy in the study—had several specific knowledge gaps in the following areas.

- Knows the numbers backward from 20
- Knows number bonds (not just the doubles)
- Knows the answer when adding units to a decade number
- Knows the answer when adding ten to a two-digit number
- Knows the answer when taking ten away from a two-digit number

Over two months, Jy's teacher planned and delivered specific instruction to address these gaps, and at the end of the process, she assessed Jy again.

Surprisingly, on the post-test, Jy could not demonstrate mastery of any of the skills that the teacher had specifically taught her, although on a delayed post-test (five months later), she did show mastery of one of the taught skills ("Knows the answer when adding units to a decade number"). However, in the post-test, she did show mastery of a number of other skills that she had not demonstrated on the pretest.

- Uses counting up, back, down strategy for take-away
- Models two-digit addition without regrouping using base-ten apparatus
- Models two-digit subtraction without regrouping using base-ten apparatus
- Models two-digit addition with regrouping using base-ten apparatus
- Bundles objects to make new groups of ten to facilitate enumeration of a collection that is partly grouped in tens and ones

The skills that Jy acquired were consistent with the hierarchies that Denvir had identified—they just weren't the skills her teacher had taught. The researchers found the same result for other students in the study (Denvir & Brown, 1986b).

This is why assessment is *the* central process in instruction—students do not learn what we teach. If they did, we would not need to keep gradebooks. We could, instead,

simply record what we have taught. But anyone who has spent any time in a classroom knows that what students learn as a result of our instruction is unpredictable. We teach what we think are good lessons, but then, after we collect our students' notebooks, we wonder how they could have misinterpreted what we said so completely.

The truth is that we often mix up teaching and learning, as the following old joke shows.

> ***Amy:*** *I taught my dog to whistle.*
>
> ***Betty:*** *Let's hear it then.*
>
> ***Amy:*** *He can't whistle.*
>
> ***Betty:*** *I thought you said you taught him to whistle.*
>
> ***Amy:*** *I did. He just didn't learn it.*

At one time, school inspectors in England claimed to be able to distinguish between the quality of teaching and the quality of learning in a classroom (Office for Standards in Education, Children's Services and Skills, 1999). However, it is hard to work out what the basis of such a distinction might be. After all, what sense does it make to talk about a lesson for which the quality of teaching was high but the quality of learning was low? It's rather like a surgeon claiming that an operation was a complete success, but unfortunately, the patient died.

In some languages, the distinction between teaching and learning is impossible to make. In Welsh and Maori, for example, the same word is used for both (*dysgu* and *ako*, respectively). In languages where it is possible to distinguish teaching from learning, however, the phrase *teaching and learning* is in many cases being replaced by the phrase *learning and teaching* (although the former still gets three times as many Google hits as the latter).

This is often touted as a good move, as if putting the word *learning* before the word *teaching* makes a difference, but it is largely a cosmetic change, and one that could actually be harmful, because it draws attention to rather trivial differences, leaving much more important issues obscured. To say that learning is more important than teaching is a bit like saying that traveling is more important than driving. Traveling is the goal, and driving is a way to achieve that goal. In the same way, student learning is the goal, and teaching is a way to achieve that goal. And in the same way that drivers achieve their goal (traveling) by driving, teachers achieve their goal (student learning) by teaching.

Everything a teacher does is teaching. That's all the teacher can do. *Learning* is the name we give to the rather mysterious process that occurs in students' heads, when

they can do things that they were not able to do before (or, in some cases, *not* do something they might have done before, like get pregnant or take drugs). Districts can have policies on teaching (what teachers do), but no district can have a policy on learning. The trap is thinking that this is the endpoint rather than a means to an end. As one teacher put it:

> Actually thinking about teaching has meant that I have been able to come up with ideas and strategies to cope with whatever has arisen and has contributed greatly to my professional development. I now think more about the content of the lesson. The influence has shifted from "what am I going to teach and what are the pupils going to do?" towards "how am I going to teach this and what are the pupils going to learn?" (Black, Harrison, Lee, Marshall, & Wiliam, 2004, p. 19)

This is, in practice, a very difficult course to steer. At one extreme, there are teachers who try to do the learning for the learners, epitomized by the old saying that schools are places where children go to watch teachers work. I visit a lot of classrooms, and in most of them, the teacher is working really hard, but the students? Not so much. That is why I often say to teachers, "If your students are going home at the end of the day less tired than you are, the division of labor in your classroom requires some attention."

At the other extreme are the teachers who use the F-word—*facilitate.* "I don't teach," they say. "I just facilitate learning." I am never quite sure what this means. Presumably, the teachers are just hanging around, hoping that some learning will occur.

Teaching is difficult because neither of these extremes is acceptable. When the pressure is on, most of us behave as if lecturing works, but deep down, we know it's ineffective. But leaving the students to discover everything for themselves is equally inappropriate. For this reason, I describe teaching as *the engineering of effective learning environments.* And sometimes, a teacher does her best teaching before the students arrive in the classroom.

Many teachers have had the experience of creating an effective group discussion task in which the students engage completely in a tricky challenge that they must resolve. The only problem is there is nothing for the teacher to do. He feels a little bored and a tad guilty that he is not doing anything, so he disrupts a group's work. This is one version of what I call the *teaching–learning trap*: I'm not doing anything; therefore, the students can't be learning anything. The other version of the trap was discussed earlier: I am working hard, so the students must be learning something.

The teacher's job is not to transmit knowledge, nor to facilitate learning. It is to engineer effective learning environments for the students. The key features of

effective learning environments are that they create student engagement and allow teachers, learners, and their peers to ensure that the learning is proceeding in the intended direction. The only way we can do this is through assessment. That is why assessment is, indeed, the bridge between teaching and learning.

Now before we conclude, I think it is important to acknowledge the limitations of formative assessment. As Paul Kirschner and his colleagues point out, learning is a change in long-term memory: "If nothing has changed in long-term memory, nothing has been learned" (Kirschner, Sweller, & Clark, 2006, p. 77). The fact that students can do something today does not mean they will be able to do it next week. But if they can't do it today, it is highly unlikely that they will be able to do it next week. The *big idea* of formative assessment is that it is better to know what students have learned before deciding what to do next.

## Conclusion

In this chapter, we learned that the regular use of minute-by-minute and day-by-day classroom formative assessment can substantially improve student achievement. Although researchers have considered many different definitions of formative assessment, the essential idea is simple: teaching is a *contingent* activity. We cannot predict what students will learn as a result of any particular sequence of instruction. Formative assessment involves getting the best possible evidence about what students have learned and then using this information to decide what to do next.

We also saw that we can conceptualize formative assessment as having five key strategies. The next five chapters probe each of these five strategies in more depth, offering details of research studies that provide evidence of their importance and a number of practical techniques that can be used to implement the strategies in classrooms.

chapter 3

# Clarifying, Sharing, and Understanding Learning Intentions and Success Criteria

It seems obvious that students might find it helpful to know what they are going to be learning, and yet, consistently sharing learning intentions with students is a relatively new phenomenon in most classrooms. This chapter reviews the importance of learning intentions and when to use them. It then explains why it is helpful to distinguish between learning intentions, the context of the learning, and success criteria, and discusses special issues in creating these constructs. The chapter makes the case for and against rubrics before providing several techniques that teachers can use to share learning intentions and success criteria with their students.

## The Importance of Learning Intentions

Back in 1971, Mary Alice White tried to imagine what she termed "the view from the student's desk":

> The analogy that might make the student's view more comprehensible to adults is to imagine oneself on a ship sailing across an unknown sea, to an unknown destination. An adult would be desperate to know where he is going. But a child only knows he is going to school. . . . The chart is neither available nor understandable to him. . . . Very quickly,

> the daily life on board ship becomes all important. . . . The daily chores, the demands, the inspections, become the reality, not the voyage, nor the destination. (p. 340)

Not all students have the same idea as their teachers about what they are meant to be learning in the classroom. For a simple example, we might ask students which is the odd one out in the following list of objects: knife, fork, hammer, bottle of ketchup. Some students will say that the bottle of ketchup is the odd one out, because the others are all metal tools. Other students will say that the hammer is the odd one out because they can find all the other objects on their kitchen table at mealtime. Of course, in an absolute sense, neither of these two answers is better than the other, but as sociologist Nell Keddie (1971) points out, schools place a greater value on the first way of thinking about the world than the second. Sometimes schools make this explicit—most students probably realize that teachers value being able to name all the presidents of the United States more than being able to name the team roster for the New Orleans Saints football team, even though both tasks are comparable in length. But often, what teachers want is not made clear, and this puts some students at a considerable advantage because *they already know.*

If I show a piece of writing to a group of third graders and ask them why I think it's a good piece of writing, some will respond with contributions like, "It's got lots of descriptive adjectives," "It's got strong verbs," or "It uses lots of different transition words." Others will suggest that I think the work is good "because it's neat and long," reminiscent of the teacher Paul Pennyfeather in Evelyn Waugh's (2001) *Decline and Fall* who instructs his class, "Meanwhile, you will write an essay on 'self-indulgence.' There will be a prize of half a crown for the longest essay, irrespective of any possible merit" (p. 38).

Several research studies highlight the importance of students understanding what they are meant to be doing. Eddie Gray and David Tall (1994) looked at the mathematical reasoning skills of seventy-two students between the ages of seven and thirteen. They found that higher-achieving students were able to work with unresolved ambiguities about what they were doing, while those students seen as lower achieving were struggling because they were trying to do something much more difficult.

To illustrate this, I often ask teachers to write $4x$ and $4\frac{1}{2}$. I then ask them what the mathematical operation is between the 4 and the $x$, which most realize is multiplication. I then ask what the operation is between the 4 and the $\frac{1}{2}$, which is, of course, addition. I then ask whether any of them had previously noticed this inconsistency in mathematical notation—that when numbers are next to each other, sometimes it means multiply, sometimes it means add, and sometimes it means something

completely different, as when we write a two-digit number like 43. Most teachers have never noticed this inconsistency, which presumably is how they were able to be successful at school. The student who worries about this and asks the teacher why mathematical notation is inconsistent in this regard may be told not to ask stupid questions, even though this is a rather intelligent question and displays exactly the kind of curiosity that might be useful for a mathematician—but she has to get through school first!

A vivid illustration of the power of understanding what one is meant to be doing was provided by a study of twelve science classrooms in two urban middle schools in the United States (White & Frederiksen, 1998). Seven of the classrooms were seventh grade, three were eighth grade, and two were ninth grade. Each class (average size of thirty students) had one forty-five-minute science period each school day.

All twelve classes followed the ThinkerTools curriculum—designed to promote thinking in the science classroom—for approximately eleven weeks, during which time they covered seven modules.

- Module 1. One-dimensional motion
- Module 2. Friction
- Module 3. Mass project (the Common Inquiry Project)
- Module 4. Two-dimensional motion
- Module 5. Gravity
- Module 6. Trajectories
- Module 7. Final project (the Chosen Inquiry Project)

Each module incorporated a series of evaluation activities. In six of the classes (selected at random to act as a control group), these evaluation episodes took the form of a discussion, once each week, about what the students liked and disliked about the topic. In the other six classes, once a week, students engaged in a process of reflective assessment. Through a series of small-group and individual activities, the teachers introduced students to the nine assessment criteria (each of which was assessed on a five-point scale) that the teachers would use to evaluate their work. At the end of each episode within a module, the students were asked to assess their performance against two of the criteria. At the end of the module, students had to assess their performance against all nine. For each assessment, they had to write a brief statement showing which aspects of their work formed the basis for their rating. At the end of each module, students presented their work to the class, and their peers used the criteria to give them feedback. The important point about the design

of the experiment is that all students, both experimental and control, got the same instructional time.

All the students in the study also took the Comprehensive Test of Basic Skills (CTBS)—a test of basic literacy and numeracy—to check that the treatment group classes and the control group classes were evenly matched in terms of prior achievement (which they were). At the end of module 3, students completed a project that was scored on a five-point scale. The students' results, classified according to whether the students in each group had scored in the upper or lower half of their class on the CTBS, are shown in table 3.1.

**Table 3.1: Project Scores for Students, by Group and CTBS Scores**

| | Average Project Score | CTBS Scores | |
|---|---|---|---|
| | | Lower half | Upper half |
| Likes and dislikes (control) | 2.6 | 1.9 | 3.4 |
| Reflective assessment | 3.2 | 3.0 | 3.5 |

*Source: Adapted from White & Frederiksen, 1998.*

There are two important features of the data in table 3.1. First, the average scores of the students in the reflective assessment classrooms are higher than those in the control group. Second, and more important, in the reflective assessment group, the achievement gap between high scorers and low scorers on the CTBS has been cut by two-thirds, compared with the likes and dislikes classes (average of 0.5 points, compared with 1.5 points).

On the final project, the students were given a score out of 100 for the skills of science inquiry they had demonstrated, which were compared with those they had shown in a pretest at the beginning of the study. The results are shown in table 3.2.

**Table 3.2: Final Inquiry Test Scores for Students, by Group and CTBS Scores**

| | Pretest | | Post-Test | |
|---|---|---|---|---|
| | Lower half | Upper half | Lower half | Upper half |
| Likes and dislikes (control) | 32 | 59 | 39 | 68 |
| Reflective assessment | 28 | 52 | 53 | 72 |

*Source: Adapted from White & Frederiksen, 1998.*

These are rather surprising results, because the *likes and dislikes* classes and the *reflective assessment* classes received similar instruction—over 80 percent of the teaching was the same for all twelve classes participating in the project—and yet the

changes were radical. In the likes and dislikes group, the high achievers improved by nine points and the low achievers improved by seven points. However, in the reflective assessment group, the high achievers improved by twenty points and the low achievers improved by twenty-five points. Everyone did better in the reflective assessment group, but the magnitude of the benefits was much greater for the lower-achieving students. Some students already know what successful work looks like, and others do not. Ensuring that all students know what quality work looks like has a profound impact on achievement gaps.

Given all this, it is not surprising that assessment expert D. Royce Sadler (1989) writes:

> The indispensable conditions for improvement are that the student comes to hold a concept of quality roughly similar to that held by the teacher, is continuously able to monitor the quality of what is being produced during the act of production itself, and has a repertoire of alternative moves or strategies from which to draw at any given point. (p. 121)

While there could be some discussion about whether, as Sadler claims, any improvement is impossible unless the conditions he specifies are met, there seems to be little doubt that it is a good idea to make sure that students understand the learning intentions behind the activities they undertake in the classroom.

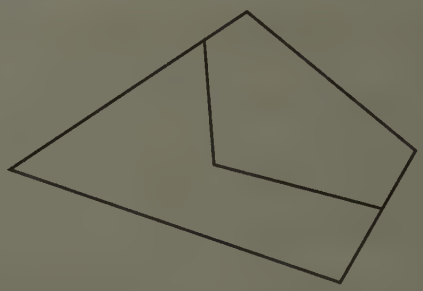
## When to Use Learning Intentions

Several U.S. school districts mandate that each period of instruction should begin with the teacher posting a learning objective. Indeed, as I have witnessed in many districts, teacher evaluation protocols specifically address this by requiring those observing teachers to record whether the teacher has clearly articulated the lesson objectives. Too often, the result is a desultory approach to the sharing of learning intentions. The teacher writes the objective on the board, the students copy the objective into their notebooks, and everyone proceeds to ignore the objective for the rest of the period—what some teachers describe as a *wallpaper objective*.

This kind of tokenistic approach to the sharing of learning intentions is most definitely not what the strategy of *clarifying, sharing, and understanding learning intentions and success criteria* intends. As Albert Einstein reputedly said, "Make things as simple as possible, but no simpler." Obviously it's a good idea for students to know where they are going in their learning, but this cannot be done in a formulaic way.

For a start, sometimes it's not even a good idea to tell the students what the lesson is about. Consider the problem in figure 3.1 (page 62), suitable for a middle school mathematics class.

Two farmers have inherited adjacent fields with a boundary that has a kink in it:



Both farmers find the crooked boundary to be an inconvenience, and so they wonder whether it is possible to divide the two fields with a straight boundary so that each farmer still has the same amount of land.

**Figure 3.1: Diagram for the "Two Fields" problem.**

There are, of course, numerous ways in which such a problem might be solved, but one key insight that simplifies the problem considerably is that triangles with equal bases and the same height must have the same area (because the formula for the area of a triangle is ½ × base × height). So, if a straight line is drawn between the points where the boundary meets the edge of the fields, and a parallel line is drawn through the kink in the boundary, any triangles with a base on the line between the boundary points and the third corner on the parallel line will have the same area (see figure 3.2).

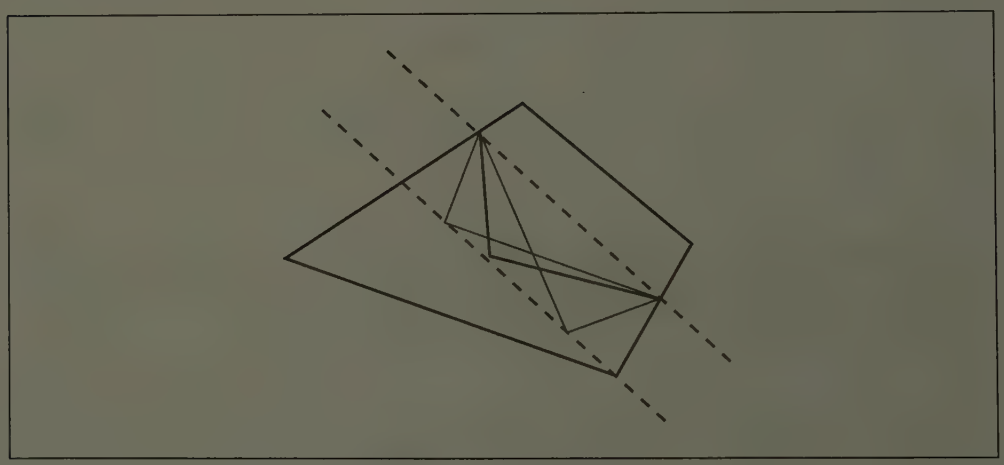


**Figure 3.2: Working for the "Two Fields" problem.**

Therefore, either of the two heavy lines in figure 3.3 will do.

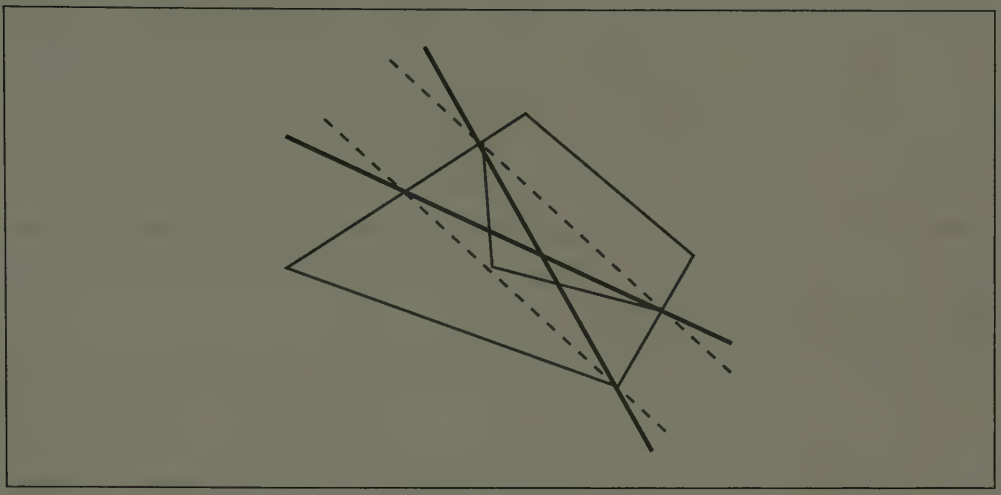


Figure 3.3: Solution for the "Two Fields" problem.

This problem would have been completely pointless if the teacher had told the students at the beginning that they were learning to solve problems involving the area of a triangle. Once one has realized that the area of a triangle is relevant, the problem is relatively simple to solve. Sometimes telling the students where they are going completely spoils the journey! Indeed, in conversations with students, many have told me that always receiving the objectives at the start of the day turns them off.

At other times, we might be quite comfortable with the idea of giving students clear guidance about what we want, but it turns out to be too difficult. For example, a teacher who is trying to develop her students' ability to engage the attention of the reader might want to spend a period developing the skill of writing an attention-grabbing opening sentence. While this is certainly something that we want students to be able to do, it is questionable whether it is of much help to the students to be told that the day's learning objective is to understand what makes an attention-grabbing opening sentence. The fact that we cannot always put into words what we know is a very profound problem that is too rarely appreciated today, even though it has been recognized for many decades.

In his groundbreaking book *Personal Knowledge*, first published in 1958, Michael Polanyi explores the problem of how people come to know things. He points out that in many important areas of human affairs, we cannot write down rules for what makes something good, but we can often summarize our perceptions of quality through maxims. However, his key insight is that while maxims are useful to those who already understand what quality means in a specific context, they are useless to those who don't understand what is meant:

> Maxims cannot be understood, still less applied by anyone not already possessing a good practical knowledge of the art. They derive their interest from our appreciation of the art and cannot themselves either replace or establish that appreciation. (Polanyi, 1958, p. 50)

More simply, Robert Pirsig, in his sequel to *Zen and the Art of Motorcycle Maintenance*, writes, "Quality doesn't have to be defined. You understand it without definition. Quality is a direct experience independent of and prior to intellectual abstractions" (Pirsig, 1991, p. 64).

The point that both Polanyi and Pirsig are making is that we sometimes use learning objectives as if they are definitions of quality when in fact they are post-hoc rationalizations of quality, familiar enough to those who know what they are doing but not helpful to those who are not yet there. A friend of mine is a serious golf player, and he told me that for weeks his coach was telling him that he needed to "quieten his lower body." He treated this as guidance about what he needed to work on but found it impossible to make any sense of it. And then, one day, it all clicked. And once it had clicked, the advice made sense. It probably didn't help him achieve the goal, but it did help him know when he had reached it.

In communicating learning intentions to students, therefore, the first thing to decide is whether we are giving, in Polanyi's language, rules or maxims. Sometimes we can be very specific, such as when we require laboratory reports to be structured in a particular way—diagrams are to be drawn in pencil and labeled, and so on. At other times, it may be that the best we can do is help the students develop what Guy Claxton (1995) calls a *nose for quality*. Rubrics have a role to play in this process, as they did in Barbara White and John Frederiksen's (1998) study. In their research, teachers shared rubrics with students but, more important, gave the students time to think through, in discussion with others, what the rubrics might mean in practice when applied to their own work.

It is often valuable to develop learning intentions jointly with the students—a process that is sometimes called *co-construction*. It is important to note that developing learning intentions or success criteria with students is not a democratic process. The teacher is in a privileged position with respect to the subject being taught and knows more about the subject than the students do, and the teacher would be abdicating his or her responsibilities to let whatever the students feel should be valued be adopted as the success criteria. The advantage of developing success criteria with the students is that doing so creates a mechanism whereby students can discuss and come to own the learning intentions and success criteria, making it more likely that they will be able to apply the learning intentions and success criteria in the context of their own work.

Shirley Clarke (2005) gives the following example. A middle school teacher is teaching a unit on food production in the developing world. She tells the class that the learning intention is to understand the impact of banana production on the banana producers themselves. The students study this topic, and at the end of the unit, the teacher conducts an end-of-unit assessment, in which she requires students to show what they have learned about the impact of banana production on banana producers. Since they have been spending an hour a day on this topic for the previous two weeks, most students are able to get high scores on the assessment. The problem with such an approach is that if we only test the students on the things we have taught them, they may do well, but so what?

Clarke (2005) points out that this kind of shallow approach to teaching and assessment is often the result of confusion between the learning intention and the context of the learning. She suggests that in this particular example, a far better learning intention would be to understand the impact of production on producers in the developing world; banana production should be the context of the learning. The success criterion—how the teacher finds out whether the learning intention has been satisfied or not—could then be whether the students can transfer what they learned about banana production to, say, sugar production.

This highlights a profound but poorly understood point. As teachers, we are not interested in our students' ability to do exactly those things we have taught them to do. We are only interested in their ability to apply their newly acquired knowledge to a similar but different context.

This is perhaps clearest in mathematics. If a teacher teaches students to add fractions with a worked example, such as ½ + ⅗, once she has done that, she is no longer interested in the students' ability to add ½ and ⅗. Most students can do that problem because they have just been shown it! What the teacher wants to know is whether the students can transfer their newly acquired knowledge to a reasonably similar but different pair of fractions. Likewise, if she teaches students how to find the area of a particular trapezoid, that individual problem is no longer interesting. What is important is whether the students can transfer their knowledge to other trapezoids. In language arts, when we get students to correct punctuation on a final draft, we're looking beyond that piece of writing; we hope that students will transfer what they have learned to other pieces of writing.

This is why, as Alfie Kohn (2006) points out, scoring rubrics that are too detailed can be counterproductive. If you specify in detail what students are to achieve, then they may well achieve it, but that is probably all they will be able to do (Kohn,

2006). The clearer you are about what you want, the more likely you are to get it, but the less likely it is to mean anything.

Table 3.3 draws on Clarke's (2005) work to provide some examples in which the teacher has confused the learning intention with the context of the learning and to present ways in which the teacher might reframe the learning objective.

**Table 3.3: Examples of Confused and Clarified Learning Intentions**

| Confused Learning Intention | Clarified Learning Intention | Context of Learning |
|---|---|---|
| To be able to write instructions on how to change a bicycle tire | To be able to write clear instructions | Changing a bicycle tire |
| To be able to present an argument for or against assisted suicide | To be able to present arguments either for or against emotionally charged propositions | Discussing assisted suicide |
| To know what the local priest does | To know the duties and responsibilities of religious leaders | Consulting the local priest |
| To produce and analyze a questionnaire about movie-going habits | To construct and analyze questionnaire data | Discovering movie-going habits |
| To design an experiment to find out what conditions pill bugs prefer | To design fair tests for scientific questions | Learning the preferred habitat of pill bugs |

Another benefit of separating the learning intention from the context of the learning is that it makes it much easier to differentiate instruction without creating a classroom in which different students are working toward different goals. All students can be working toward the same learning intention; the differentiation comes in the success criteria. One particularly effective way to do this is to differentiate the success criteria by how far students are able to transfer their learning to novel contexts. All students should be able to transfer what they have learned to very similar contexts, while others can show how far they can transfer what they have learned.

Also, we must be able to distinguish between the intended learning outcomes and the instructional activities that we hope will result in those outcomes, and this is a distinction that many teachers find hard to make. I often ask teachers, "What are your learning intentions for this period?" Many times, teachers respond by saying things like, "I'm going to have the students . . ." and then specify an activity. When I follow up by asking what the teacher expects the students to learn as a result of the activity, I am often met with a blank stare, as if the question is meaningless or trivial. This is why good teaching is so extraordinarily difficult. It is relatively easy to think up cool stuff for students to do in classrooms, but the problem with such an activity-based approach is that too often, it is not clear what the students are going to learn. It is also relatively easy, on the other hand, to stand at the front of the class

and lecture students on what you want them to learn, but many students do not learn well with such an approach. Teaching is hard because, as Grant Wiggins and Jay McTighe (2000) point out, it has to be designed backward.

## Issues in Constructing Learning Intentions and Success Criteria

There are many sources of advice for teachers about the construction of learning intentions and success criteria—the works of Shirley Clarke (2005) in the United Kingdom and Rick Stiggins (2001) and his colleagues at the Assessment Training Institute in the United States are particularly powerful in this regard—but the development of good learning intentions is more craft than science and will always depend on the creativity of teachers. Nevertheless, there are three issues in the development of learning intentions and success criteria that may be useful to think about.

1. Task-specific versus generic scoring rubrics
2. Product-focused versus process-focused criteria
3. Official versus student-friendly language

### Task-Specific Versus Generic Scoring Rubrics

A scoring rubric—which is really just a way of presenting success criteria—can be task-specific, so that it applies to just a single task, or it can be generic, so that the same rubric can be applied to a number of different tasks (see Stiggins, 2001, pp. 314–322). Task-specific rubrics are beneficial in that they can be written in very clear language, so they are good for communicating accurately to students what is required for that particular task.

However, this strength of task-specific rubrics is also a weakness, in that they focus only on that specific task. This harks back to the same issue that arose earlier in the discussion of contextualized learning intentions—by being specific about what we want, we focus the students' learning too much. An additional weakness of task-specific rubrics is that students need to adjust to a new rubric for each task.

Judith Arter and Jay McTighe (2001) suggest that task-specific rubrics are generally more appropriate for summative assessment. When we want students to know exactly what we want them to show us they can do, then specific criteria are very useful. They make sure that students know what we are looking for, and thus they are useful at the end of learning. During learning, however, it is useful to build a degree of generality into scoring rubrics so as to promote transfer. Another advantage

of generic rubrics is that they do not need to be rewritten for every new assignment, which saves the teacher time. (For more on the characteristics of good rubrics, see Relearning by Design, 2000.)

## Product-Focused Versus Process-Focused Criteria

Most learning intentions and success criteria focus on the outcomes of learning—for example, by stating what students are expected to be able to do at the end of a period of instruction. This is very natural, because, as noted previously, the best learning needs to be designed backward from the intended destination. However, in the same way that during a car journey it is helpful (and reassuring) to know that one is on the right track (for example, "You will pass a gas station on the left"), process criteria, or criteria that indicate that one's work is progressing toward these success criteria, can be useful, too.

Formative assessment expert and former teacher Shirley Clarke (2005) gives the following example:

> Learning intention: to write an effective characterization
>
> Product success criterion: the reader will feel as if they know the person
>
> Process success criteria: the characterization includes at least two of the following:
>
> - the character's hobbies and interests
> - the character's attitudes toward self and others
> - examples of the character's extrovert or introvert personality
> - examples of the character's likes and dislikes (p. 31)

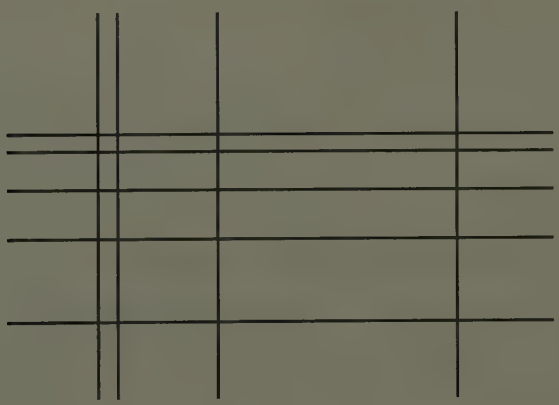
Physical education teachers and athletics coaches are often very good at developing process criteria; they break complex skills down into simpler ones and then reassemble them. To those who have never had to teach throwing a ball, it just looks like one action, but coaches know that most children learn to do this better when it is broken down into a number of steps that are practiced individually and then integrated into one fluid action.

Writing frames are also examples of process criteria, because they structure the student's response. However, in the same way that leg braces are useful to individuals with weak legs but are a constraint for those who are healthy, writing frames can help some students but will prevent others from responding creatively.

For example, if a group of students is working on writing a suspense story, it might be useful to begin by having them read some suspense stories produced by other students of the same age and, through the process of co-construction as described

previously, decide which ones are better, and why they are better. The group may then decide that the best suspense stories have four main phases: character establishment, a period of rising tension, a climax, and a resolution. This does, of course, indeed represent the standard model of a suspense story. But it can become rather formulaic—almost a straitjacket—which is why many TV shows now start with the climax and then, after a few minutes, go back to the beginning of the story introduced with the phrase "a few hours (or days) earlier." In fact, so many TV shows have used this device that it has become somewhat of a cliché itself, but this illustrates the point that structures, templates, and guidelines are potentially both supports and constraints. This does not mean that we should not give students supports. But we should also make sure that our students realize that, if they feel confident that they know what they are doing, they can throw the supports away. Much of the time, the result will be that a student will fall flat on her or his face, but, as Samuel Smiles (1862) said over a century and a half ago, "he who never made a mistake, never made a discovery" (p. 275).

An illustration of the creativity that can occur when students ignore teachers' recommendations about how to solve problems comes from an episode early in my teaching career. I was teaching a class of ninth-grade students and assigning them the task of working out how many rectangles were produced by the intersection of a number of horizontal lines and a (possibly different) number of vertical lines.



I had developed a set of process criteria that encouraged students to "try simpler cases" as recommended by George Polya (1945)—for example, by looking at just two horizontal lines and then counting the rectangles formed by two, then three, then four vertical lines, and so on. Within a matter of minutes, one of the students had written down the following mathematical formula.

$$\frac{m(m-1)}{2} \times \frac{n(n-1)}{2}$$

I asked to see his scratch work, and he said, "What scratch work?" He patiently pointed out to me that each rectangle was formed by a choice of a pair of horizontal lines and a choice of a pair of vertical lines. If there were $m$ horizontal lines, then there were $m$ choices for the first of the two horizontal lines and, therefore, $m - 1$ choices for the second horizontal line, giving a total of $m \times (m - 1)$ pairs in all. But then each pair of lines would be counted twice, so to get the number of choices of horizontal pairs, $m \times (m - 1)$ needed to be divided by 2. Applying a similar argument to the vertical lines and multiplying the two expressions together gave the solution.

For this student, my careful planning frame (had he paid any attention to it!) would have prevented him from finding a truly insightful solution that I had not seen. We need to realize that process criteria are, at the same time, both constraints *and* affordances for our students, and so we need to be thoughtful about how we frame them and how we use them.

Having said this, process criteria are particularly important in helping students become owners of their own learning (see chapter 7). If the teacher has spent some time helping students develop a "nose for quality," as described previously, then the students will be all too aware of the gap between where they are and where they want to be. What they will not know is how to move forward. Providing them with process criteria breaks up the long journey from where the learners are to where they need to be into smaller steps, making it more manageable.

When we are in summative mode, the purpose of success criteria is to determine the extent to which students have been successful. When we are in formative mode, the purpose of success criteria is to bring about that success.

## Official Versus Student-Friendly Language

State and provincial standards are often written in rather formal language, and it is not particularly easy for students (or teachers!) to work out what they mean. As a result, some authors advocate that teachers should adapt the state or provincial standards and present them to students in student-friendly language (see, for example, McEldowney & Henry, 2017). There are merits in such an approach, but it is also worth remembering that the language used in state standards is often characteristic of the discipline that the standards represent. Students might not understand what the term *sense of audience* means, and one response might be to try to express the meaning in simpler words, but it is also important that students come to understand the phrase *sense of audience* as a term of art in language arts. Student-friendly

language can be useful when introducing students to a discipline, but it is also important to help students develop the habits of mind that define the discipline, and coming to terms with the "official" language is part of that process.

## The Case for and Against Rubrics

Readers may be surprised that this chapter has so far made little mention of perhaps the most common way that teachers in North America communicate learning intentions to their students, and that is through the use of rubrics. This is not an oversight.

As I hope I have made clear in this chapter, I strongly believe that learners of all ages should, as far as possible, understand what it is that they need to learn and be able to monitor their progress toward their goals. However, whether rubrics are a good way to do this is far from clear. Many studies of rubrics have focused not on whether they improve learning but on whether they improve the consistency of scoring, and here the evidence is that they do (Jonsson & Svingby, 2007). Where the studies have looked specifically at whether they help students learn more, the findings are more equivocal (Panadero & Jonsson, 2013). While some studies have shown that rubrics can have a substantial impact on student achievement, most of these findings are from studies with a relatively short-term focus. If a rubric identifies particular features that are needed in a piece of work, then students are more likely to ensure those features are present, but whether this produces any greater learning in the long term is less clear. Where rubrics intend to improve learning, rather than just to improve the accuracy of assessment, there are at least three issues that need careful consideration.

First, rubrics rely on descriptions of quality, and as such are often imperfect representations of what we mean by quality even if we can agree about what quality work looks like. As the writer Roger Shattuck says, "Words do not reflect the world, not because there is no world, but because words are not mirrors" (as cited in Burgess, 1992, p. 119). The philosopher Stephen Toulmin (2001) provides a stark illustration of this in his book *Return to Reason*. He describes his work while serving on the National Commission for the Protection of Human Research Subjects, where he had been assigned the task of making recommendations about the ethics of using young children in biomedical and behavioral research. After six months of deliberation and numerous public meetings, the commission was able to identify a set of proposals that commanded, among the commission's members, almost unanimous agreement. However, when they were asked for their respective reasons for the proposals,

Toulmin (2001) says that "Babel set in" (p. 132). It seems that the commissioners could agree about what the recommendations should be, but beneath that surface consensus, there was considerable disagreement about what the words meant. If this is true for experts in a field, then it is quite likely that novices might understand terms in a rubric rather differently from experts.

For example, in responding to literary text, it might be quite reasonable to expect the best work to be *sustained and concise*, but for some students, this is confusing, since *sustained* means *long* and *concise* means *short*. Indeed, their immediate response might be, "Make up your mind. Do you want *sustained* or *concise*?" because they cannot see how a piece of work could be both. For the subject expert, these two terms are quite compatible, and so there is no inherent conflict in asking for work that is both sustained and concise. This is the point that Michael Polanyi (1958) was making when he said that words could not convey to a novice an understanding of quality.

Second, rubrics seem to focus more on grading rather than improvement. As noted, provided the samples are not too challenging, high-quality work can be very effective in getting students to understand quality. Whether there is value in providing descriptions of less-good work is not clear. For a concrete example, consider Education Northwest's (2016) widely used 6 + 1 writing rubric. For the highest level (level 6, or *exceptional*), the key question "Does the author's choice of words convey precise and compelling meaning and/or create a vivid picture for the reader?" has the following description in the rubric:

> Creates vivid imagery, with powerful and engaging vocabulary; uses precise words and/or figurative language to enhance meaning in interesting, natural ways. (Education Northwest, 2016, p. 7)

This seems entirely appropriate. What is less clear is whether it is useful to provide, in addition to this description, a characterization of work that is not as good as this. For example, the description of work at level 3 of this dimension of the rubric (*developing*) reads as follows:

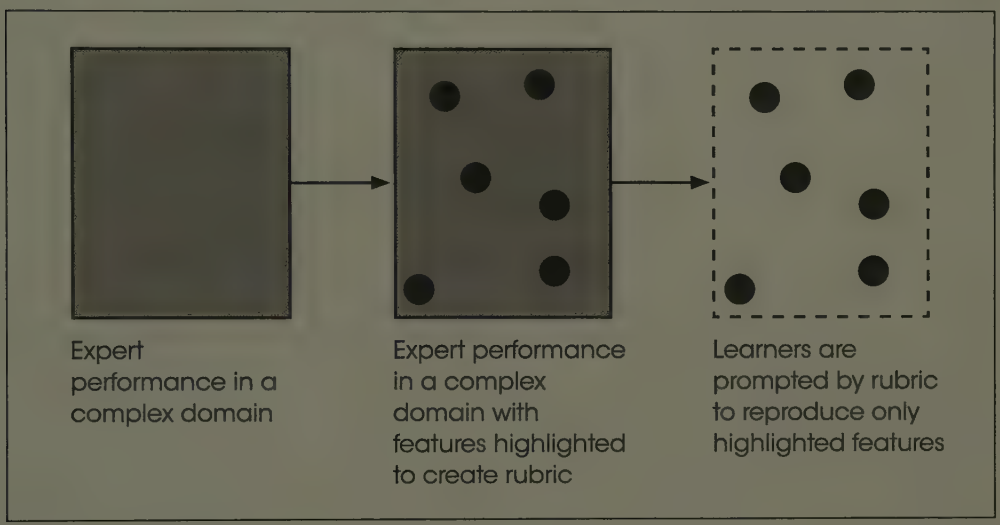
> Has vocabulary that is understandable yet lacks energy and imagination; some interpretation may be needed to understand parts of the piece. (Education Northwest, 2016, p. 7)

Characterizing less-than-excellent work in this way is certainly useful for grading purposes, but it may direct attention away from improving the work.

Moreover, in many rubrics, such descriptions tend to be rather formulaic, involving changes of a few words to indicate differences in quality. Indeed, I have heard people who construct rubrics admit to using a thesaurus to generate near-synonyms

that they can use to differentiate work at one level from work at another. And then, of course, students embark on adjective hunts to see which words have changed from one level to the next. None of this does anything in particular to enhance student learning.

Third, rubrics identify particular aspects of quality and elevate them to a kind of special status. It could be that improving performance from one level to the next involves small or even minimal changes in a number of complex attributes, but because listing all these would make the rubrics far too cumbersome, only certain aspects are mentioned, and the development focuses on these particular aspects of quality and prioritizes them at the expense of others. Greg Ashman (2015) summarizes it as follows in figure 3.4.




*Source: Ashman, 2015.*

**Figure 3.4: How rubrics fail.**

Rubrics clearly have a role to play in helping students, from preK to college, understand what they are meant to be learning, but it is important to remember that rubrics are really just collections of success criteria, organized into a rectangular grid. The arrangement is certainly convenient, but it may also be constraining, and even misleading, since the rubrics rarely undergo any rigorous calibration. With a large and complex rubric, it would be impossible to ensure that level 4 on one dimension of the rubric was equivalent in value or in difficulty to level 4 on another dimension of the rubric. Rubrics have a role to play, but the important thing with rubrics, as with all other ways of sharing learning intentions and success criteria with students, is that they communicate effectively to students what they should be learning and

the basis on which teachers will assess their work. What matters most is not what teachers put into the rubric but what the students get out of it.

## Practical Techniques

The following techniques may be helpful for helping students understand and achieve learning intentions in the classroom: strengths and weaknesses discussion; model papers; What Not to Write; immediate and delayed post-tests; test-item design; daily sign-in; choose-swap-choose; and WALT, WILF, and TIB. We will now discuss each of these in detail.

### Strengths and Weaknesses Discussion

An important technique for helping students understand learning intentions and success criteria is asking them to look at examples of other students' work and to engage in a discussion about the strengths and weaknesses of each. For example, a middle school science teacher is teaching a class how to write laboratory reports. Before students are asked to write their own reports, she chooses five anonymous examples of work from one of last year's classes, and groups of students decide whether some of these are better than others. If they decide that some are, indeed, better than others (as they always do!), they rank the pieces of work in terms of their quality, and then each group reports back in a whole-class session. Once each group has shared its responses, the teacher invites students to give reasons for their views, and these are then used to co-construct a scoring rubric for laboratory reports. As noted previously, this is not a democratic process. The teacher uses her own subject knowledge to shape the discussion to ensure that the scoring rubric faithfully captures developing notions of competence in the subject.

To this end, it can be useful for the teacher to have prepared a model of progression that can underpin a scoring rubric. For example, a group of geography educators propose the following model of progression in geographical writing (Davies, Durbin, Clarke, & Dale, 2004):

- Stage 1 *Extremes*: using accurate place names to identify locations and using extremes like hard/soft, hilly/flat, wet/dry, rich/poor to describe features and places.
- Stage 2 *Different types*: recognizing different kinds of places using words like warm, cool, freezing and very hot as well as hot and cold; using terms like detached, semi-detached, terraced and flats to describe different dwellings.

- Stage 3 *Comparisons*: using number to compare features (for example, twice as many people, half the range of temperature) and places, and describing the differences within places.
- Stage 4 *Ratios and patterns*: grouping descriptions to give a sense of a whole place and using terms that combine ideas such as population density, converge and diverge, and humidity to describe features and places. (p. 23)

While it would probably not be helpful to give the students a model such as this, it is helpful for teachers to have thought about models of progression in their subjects ahead of time.

## Model Papers

A variation on the strengths and weaknesses technique described previously is using student sample papers to exemplify excellent work. For example, a high school teacher assigns his class the task of writing a paper. The teacher collects the papers and assigns each paper a provisional grade in his gradebook but does not write anything on the papers. The following day, students receive their own essays, together with copies of what the teacher judges to be the three best essays in the class, and the teacher asks the students to read them before the next lesson. (A key difference between this technique and the strengths and weaknesses discussion is that, here, students are told which examples are good.) During the next lesson, the class discusses what it thinks are the important features of the three essays judged as good. The teacher then invites all students (including those who had written the three selected essays) to redraft and resubmit their essays.

Two features of this technique are worth noting. First, it exemplifies excellent work in a concrete way. Second, and probably more important, the students whose essays are not selected have to do intellectual work to compare their own work with the samples the teacher provides, so the feedback is more engaging than it would be had the students been given feedback someone else provided. There are, however, a number of cautions that need to be taken into account in sharing examples of work with students.

The first is that it is important that students believe that the model papers are within their ability to achieve. In one study of 150,000 people who registered for a Massive Open Online Course (MOOC), a total of 5,740 got as far as submitting the first assignment, and then assessed the work of their classmates (Rogers & Feller, 2016). Students who read classmates' essays that were much better than their own were less likely to continue with the course, and this effect was strongest for those who read the very best essays. However, there was no complementary effect for those

who read work that was worse than their own. Those who read work much worse than their own completed the course at the same rates as those who read work at about the same level as their own (Rogers & Feller, 2016).

The second danger is that when a single example of the highest-quality work is shown to students, this example may be seen as a model to be copied rather than just an example. In the social studies example, it is significant that the teacher shared three (very different) examples of the best quality work with the class, rather than a single example. Students could see that there was not a single "royal road" to success, but in fact many ways of getting an A.

The third caution is that student learning is maximized when we direct students' attention to the things we are focusing on. For example, many people are familiar with dummy text passages such as this:

> Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

This particular passage of text has been in use since the 16th century, when an unknown printer scrambled a galley of type in order to produce samples of typeface. The important idea here is that if the passage were text that the reader could understand, the reader would read the text, rather than looking at layout and typeface. When the text is unintelligible, the reader has to focus on the things he or she is meant to be focusing on—typeface and layout. In a similar way, if we choose examples of student work where all aspects of the work are excellent, rather than the specific features on which the teacher is focusing, then the reader is less likely to draw the intended conclusion. So, for example, if a really good essay on checks and balances in the U.S. Constitution is also correctly spelled, and neatly handwritten, students may focus on the handwriting and the spelling as well as the points being made about checks and balances. Those reading the piece are likely to become clearer about what makes a good essay about checks and balances where this is the only notable attribute of the work. Now, of course, we do not want our students to submit incorrectly spelled and poorly handwritten work, but when we need students to focus on just one thing, it is important that the one thing we want our students to focus on is distinctive.

Some people believe that this kind of process only works with older students (those in secondary school and higher education), but in fact, students can engage with ideas of quality in work at a very young age. For example, in one example I witnessed, a kindergarten class was painting with watercolor paint, and after about fifteen minutes, the teacher held up one girl's painting and asked the class to suggest why the teacher thought this was a particularly good example of painting. A boy looked at the painting that was being held up and then back at his own painting, and after some moments, he said, "Because it's not all brown." The boy realized that he had been too impatient in his painting and had not waited for the paint to dry before using a different color. It was only when comparing his work to a model that he was able to make this cognitive leap. But he *was* able to make that leap—and at a young age. All it took was the comparison.

## What Not to Write

Some teachers wonder why a class should spend time looking at other students' work when they could be doing their own work, but as many teachers have discovered, students are much better at spotting errors and weaknesses in the work of others than they are in their own. Once students have pointed out such errors or weaknesses, they are more likely to avoid repeating them in their own work.

One way of drawing together a session in which the class determines weaknesses in other students' work is to compile a list of *What Not to Write* (based on the TV show *What Not to Wear*), in which students share pitfalls that they recommend others should avoid. We will deal with misconceptions in more detail in the next chapter, but for now it is important to note that presenting students with both correct and incorrect examples seems to be more effective than just presenting students with correct examples.

## Immediate and Delayed Post-Tests

Another technique is conducting two post-tests—one immediately after teaching the material, and one delayed. In one study, seventy-four fourth- and fifth-grade students participated in a twenty-five-minute teaching session in which they learned about placing decimal numbers on a number line and ranking decimals in order of magnitude (Durkin & Rittle-Johnson, 2012). The students were divided into two groups at random. Half of the students studied only correct examples, and half of the students studied a mixture of correct and incorrect examples. At the beginning of the experiment, the students completed a twenty-minute test of their decimal knowledge that tested both their procedural and conceptual knowledge of decimals. Procedural knowledge was tested with questions that asked students to place

a number on a number line (for example, placing 0.9 on a number line that extends from 0 to 1), and conceptual knowledge was tested by asking students to identify equivalent decimals (for example, by asking students to circle which of 0.5100, 0.051, 0.510, and 51 were equivalent to 0.51). At the end of the teaching session, students took a second test equivalent to the first, and two weeks later, the students completed a delayed post-test.

As might be expected, all the students scored higher on the immediate post-test than they had on the pretest, and did just as well two weeks later on the delayed post-test (see table 3.4). However, while the scores of students just looking at correct examples went up by 7 percentage points for conceptual knowledge, and by 19 percentage points for procedural knowledge, for those students looking at both correct and incorrect examples, the improvement was larger: 14 and 26 percentage points, respectively, with similar benefits for low- and high-achieving students. In terms of standardized effect size, these improvements are, respectively, 0.35, 1.01, 0.72, and 1.05 standard deviations, equivalent to moving an average student up to the 64th, 84th, 76th, and 85th percentile. Moreover, students looking at both correct and incorrect examples made fewer misconception errors in the delayed post-test; the proportion went down by 1 percentage point in the "correct only" group, and by 8 percentage points in the "correct and incorrect" group (Durkin & Rittle-Johnson, 2012).

**Table 3.4: Post-Test Scores Following "Correct Only" Versus "Correct and Incorrect" Examples**

| | Conceptual Knowledge | | Procedural Knowledge | |
|---|---|---|---|---|
| | Immediate post-test score increase | Effect size | Immediate post-test score increase | Effect size |
| "Correct only" group | 7 percent | 0.35 | 19 percent | 1.01 |
| "Correct and incorrect" group | 14 percent | 0.72 | 26 percent | 1.05 |

*Source: Adapted from Durkin & Rittle-Johnson, 2012.*

## Test-Item Design

For older students (say, fourth grade and up), one technique that is particularly useful both for getting students to clarify, share, and understand learning intentions and for informing the teacher about the students' level of understanding is to have the students design test items, with correct answers, about what they have been learning. In a study involving several experiments with 260 college students, this was shown to be more effective than administering practice tests, giving students study guides, or leaving them to prepare for tests in their own way (Foos, Mora, & Tkacz, 1994).

In one of the experiments, students who created their own outlines of what they had been learning and then generated study questions based on these outlines outperformed students who received materials designed by others, but only on those items that assessed aspects of the course for which the students had generated items. The problem with this study is that it is not clear whether the increased achievement was due to the creation of an outline or the generation of study questions. A second experiment showed that it was the fact that students generated their own items that had the effect—students who generated their own items outperformed students given items generated by other students (Foos et al., 1994).

This can be a particularly effective strategy with disaffected older students, who often feel threatened by tests. Writing a test for the topic they have completed, and knowing that the teacher is going to grade the questions rather than the answers, can be a hugely liberating experience for many students.

The other benefit of having students generate their own questions is that the teacher can find out what the students think they have been learning. This is often different from what the teacher thinks they have been learning, but what the teacher thinks the students have been learning is irrelevant! A good illustration of this was an inquiry science lesson I observed, in which the teacher asked the students to design an experiment to find out what kind of habitat pill bugs preferred, mentioned in table 3.3—specifically, whether they preferred dry or moist conditions and warm or cool conditions. At the end, the students were asked to devise a question that the teacher could use to find out whether the activity had been successful, and the vast majority of students wrote questions that asked what kind of habitat was preferred by pill bugs. This showed the teacher that the students had largely missed the point of the activity—that of devising a fair test of a scientific question or proposition—no doubt in part because of the way the teacher had framed the learning intentions (see table 3.3, page 66).

## Daily Sign-In

This technique is great for engaging kindergarten students. In daily sign-in, each week, the teacher prepares a large sheet of paper with the students' names down the left-hand side of the sheet and columns headed with the days of the week. Each day, when the students arrive, they sign their names in the appropriate place. Obviously at this age, some students are able to write their names quite legibly, while others are placing little more than squiggles in the box. On Friday of each week, each student goes to the sign-in sheet with a peer, and students must agree which of their five sign-ins is the best.

### Choose-Swap-Choose

Another technique to engage grades K–2 students is *choose-swap-choose*. Choose-swap-choose works similarly to daily sign in. For example, a teacher asks each student in the class to write the letter *d* ten times. Each student then chooses which of his or her ten *d*s he or she thinks is the best and circles it. The students then swap with a neighbor, and the neighbor circles which of his or her peer's *d*s he or she thinks is the best. If the neighbors disagree, they discuss why.

The opportunities for students to talk about quality work appear to be practically unlimited. A modern languages teacher I once observed wanted her students to understand what a good French accent sounded like, so she organized her class into groups of five students and gave each group a card with the same short passage of French. In each group, students took turns reading the passage aloud, and then, after all the students in the group had read the passage, the group had to decide which of them had the best French accent. The class then heard from each of the chosen group representatives, and the teacher led a whole-class discussion about the strengths and weaknesses of each of the accents the class had heard.

### WALT, WILF, and TIB

Of course, there will be times when it is appropriate simply to present the learning intentions and success criteria to the students. In her groundbreaking book *Unlocking Formative Assessment*, Shirley Clarke (2001) introduced elementary school teachers to the acronyms WALT (We are learning to), WILF (What I'm looking for), and TIB (This is because). While there is no doubt that the use of such words can become formulaic, for many students in the elementary grades, the use of WALT, WILF, and TIB can be a useful starting point. For example, if a kindergarten teacher was introducing students to the conventions of sentence construction, the learning intention might be, "We are learning to write grammatically correct sentences." The teacher could then tell the students about the success criteria: "What I'm looking for is a capital letter to start each sentence, finger-sized spaces between the words, and a sentence that ends with a stop sign (period, question mark, or exclamation point)." She might then go on to say, "This is because it makes it easier for me to read your writing."

## Conclusion

It seems obvious that to get anywhere, it helps to be clear about where you are going, and yet, teachers have not traditionally regarded sharing learning intentions and success criteria with students as important. Unfortunately, in many districts, the

pendulum has swung too far the other way: a lesson is considered a bad lesson if the teacher fails to post a learning objective at the start.

This chapter reviewed research evidence that shows it is important that students know where they are going in their learning and what counts as quality work, but there cannot be any simple formula for doing this. Like everything else in teaching, there are no simple rules, and it is up to the teacher to exercise professional judgment in how best to communicate learning intentions and success criteria to students. While there cannot be any simple formula, there are a number of practical techniques that many teachers have found useful, some of which have been summarized in this chapter.

Once teachers and students are clear about where they are headed, the next step, of course, is to understand whether they are on track, and for that, it is necessary to collect evidence about where students are in their learning. This is the subject of the next chapter.

chapter 4

# Eliciting Evidence of Learning

We discovered in chapter 3 the importance of being clear about what we want students to learn. Once we've accomplished that, we need to ascertain where the students are in their learning. Years ago, psychologist David P. Ausubel (1968) argued that the most important factor influencing learning is what the learner already knows, and that the teacher's job is to ascertain this and to teach accordingly. In many classrooms, the process of eliciting such evidence is done mainly on the fly—teachers almost always plan the instructional activities in which they will engage their students, but they rarely plan in detail how they are going to find out where the students are in their learning. This chapter emphasizes the importance of planning this process and provides guidelines on using questions to find out what students know and discover where students' ideas come from. It delves into what makes a good question, as well as some alternatives to questions. The chapter also offers some practical guidance on how to use questions effectively to adjust instruction to meet students' needs.

## Finding Out What Students Know

Figure 4.1 displays two questions from the Third International Mathematics and Science Study (TIMSS) quoted by Shlomo Vinner (1997).

| Question 1: Which fraction is the smallest? | Question 2: Which fraction is the largest? |
|---|---|
| A. $\frac{1}{6}$ | A. $\frac{4}{5}$ |
| B. $\frac{2}{3}$ | B. $\frac{3}{4}$ |
| C. $\frac{1}{3}$ | C. $\frac{5}{8}$ |
| D. $\frac{1}{2}$ | D. $\frac{7}{10}$ |

*Source: Vinner, 1997.*

**Figure 4.1: Example questions on fractions.**

These two items appear to be quite similar, although the second question is obviously somewhat more complex because of the larger numbers involved. However, the second question turns out to be *much* more difficult than the first. For middle school students in Israel, for example, the success rate on the first item was 88 percent, but for the second, only 46 percent (Vinner, 1997). In other words, only about half of the students who answered the first question correctly could also answer the second.

When I ask teachers why this might be, a number of possible reasons emerge. Some suggest that the first item is easier than the second because the fractions in the first question can be easily visualized. Others suggest that apart from option B, which is obviously bigger than C, they are all unitary fractions (fractions with a numerator of 1), which makes comparison easier. A third common suggestion is that the lowest common multiple is already present in the first question but not in the second and so has to be calculated, making this a two-step problem, which is obviously more complicated—although it is interesting to note that many adults (including mathematics teachers!) prefer to answer the second question by converting each fraction to a percentage rather than by using the method that students are typically taught in school.

All these explanations are credible, but a little more detective work reveals something rather interesting. In answering the second question, 39 percent of the students chose B. Since this question was answered correctly by 46 percent of the students, it was answered incorrectly by 54 percent, but 39 percent chose the *same* incorrect answer. In other words, almost three-fourths of the students who answered incorrectly chose the *same* incorrect answer.

This is highly significant. After all, if students' errors were random, then we would expect each of the incorrect answers to be chosen as often as the other incorrect responses, so that approximately 18 percent of students would choose each of B,

C, and D. Instead, option B was chosen more than twice as often as the other two incorrect responses combined. This suggests that students' errors on this item are not random but systematic.

The most plausible explanation is that the choice of option B is related to how students learn about fractions. Although confused at first, students come to realize that one-fifth is *less* than one-fourth because the whole has been divided into more pieces. From this, many students conclude that the largest fraction is the one with the smallest denominator, and the smallest fraction is the one with the largest denominator. This rule works perfectly well for unitary fractions but does not work in general.

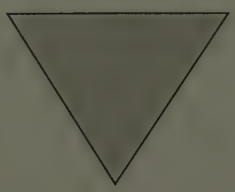
When attempting the first question, a student with this incomplete understanding of fractions will simply look for the largest denominator, find 6, and choose A—the correct answer. However, the same strategy applied to the second question leads the student to look for the smallest denominator, and, finding 4, the student chooses B—an incorrect answer.

While we cannot be sure that this is the only reason for the differences in responses for the two questions, it is significant that adding the percentage of students who chose the correct answer to the second question (46 percent) to that of those who chose the answer generated by the naïve strategy (39 percent) gives us 85 percent, which is very close to the 88 percent correct figure for the first question. In other words, there is strong evidence that many students who got the first question right got it right for the wrong reason.

Why does this matter? Because when we, as teachers, ask students a question, and we get the answer we were hoping for, we tend to conclude that the students' learning is heading in the right direction. If, however, our questions are more like the first question than the second question, then there is a real danger that we will assume that our students' learning is on track when it is, in fact, heading off in a completely different direction.

## Discovering Where Students' Ideas Come From

Many students, when asked to describe the following shape, say it's an "upside-down triangle."



It is common to regard this as a misconception since it appears that the students do not realize that shape and orientation are independent of each other; a triangle remains a triangle no matter which way it is oriented. However, some students describe this shape as an upside-down triangle even though they know that the orientation does not matter for the naming of the shape, because they are using vernacular, rather than mathematical, language.

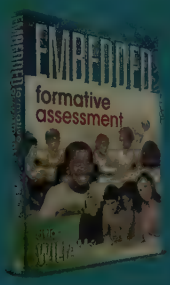
A good example of this distinction is the word *square*. In the mathematics classroom, this word means a quadrilateral with four equal sides and four equal angles. In the world outside the mathematics classroom, however, people often use the word *square* to describe orientation rather than shape (for example, a picture can be "square" on a wall even when its sides are not the same length). Additionally, we describe shapes that are, in fact, squares in ways based on their orientation rather than their shape (a baseball "diamond" is, after all, a square).

What seems like a misconception is often, and perhaps usually, a perfectly good conception in the wrong place. Describing a triangle as inverted is obviously an error in the mathematics classroom but a helpful aid to communication in other contexts (try googling the phrase "inverted triangle" to see what I mean).

The same is true in other school subjects. When a child says, "I spended all my money," this could be regarded as a misconception, but it makes more sense to regard this as overuse of a general rule. Adding *d* or *ed* to a verb to form the past tense works most of the time, so the student may be guilty of nothing more than playing the odds!

Some people have argued that these unintended conceptions are the result of poor teaching. If only the teacher had phrased his explanation more carefully, had ensured that no unintended features were learned alongside the intended features, then these misconceptions would not arise. But this argument fails to acknowledge two important points. The first is that overgeneralization is a fundamental feature of human thinking. A child might initially associate the term *dog* with a particular fluffy toy that has ears and four legs and a tail. Pretty soon, as the child learns that *dog* applies to other things than just that particular stuffed toy, the word *dog* becomes associated with a range of fluffy toys with ears, four legs, and tails—some of which are not dogs—until the child learns to discriminate between dogs, cats, and horses.

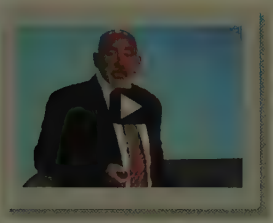
The key insight here is that students are active in the construction of their own knowledge. They literally *make sense* of the things that go on around them, including what we teach, and sometimes, the sense they make is not what we intended. Perhaps the best example of this is the answer you get when you ask children between the ages of four and seven, "What causes the wind?" "God," which is what most adults




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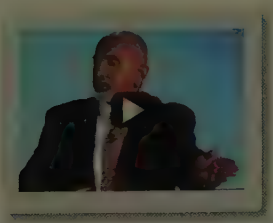


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*5:05*

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## CHAPTER 3 *Clarifying, Sharing, and Understanding Learning Intentions and Success Criteria*



**Intended, Implemented or Achieved Curriculum** – *Dylan Wiliam*
*2:53*

**Sharing Learning Intentions** – *Dylan Wiliam*
*7:07*

## CHAPTER 4 *Eliciting Evidence of Learning*



**Instructional Strategies to Engage Students in Learning** – *Dylan Wiliam*
*6:24*

**All-Student Response Systems** – *Dylan Wiliam*
*6:11*

## CHAPTER 5 *Providing Feedback That Moves Learning Forward*



**Feedback to Move Learning Forward** – *Dylan Wiliam*

*4:00*

**Giving Corrective Feedback to Students** – *Mike Mattos*

*1:23*

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## CHAPTER 6 *Activating Learners as Instructional Resources for One Another*



**Activating Learners' Intellectual and Social Capital** – *Cassandra Erkens*

*3:48*

**Unstoppable Learning: Consolidating** – *Douglas Fisher*

*4:54*

## CHAPTER 7 *Activating Learners as Owners of Their Own Learning*



**Students as Owners of Their Learning** – *Dylan Wiliam*

*6:22*

**Sharing Learning Targets and Criteria for Success with Students** – *Sharon V. Kramer*

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assume children will say, is a common response, but an equally popular answer is "trees." This is not likely to be something that they have been taught and misremembered. Children have observed that trees sway when the wind blows and have concluded that a correlation implies causation.

The second point is that even if we wanted to, we are unable to control the students' environments to the extent necessary for unintended conceptions not to arise. For example, many students believe that the result of multiplying 2.3 by 10 is 2.30. It is highly unlikely that they have been taught this. Rather, this belief arises as a result of observing regularities in what they see around them. They see that $7 \times 10 = 70$, $8 \times 10 = 80$, and so on, so they conclude that the result of multiplying whole numbers by ten is just to add a zero. We could make such a "misconception" less likely to arise by introducing decimals before teaching multiplying single-digit numbers by ten, but that would be ridiculous.

Teachers must acknowledge that what their students learn is not necessarily what they intended, and this is inevitable because of the unpredictability of teaching. Thus, it is essential that teachers explore students' thinking before assuming that students have understood something. However, generating questions that provide these powerful insights into students' thinking is far from straightforward.

Consider the following pair of equations.

$$3a = 24$$

$$a + b = 16$$

When asked what $a$ and $b$ are, many students respond that the equations can't be solved. The teacher might conclude that they need some more help with equations of this sort, but often, the reason behind the difficulties with this item is not the students' mathematical skills but their *beliefs* (Schoenfeld, 1989). When students are encouraged to talk about their difficulty, they often say things like, "I keep on getting $b$ is eight, but it can't be because $a$ is." The reason that many students have developed such a belief is that before they were introduced to solving equations, they practiced substituting numbers into algebraic formulas, where each letter represented a different number. Although the students were not taught that each letter must represent a unique number, they have generalized implicit rules from their previous experience; they talk of "upside-down triangles" because we always show them triangles where the lowest side is horizontal and then compound the error when we teach them that the area of a triangle is half the base multiplied by the height (if there is no "right way up," how can a triangle have a base?).

The point here is that had the sixteen in the second equation been any other number at all, provided they had the necessary arithmetical skills, students would have solved these equations, and the teacher would have, in all likelihood, assumed that the class's learning was on track.

Questions that give us insight into student learning are not easy to generate and often do not look like traditional test questions. Indeed, to some teachers, they appear unfair. That is certainly the reaction of many mathematics teachers to the following question, taken from the Chelsea Diagnostic Test for Algebra (Hart, Brown, Kerslake, Küchemann, & Ruddock, 1985).

Simplify (if possible): $2a + 5b$

The question is perceived as unfair because students "know" that in answering test questions, you have to do some work, so it must be possible to simplify this expression; otherwise, the teacher wouldn't have asked the question—after all, you don't get points in a test for doing nothing. Such a question may well fall short of standards and fairness guidelines established for items in a high-stakes test (for example, Educational Testing Service, 2002). But for finding out whether students understand a key principle in algebra, it is a useful question. If a student can be tempted to simplify $2a + 5b$, then the teacher should want to know that, because addressing this misunderstanding will be essential before the student can make progress in algebra.

Similar issues occur when asking students which of the following two fractions is the larger.

$$\frac{3}{7} \text{ or } \frac{3}{11}$$

Middle school students generally do not answer this item correctly. Many teachers see it as a trick question, and for that reason it probably should not be used in a high-stakes test. But it is a very good question for generating classroom discussion. The fact that this item is seen as a trick question shows how deeply ingrained into our practice is the idea that assessment should allow us to sort, rank, and grade students, rather than inform the teacher what needs to be done next.

In another example I witnessed, a teacher conducted a lesson about the molecular structure of matter. She explained to her students that water is composed of molecules, each of which has two atoms of hydrogen linked to an atom of oxygen, and asked the students to draw a sketch of what this might look like. The students in the class produced appropriate sketches, but when the teacher asked the class what was between the molecules, every single student said "water"—the students thought that the molecules were *in* the water, not that they *were* the water.

Whether to go over something one more time or to move on is a professional decision that the teacher must make, and in doing so, she needs to take a number of factors into account. For example, some aspects of the curriculum may be important but not consequential in terms of learning progressions. In other words, if a student fails to learn something at that point, he can still make progress. On the other hand, some things are absolute prerequisites for any meaningful progress. For example, there is no point in trying to teach adding fractions if students are not able to generate equivalent fractions for a given fraction. It is also necessary to consider the wider context. For example, a score of four or five on one of the College Board's advanced placement examinations confers college credit, whereas scores of one and two do not. So, a teacher might judge that moving on at a particular point is the right thing to do even though many students have not understood something, on the grounds that for a student, getting a four rather than a three represents a real difference, whereas getting a two rather than a one does not. I am not saying that this is right or wrong. It is a professional decision that each teacher needs to make for him or herself. What is unprofessional is to make that decision without finding out what the students in the class know. Questions that provide a window into students' thinking are not easy to generate, but they are crucially important if we are to improve the quality of students' learning.

One common objection at this point is that teachers do not have time to develop such questions, not least because they are too busy grading, but this just shows how inefficient many of our standard classroom routines are. Every teacher has had the experience of writing the same thing on fifteen or twenty students' notebooks because the students could leave the classroom before the teacher discovered that the students had failed to understand some crucial point. So, the important issue is this: Does the teacher find out whether students have understood something when they are still in the class, when there is time to do something about it, or does the teacher only discover this once she looks at the students' notebooks? Viewed from this perspective, grading can be seen as the punishment given to teachers for failing to find out that the students did not achieve the intended learning when the students were in front of them. Does the teacher get the class's learning back on track with the class in front of her, in one go, and when the meanings of students' responses and the teacher's questions can be negotiated, or does the teacher do it one student at a time, after they have gone away, and in writing?

There is no doubt that in the United States, most teachers spend a majority of their lesson preparation time grading student work, and almost invariably doing so alone. In other countries, much, if not the majority, of lesson preparation time is spent planning how new topics can be introduced and which contexts and examples will

be used, and teachers work in groups to devise questions to find out whether their teaching has been successful. As noted previously, questions that give us this window into students' thinking are hard to generate, and teacher collaboration will help to build a stock of good questions.

## Practical Techniques

Teacher-led classroom discussion is one of the most universal instructional practices. Typically, this follows the *initiation-response-evaluation* (I-R-E) model (for example, see Mehan, 1979): the teacher asks a question, selects a student to answer the question, and then responds to the student's answer, which is generally some kind of evaluation of what the student said.

Obviously, teacher talk dominates such interactions, but there are significant differences among countries in the relative proportions of teacher and student talk. American teachers seem to talk *less* than teachers in countries with higher average levels of achievement. For example, the 1999 TIMSS video study finds that in U.S. middle school mathematics classrooms, there are eight teacher words for every student word. In Japan and Hong Kong, the figures are thirteen and sixteen, respectively (Hiebert et al., 2003). So although many people assume that American teachers talk too much, they actually talk less than teachers in countries with higher performance. It would appear that how much students learn depends much more on *what* it is that teachers are saying, rather than *how much* they are saying—the quality, rather than the quantity, of teacher talk.

As part of a larger study of elementary school classrooms, Ted Wragg and his colleagues analyzed one thousand teacher questions they had recorded (Brown & Wragg, 1993). Over half the questions (57 percent) were managerial questions, such as, "Who has finished all the questions?" or "Have you got your books?" Another one-third of the questions required only recall of previously provided information, such as, "How many legs does an insect have?" Only 8 percent of the questions asked by teachers required the students to analyze, to make inferences, or to generalize, such as, "Why is a bird not an insect?" (Brown & Wragg, 1993). Less than 10 percent of the questions that were asked by teachers in these classrooms actually caused any new learning. The others were rehearsing things the students already knew or were about classroom management. Addressing this seems like an obvious way to improve student learning.

I suggest there are only two good reasons to ask questions in class: (1) to cause thinking and (2) to provide information for the teacher about what to do next. As an

example of the former, a sixth-grade teacher I once observed asked a class whether a triangle could have two right angles, and students were discussing their answers in groups. In one group, one student believed that it could, by building a really long, thin triangle, because he had remembered being told by someone that parallel lines meet at infinity. Another student in the group knew that the three angles had to add up to 180° and realized that by having two right angles, he had already reached that total, and so he wondered whether it was possible to have an angle of 0°. A girl in the group was confident that she could create a triangle with three right angles, if she was allowed to place one corner at the North Pole and the other two on the equator (which the first two students dismissed, saying that it wasn't a proper triangle). Although this question looks like a closed question, answering it was a valuable activity for these students because it caused thinking.

The other reason to ask questions is to collect information to inform teaching. For example, when a science teacher asks a class, "Which way does light travel? From my eye to the object or from the object to my eye?" this, too, is a closed question. After all, there are two answers, and one is incorrect. But it is a valuable question to ask, because many students believe that light travels from the eye to the object they are viewing, rather than the other way around.

Another finding of the TIMSS video study is that American classrooms are characterized by relatively low degrees of student engagement (Hiebert et al., 2003). This is consistent with other studies (for example, Rowan, Harrison, & Hayes, 2004; Weiss, Pasley, Smith, Banilower, & Heck, 2003). To begin our discussion of practical techniques that help elicit evidence of learning, we will focus first on student engagement techniques. We will then discuss wait time, evaluative and interpretive listening, question shells, hot-seat questioning, all-student response systems, ABCD cards, mini whiteboards, exit passes, discussion versus diagnostic questions, and alternatives to questions.

## Student Engagement Techniques

In his book *Outliers*, Malcolm Gladwell (2008b) presents data on the twenty-five members of a Junior Hockey League team in Canada called the Medicine Hat Tigers, including height, weight, date of birth, position, and so on. He then invites the reader to look at the data for unusual features. The remarkable feature of the data, which few people notice until someone points it out to them, is that, although the players are of different ages, eight of the twenty-five players were born in January and another six were born in February or March. The leagues are organized by age group, with those born in the same calendar year playing together. When children

begin to play hockey in competitive leagues, those born in the early part of the year are a little stronger, faster, and bigger than the rest, and so they are more likely to get picked for the team. As a result, they get more time on the ice and are also likely to get more coaching. Over time, this builds up into a significant advantage.

The same effects have been noticed in other sports, too—those born soonest after the date used to divide age groups are more likely to be playing at the highest levels. It has been termed the *Matthew effect* (Stanovich, 1986): "For to all those who have, more will be given, and they will have an abundance; but from those who have nothing, even what they have will be taken away" (Matt. 25:29, New Revised Standard Version). Why does this matter? Because we create the same kind of effect in our classrooms every day.

In almost any classroom, some students nearly dislocate their shoulders in their eagerness to show the teacher that they have an answer to the question that the teacher has just asked. In the same classroom, however, other students try to stay below the radar and avoid being called on. One teacher described his classroom thus:

> I'd become dissatisfied with the closed Q&A style that my unthinking teaching had fallen into, and I would frequently be lazy in my acceptance of right answers and sometimes even tacit complicity with a class to make sure none of us had to work too hard. They and I knew that if the Q&A wasn't going smoothly, I'd change the question, answer it myself or only seek answers from the "brighter students." There must have been times (still are?) where an outside observer would see my lessons as a small discussion group surrounded by many sleepy onlookers. (Black et al., 2004, p. 11)

High-engagement classroom environments appear to have a significant impact on student achievement. In one study, 191 students in seven fourth-grade classrooms followed the Thinking Together program (Dawes, Mercer, & Wegerif, 2000) in which teachers were provided with twelve lesson plans that were designed to help students develop their ability to use language as a tool for thinking about science and mathematics, both individually and in collaboration with other students. The researchers found that these students outperformed controls in similar schools on both teacher-constructed measures and standardized science achievement tests (Mercer, Dawes, Wegerif, & Sams, 2004). More surprising, these students outperformed controls on Raven's Progressive Matrices, which is a purely spatial test of intelligence. Engaging in classroom discussion really does make you smarter.

So, when teachers allow students to choose whether to participate or not—for example, by allowing them to raise their hands to show they have an answer—they are actually making the achievement gap worse, because those who are participating

are getting smarter, while those avoiding engagement are forgoing the opportunities to increase their ability. Teachers can, however, utilize some strategies to encourage students to participate while remaining engaged. The most common revolve around selecting students at random and implementing the "no hands up" rule.

Many teachers now employ a rule of "no hands up except to *ask* a question" in their classrooms (Leahy et al., 2005). The teacher poses a question and then picks on a student at random. One middle school teacher from Greenwich in southeast London describes this technique as *pose-pause-pounce-bounce.* She poses the question, pauses for at least five seconds (sometimes, to help her measure the time, she mutters, under her breath, "One, two, three, four, got to wait a little more"), "pounces" on one student at random for the answer, and then "bounces" that student's answer to another student, again at random, saying, "What do you think of that answer?"

Many teachers believe they can choose students at random without any help, but after some reflection, they generally realize that when they are in a hurry to wrap up a discussion so that the class can move on, they are often drawn to one of the "usual suspects" for a good answer. This is why many teachers have found that using some sort of randomization device is useful. Examples of these can be downloaded from the Internet, and randomization routines are often included in the software supplied with interactive whiteboards. There are now even apps for the iPhone into which teachers can enter student names and the app will draw a name at random. However, a beaker of Popsicle sticks (or for science teachers, tongue depressors) on which the students' names are written does the job just as well. The Popsicle sticks are also much more flexible; one teacher deals with inattentive students by writing their name on ten additional sticks and adding them to the beaker. If there is any concern that the teacher is "rigging" the choice of sticks, many teachers have resolved the issue by giving the beaker of sticks to a student to be the selector for the day.

The major advantage of Popsicle sticks can also be a disadvantage. It is essential to replace the sticks to ensure that students who have recently answered know they need to stay on task, but then the teacher cannot guarantee that all students will get a turn to answer. One way around this is to have a dividing screen in the beaker, and when teachers select sticks, they place the selection in the "discards" side of the beaker. That way, if students who have just answered a question seem off task, the teacher can select a stick from the discards side of the beaker to reinforce the idea that just because one answers a question does not mean one can relax for the next thirty or so questions.

Selecting students at random is a radical change in the *classroom contract* (Brousseau, 1984)—the network of norms and expectations about how the classroom

will operate and the roles of teachers and students. Moreover, it seems fair to say that just about everyone dislikes *no hands up* when it is first introduced. Teachers dislike it because it involves a change to well-established routines, and students accustomed to classrooms where participation is optional dislike it too. Most teachers realize that being called upon at random will be a shock for students unused to participation in classrooms. Like any technique in teaching, educators should apply random questioning with sensitivity and discretion. Many students are very shy and do not want to participate, and teachers need to be sensitive to this. However, deciding that the best response to such a situation is never to call on that student may be just as bad, since that would allow the student not to engage in any of what is going on. Some teachers allow students to place a forefinger on their chins to show they are still thinking. The fact that this is a clear visual sign allows the teacher to monitor whether some students are using this all the time.

It is also important to note that using *no hands up* is not intended as a "gotcha" to embarrass students; rather, it is intended to ensure that all students realize that participation is not optional. Students are much less likely to be tongue-tied if they have been given a few minutes to think about the answer to the question and to discuss this with a neighbor before the teacher chooses a student to respond. Requiring students to talk to a neighbor has two advantages—it allows students to assemble their thoughts, and it also provides an opportunity for students to rehearse the vocabulary that they might use in a response. Another good way to lower the stakes with *no hands up* is, when students are selected at random, they are not asked to say what they themselves thought but rather are asked to report what their talk partner said. This is less emotionally charged, because students are asked to relay what someone else has said, and of course this has the additional advantage of making students listen to each other.

However, moving to random selection can also be unpopular with students who participate regularly. For some, this is because they are no longer able to show the teacher that they have an answer. One way to tackle this issue is to ask two students at random for responses to a question, but for the third answer, ask the remainder of the class if anyone has anything else to contribute. It is also essential to ask the question first, give students time to think, and perhaps talk with a neighbor, and then pick the student or students to answer. Doing it the other way around guarantees that all but the selected student know they need not pay attention.

For other students, random questioning is unwelcome because they are unable to control when they are asked questions. In one classroom, three students who were perceived by their teachers as having their hands up in response to every question

actually removed their Popsicle sticks from the beaker when the teacher wasn't looking (Barry & Hardy, 2010). When this was discovered, they revealed that they only raised their hands when they had an answer, which was most of the time. However, they found the prospect of being asked to respond when they did not have an answer threatening to their image of themselves as high-performing students, and this feeling was so strong that they preferred never answering questions to the risk of being asked a question to which they did not know the answer (Barry & Hardy, 2010). This is a perfect example of what psychologists call a *performance orientation* to their work (Dweck, 1986), which is discussed in greater detail in the following chapter. Students' reactions to *no hands up* illustrate a perhaps surprising effect of the technique, and this is that it tends to make classes more cohesive and lead students to be more supportive of each other as learners. Where high-achieving students participate only when they are absolutely certain they are correct, then the impression given is that they know everything. When they are selected at random, they make mistakes, and this sends the message to other students in the class that everyone makes mistakes. Moreover, high-achieving students become more appreciative of their classmates. As one high achiever said, after his teachers had been using *no hands up* for ten weeks, "I never knew my classmates were so smart." When students are allowed to raise their hands to signal a desire to participate, the classroom conversation tends to be dominated by the students who are quickest, and not necessarily those with the most important, thoughtful, or interesting thing to say.

Perhaps what is most surprising to many teachers is how quickly students get used to a *no hands up* classroom. Even if they don't like the idea, they accept that it is fairer, and they want to make it work. This was neatly illustrated by an incident from Båtsmansskolan, an elementary school in Haninge, Sweden. The teacher, Susanne Holmberg, described what happened when a new student joined the class well after the new school year had begun:

> I got a new pupil two weeks ago. And when I picked up the cup with lollipop sticks to get the pupils' answers, I discovered that there was a stick with the new girl's name on it (and it wasn't me who had put it there). And then I discovered there was another stick with her name on it. Two other pupils had done something I had forgotten—to put in a stick for the new girl. (as cited by E. Hartell, personal communication, March 17, 2015)

At this point, it is worth pointing out that it is not students raising their hands that is the problem; it is teachers selecting only those who have raised their hands. It is fine to allow students to raise their hands to show they want to respond as long as the teacher does not restrict her or his choice of respondent to those who have their

hands raised. The problem is that when you have a sea of waving arms, then it seems to be harder to select a student who has not raised a hand.

When teachers do call on students at random—what Doug Lemov (2010) terms *cold calling*—many students will still resist engagement by responding with, "I don't know." How the teacher responds to this is crucial. Some teachers accept this and move on to ask the same question of another student, but this, in effect, allows the student to opt out. By adopting a rule of "no hands up except to ask a question," the teacher has implied that the classroom should be one in which all students are engaged. However, the student has indicated that she does not wish to engage, and the teacher has, in turn, accepted this. In other words, the student has successfully resisted the teacher's attempt to change the classroom contract. A better response is for the teacher to say, "OK; I'll come back to you," and then go around the classroom and get a number of responses to the question, after which the teacher comes back to the original student and says, "Now, which of those answers do you like best?" Even when the other students have answered with a single correct response, it is still valuable to go back to the original student simply to get her to repeat the correct response, because it emphasizes that this is a classroom in which there are no opt-outs (Lemov, 2010).

Often, students will say, "I don't know," not because they do not know, but because they cannot be bothered to think. Henry Ford once said that we know thinking is difficult because of the lengths people go to in order to avoid doing it. Although the teacher first needs to be sure that this *is* the reason for the student's response, in such situations, one teacher suggests that teachers could ask, "OK, but if you did know, what would you say?" (E. Keene, personal communication, March 7, 2011). Other practices that support the students without letting them off the hook are allowing students to *phone a friend*, or for multiple-choice items, they can *ask the audience* or ask to *go fifty-fifty*, where the teacher removes two incorrect responses. All these strategies derive their power from the fact that classroom participation is not optional, and even when the student resists answering a question, the teacher looks for ways to maintain the student's engagement.

## Wait Time

How much time a teacher allows a student to respond before evaluating the response is also important. We have known for many years that teachers do not allow students much time to answer questions (see, for example, Rowe, 1974), and if they do not receive a response quickly during this "thinking time," they will often "help"

the student by providing a clue, weakening the question in some way, or moving on to another student. However, the amount of time between the student's answer and the teacher's evaluation of that answer—what we might call *elaboration time*—is just as, if not more, important, because when the teacher does not evaluate the student's response immediately, then students are likely to extend their responses. When they are talking, they are thinking, and when they are thinking, they are learning.

Of course, when the question is a simple matter of factual recall, allowing a student time to reflect and expand upon the answer is unlikely to help much. If you don't know the capital of Connecticut, extra time isn't going to help. But when the question requires thought, increasing the time between the end of the student's answer and the teacher's evaluation from the average wait time of less than a second to three seconds produces measurable increases in learning, although, according to Kenneth Tobin (1987), increases of thinking time beyond three seconds have little effect and may cause lessons to lose pace.

While increasing wait time might be good advice in the course of a typical question-and-answer session in a classroom (and particularly a higher education lecture theater), what matters far more is what students are doing during the wait time. Long periods of "dead" time are obviously uncomfortable, but if a teacher has asked a question of the class, and no response is forthcoming, then rather than providing the answer herself, the teacher would be better advised asking the students to discuss the question with a neighbor (a technique often called *think-pair-share*). Students' conversations are likely to be more focused when they know that the teacher may well select a student at random to respond at the end of the conversation.

## Evaluative and Interpretive Listening

John Wooden, head coach at the University of California at Los Angeles, was one of the greatest college basketball coaches of all time—some have suggested that he was the greatest coach of any sport (Serwer, 2010). He was once asked why other coaches were not as successful, and he said, "They don't listen. Listening is the best way to learn. You have to listen to those who you are supervising" (Serwer, 2010). But, of course, what matters is *how* you listen.

When teachers listen to student responses, many focus more on the correctness of the answers than on what they can learn about the students' understanding (Even & Tirosh, 1995, 2002; Heid, Blume, Zbiek, & Edwards, 1999). It is easy to identify such teachers because when they get incorrect answers from students, they respond by saying things like, "Almost," "Close," or "Nearly; try again." What the teacher is really

saying is, "Give me the correct answer so that I can get on with the rest of my script for the lesson." Brent Davis (1997) calls such teacher behavior *evaluative listening.*

Teachers who listen evaluatively to their students' answers learn only whether their students know what they want them to know. If the students cannot answer correctly, then the teachers learn only that the students didn't get it and that they need to teach the material again, only, presumably, better. However, when teachers realize that there is often information about *how* to teach something better in what students say—and thus how to adjust the instruction to better meet students' needs—they listen *interpretively.* What such teachers seek to learn from the students' responses is not, "Did they get it?" but rather, "What can I learn about the students' thinking by attending carefully to what they say?" A seventh-grade student eloquently summarized the shift from evaluative to interpretive listening by, when asked if she had noticed any change in her teacher over recent months, saying, "When Miss used to ask a question, she used to be interested in the right answer. Now she's interested in what we think" (Hodgen & Wiliam, 2006, p. 16).

## Question Shells

There are many general structures that can help frame questions in ways that are more likely to reveal students' thinking. One is the general structure of "Why is _____ [or *not*] an example of _____?" We saw one of these earlier ("Why is a bird not an insect?"). Similarly, rather than asking, "Is magnesium a metal?" more thoughtful and reasoned responses are likely to be generated by asking, "Why is magnesium a metal?" Other examples are shown in table 4.1.

Table 4.1: Examples of the Use of Question Shells

| Original | Reframed |
|---|---|
| Is a square a trapezoid? | Why is a square a trapezoid? |
| Is carbon a metal? | Why is carbon not a metal? |
| Is *être* a regular verb? | Why is *être* an irregular verb? |
| Is this a sentence or a clause? | Why is this a clause rather than a sentence? |
| Is slate a metamorphic rock? | Why is slate a metamorphic rock? |
| Is *The Merchant of Venice* a comedy [or a tragedy]? | Why is *The Merchant of Venice* a comedy [or a tragedy]? |
| Is 23 prime? | Why is 23 prime? |
| Is photosynthesis an endothermic reaction? | Why is photosynthesis an endothermic reaction? |

Another technique is to present students with a contrast and then ask them to explain the contrast, as shown in table 4.2.

Table 4.2: Examples of Questions Reframed in Terms of Contrasts

| Original | Reframed |
|---|---|
| What is a prime number? | Why is 17 prime and 15 not? |
| What was life under apartheid like? | How were the lives of blacks and whites different under apartheid? |
| Is a bat a mammal? | Why is a bat a mammal and a penguin not? |

## Hot-Seat Questioning

Another useful technique for deepening classroom discussion is *hot-seat questioning*. In the typical classroom, the teacher scatters questions around the class, which can create student engagement, but tends to lead to a rather flat discussion in which there is little development of the subject matter. In hot-seat questioning, the teacher asks a student a question and then a series of follow-up questions to the same student to probe the student's ideas in depth. Other students in the class pay close attention because they know that at any minute, the teacher can turn from the student in the hot seat to anyone else in the class (chosen with a Popsicle stick!) and say, "OK, summarize for me what Jamilla just said." When a teacher is having a discussion with one particular student, it is not enough that other students are quiet. If we want to create a community of learners, then it is important that when one student is speaking, the other students are listening appreciatively to what that student and teacher are saying.

The techniques discussed so far are very effective ways of creating classroom engagement, but they are not necessarily good sources of evidence for teachers' instructional decision making. When I was teaching full-time, the most common decision I made every day was, "Do I need to go over this one more time, or is it OK to move on?" Like most teachers, to answer the question, I would make up a question there and then, I would ask the class, and perhaps half a dozen students would raise their hands. I would then select one of those students, and if the student responded correctly, I would say, "Good," and move on. Implicitly, I was assuming that the response of that one student was a good guide to the learning of the whole class. Now I am fairly sure that if someone had asked me back then, "Is the response of that student a good guide to the level of understanding of the rest of the class?" I would have said that it was not. But no one asked me that question. So, like most teachers, I carried on making decisions about the learning needs of a diverse group of thirty or more students based on the response of one confident volunteer.

Now, as noted previously, asking a random student rather than a volunteer would be better, because a student chosen at random is more likely to be representative of the needs of the group than a confident volunteer, but the evidence base for the decision is still pretty thin. If we want to make decisions that reflect the learning needs of all our students, then we need to broaden the base of our evidence, and this is why formative assessment really is, at its heart, assessment.

This is an important observation because many people have argued that the term *formative assessment* is unhelpful since, when the word *assessment* is used, people think about quizzes, tests, and examinations—the formal mechanisms for finding out what our students know. This is true, but *assessment* is the right term for this process, because as Lee Cronbach (1971) pointed out many years ago, an assessment is simply a procedure for drawing inferences. We give students things to do, we look at what they do, and based on the evidence we identify, we draw conclusions. Thinking about teachers' instructional decision making as an assessment process draws attention to the adequacy of the evidence that we have for the conclusion that we want to draw. And if you're drawing conclusions about the learning needs of thirty or more students based on evidence from one of them, you aren't going to draw good conclusions no matter how good a teacher you are, because the quality of your evidence is poor. If teachers are to harness the power of high-quality questioning to inform their instructional decisions, they need to find ways of broadening the evidence base for their instructional decisions. That means routinely using all-student response systems when teaching a class.

## All-Student Response Systems

The idea of an all-student response system is very simple: the teacher asks a question in such a way that allows her to get a response from every student in real time. Some teachers employ class polls—going around the class and asking each student for his or her view on a question, such as, "Global warming: natural or anthropogenic?" or, "Macbeth: mad or bad?" but these work well only when every student expresses a view, and getting around the whole class can cause a considerable loss of momentum. That is why it is best to collect the information from every student at the same time.

Many teachers collect information from the whole class simultaneously with techniques like thinking thumbs or fist to five, where students indicate their confidence in their understanding either by the position of the thumb (pointing up: confident; horizontally: not sure; down: still confused) or with a number from zero to five by holding up a clenched fist (zero: do not understand at all) or the appropriate number

of fingers (up to five fingers: confident). The problem with such techniques is that they are self-reports, and, as we know from literally thousands of research studies, self-reports are unreliable.

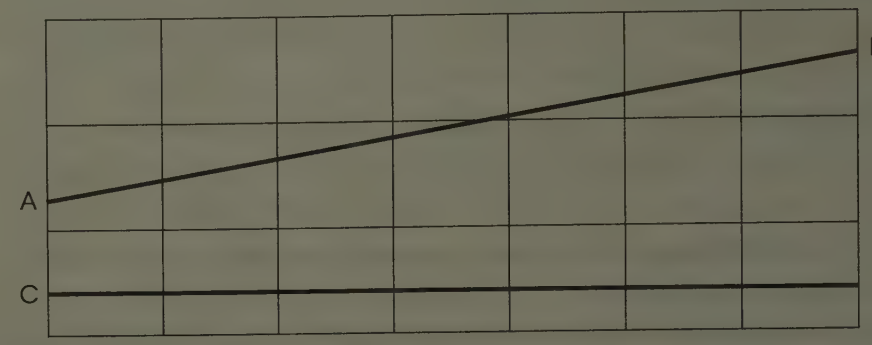
However, a very small change can transform useless self-reports into a very powerful tool. Simply make sure that the question the teacher is asking is cognitive rather than affective—in other words, that it is asking about thinking, not a feeling. For example, I observed an elementary school teacher teaching her students when the word *its* needed an apostrophe and when it didn't. On the board, she wrote the sentence, "Its on its way" and invited students to come up to the board and add any necessary punctuation. One student came up and added a period (presumably on the grounds that one should always pick the low-hanging fruit first); a second added an apostrophe to the first *its*; and a third student added an apostrophe to the second *its*. The teacher then asked, "Is that now correct?" and every student had to respond by holding out a thumb pointing up or down. Just this small change in the way the question is posed creates a situation in which there is no place for the students to hide. If they signal that it is correct when it is not, then they reveal that they do not understand. On the other hand, if they signal that it is incorrect, then the teacher is likely to ask them to go to the board to correct the error.

The same basic idea can be used in every subject and at every level. In a high school chemistry class I once witnessed, for example, students were learning how to balance chemical equations. On the board, the teacher wrote the basic unbalanced equation for the reaction of mercury hydroxide with phosphoric acid to produce mercury phosphate and water.

$$Hg(OH)_2 + H_3PO_4 = Hg_3(PO4)_2 + H_2O$$

Students were invited to go up to the board to suggest changes to help balance the equation, and after there were no more volunteers, the teacher asked the students to indicate, by thumbs up or down, whether the equation was now balanced. Similarly, in an undergraduate economics lecture, a professor could ask whether the effect of a policy change has been correctly indicated on a supply-demand graph.

Of course, this basic technique can be enhanced by expanding the number of choices to three or more. In an elementary school mathematics class, a teacher wanted to check on students' understanding of length and asked the students to indicate, by holding up one, two, or three fingers, which of the following statements were true of the lines shown in figure 4.2 (page 102).




Which of the following statements is true?

1. AB is longer than CD.
2. AB is shorter than CD.
3. AB and CD are the same length.

*Source: Adapted from Hart, 1981.*

**Figure 4.2: Example of a question with multiple answer options.**

I once observed a middle school science teacher teaching students to distinguish between different kinds of levers. After explaining that the key principle of the classification of levers concerns the relative arrangement of the load, the effort, and the fulcrum, she illustrated the principle with three examples: a seesaw (type 1), a wheelbarrow (type 2), and a deep-sea fishing rod (type 3). To check the students' understanding, she asked the class how it would classify a pair of tweezers, requesting that each student hold up one, two, or three fingers to indicate his or her response. She was surprised when most of the students indicated that they thought the tweezers were a type 2 lever. When she asked them why, the students replied that there were two arms to the tweezers. She realized that she needed to get the students to understand that it is the relative distribution of the effort, load, and fulcrum that is important, not the number of components, by introducing more examples, such as a pair of scissors and a nutcracker.

Maureen Wellbery teaches second grade at the American International School of Vilnius, in Lithuania. After hearing about the value of using the all-student response system at a workshop, she decided to try it out the very next day. She took the basic idea of "Is this now correct?" described previously (the two examples of punctuating "its" and balancing chemical equations) and gave it a neat twist. Instead of just indicating whether a response was correct, students had to indicate, with their fingers, the *number* of errors.

> This morning during Language Arts I typed a sentence on the Smart Board with six grammatical errors. I had five choices of answers, ranging from no mistakes to six (which was the correct answer).
>
> The formative assessment was so revealing. One girl said there were no mistakes, only one student was correct, and the others were split among the other choices. I invited two students to come to the board and explain their reasoning, and found everyone was very focused and attentive.
>
> No one laughed at any incorrect responses, and the most interesting comment came from the boy who did choose the answer of six mistakes. He said that although he did choose the correct number of answers, he was wrong on a correct answer and incorrect on what should have been the final error. He was so honest, but didn't need to be. He could have just sat back basking in the limelight of being the only one who solved the problem. (M. Wellbery, personal communication, March 14, 2016)

As a final couple of examples, in order to illustrate how such questions can be used in higher education, here are two questions that I use routinely with my graduate classes. The first, in figure 4.3, is a question that I use to check whether my students have actually read a piece of required reading on the work of Piaget and Vygotsky for a master's course on educational psychology.

Which of the following is the most important difference between the theories of Piaget and Vygotsky?

A. Piaget places greater importance on the role of conservation in cognitive development.
B. Vygotsky places greater importance on the role of cultural artifacts in cognitive development.
C. Vygotsky did not believe in distinct stages of cognitive development.
D. Piaget was a social constructivist, while Vygotsky placed greater emphasis on cultural-historical activity theory.

**Figure 4.3: Example question checking understanding of differences between Piaget and Vygotsky.**

Students who have not completed the reading assignment tend to choose option C, since they know that Piaget did believe in distinct stages of development, and gave them names such as *sensorimotor stage* and *concrete operational stage*. However, as the reading that I had assigned makes plain, Vygotsky also believed that learners

move through a series of stages of development but did not give them labels. The best answer is option B, because Vygotsky placed a much greater weight than did Piaget on the role of cultural artifacts, such as language, in a child's development.

The second example, seen in figure 4.4, comes from a research methods course at doctoral level, and concerns a central principle of statistical inference—the logic of null hypothesis significance testing.

An experimental study of a new method of teaching reading reports that a result was significant ($p < 0.05$). This means that:

A. The experimental group outperformed the control group by 5 percent.

B. There is a 5 percent chance that the experimental group did not outperform the control group.

C. There is a 5 percent chance that there is no difference between the experimental group and the treatment group.

D. There is only a 5 percent chance that the observed result would have happened if the experimental and control groups had the same achievement.

**Figure 4.4: Example question testing understanding of the null hypothesis.**

Options A, B, and C are all incorrect, and relate to standard errors that many students (and even some researchers!) make. The only correct answer is D, and if the students choose this option without copying others, it's a pretty good indication that they have understood this point and are ready to move on.

In each of these examples, the teacher is able to ensure student engagement—after all, it is very easy to tell if a student has not voted—and is able to tailor his or her response to the students' needs. These two principles—engagement and responsiveness—are at the heart of effective formative assessment.

In higher education, the use of such questions has been advocated by Harvard physicist Eric Mazur for many years (Crouch & Mazur, 2001), and also by Nobel Prize winner Carl Wieman (2014). A review of 225 studies of the use of such questions in college courses in science, technology, engineering, and mathematics found that students on courses using traditional (non-interactive) lectures were 1.5 times as likely to fail compared with students on courses where teachers used all-student response questions in order to promote active learning (Freeman et al., 2014).

In large classes, such as those in higher education, it makes sense to use electronic voting systems (sometimes called *classroom clickers*) to collect students' responses. Vendors of these systems point out that with such electronic devices, or with Nolan

Amy's paper-based version called Plickers (see www.plickers.com), it is easy to collect every single student's response, but with younger students, that may not be a good thing. If we want to create a classroom environment in which students are happy to make mistakes, then the last thing we should do is record every single one of them. The idea that every single error one has ever made in a classroom is recorded in a spreadsheet forever is somewhat unappealing, which is why I generally advocate systems where the responses of students are not routinely recorded, but rather used by the teacher to help decide what to do next. A further limitation of electronic voting systems is that most such systems only allow for a single correct response, which is why many teachers prefer to use ABCD cards.

## ABCD Cards

The idea is simple. Each student has a number of cards, each of which bears a single letter. Some teachers just use A, B, C, and D, while others use sets with nine cards: A, B, C, D, E, F, G, H, and T (for true-or-false questions). These can be used just like fingers, but they can also be used with questions for which there is more than one correct answer, as in figure 4.5, or when different views may be present.

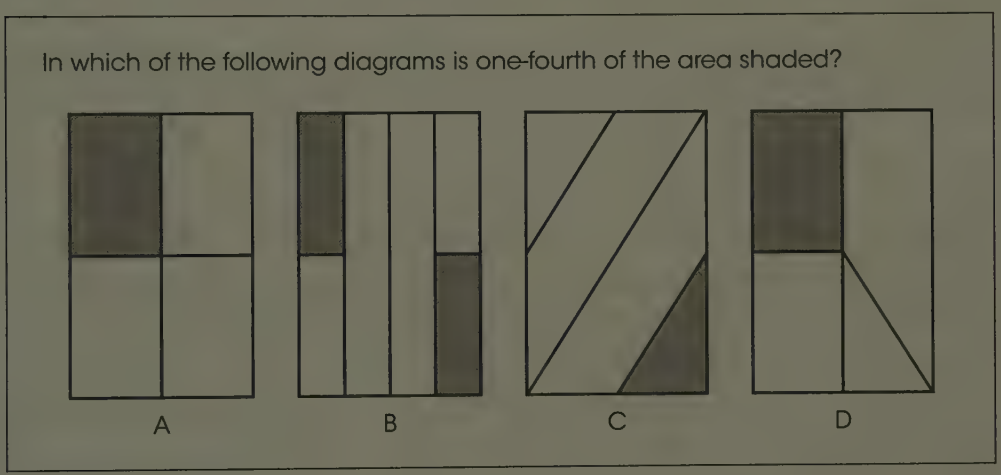



**Figure 4.5: Example question using ABCD cards.**

Most students should recognize that A and B represent one-fourth, and hopefully, most students will also realize that in diagram C, the fraction shaded is not one-fourth. However, many teachers tell students that if one out of four regions in a diagram is shaded, the four regions must have the same area for this to represent one-fourth, and as a result, many students come to believe that the fraction shaded in D is not one-fourth, because the regions are not the same area.

The use of multiple correct answers allows the teacher to incorporate items that support differentiation, by including some responses that all students should be able to identify correctly but others that only the ablest students will be able to answer correctly. Such differentiation also helps keep the highest-achieving students challenged and, therefore, engaged.

ABCD cards can also be used when there are no right and wrong answers but different views. A teacher was teaching a lesson on media reporting of emotionally charged events, and had chosen the 1985 Heysel Stadium tragedy, during which thirty-nine people died and over six hundred people were injured in rioting before a soccer game between Liverpool and Juventus. The students read various press clippings of the reporting of the event, and then the teacher asked the students to indicate, by holding up cards, who they thought was most to blame for the tragedy.

A. The Liverpool fans

B. The Juventus fans

C. The police

D. The match organizers

E. The builders of the stadium

During the discussion, students displayed the cards indicating their views on their desks, and the teacher brought them into the discussion at the appropriate time (students who changed their minds during the discussion were encouraged to change the letters they were displaying). At the end of the discussion, the teacher asked the students to vote again, and she was pleased to see that the class now had a much more complex view of the tragedy, with each student displaying at least two cards. One student, a girl named Katie, was waving all five cards in the air, "Because everyone here had some responsibility." What is interesting here is that these complex and nuanced responses by the students would not be possible with most electronic voting systems, where only one correct response is possible.

One elementary school teacher takes the idea of cards one step further with what she calls *letter corners*. She uses multiple-choice questions with four options (A, B, C, and D), and if each option has at least three students supporting it, she sends the students to the four corners of the room, which she has labeled A, B, C, and D. The task of the students in each corner is to work out how to convince the students in the other corners that their choice is the best. Occasionally, students sneak from one corner to another. They move surreptitiously because they worry that the teacher will think that they are cheating, whereas, of course, the teacher celebrates such changes

of mind as indicating that the students have learned something through discussion with peers.

A major difficulty with ABCD cards is that they generally require teachers to have planned questions carefully ahead of time, and so they are less useful for spontaneous discussion, which is where mini whiteboards come in.

## Mini Whiteboards

The mini whiteboard, sometimes touted as a modern invention, is really just the latest reincarnation of the slates used in 19th century classrooms. Whiteboards are powerful tools in that the teacher can quickly frame a question and get an answer from the whole class, whether through asking elementary school students to write down a four-letter word with a short *i* sound or, as we saw in chapter 2, asking students in an AP calculus class to sketch the graph of $y = \frac{1}{(1+x)^2}$.

One teacher wanted to use whiteboards, but there was insufficient money in the school's budget to acquire these, so instead she placed sheets of letter-sized white card stock inside page protectors to provide a low-cost alternative. She realized that this was actually far more flexible than the whiteboards, because she was able to preprint different images on the inserts for specific lessons. When she was teaching mathematics, the insert could be a sheet of graph paper. When she was teaching geography, it could be a map of the United States.

Mini whiteboards can also be used to bridge two lessons. A middle school mathematics class was learning to solve equations with a single unknown, and five minutes before the end of the lesson, the teacher asked the students to work out the answers to six questions (see figure 4.6) and write their responses on a mini whiteboard.

1. $3x + 3 = 12$
2. $5x - 1 = 19$
3. $12 - 2x = 3$
4. $4 = 28 - 3x$
5. $4x - 3 = 2x + 7$
6. $3 - 2x = 17 - 4x$

Figure 4.6: Example question using mini whiteboards.

Four of the questions related to what the class had done that lesson. As the students held up cards indicating their responses to these questions, the teacher was pleased to see that most of the students answered correctly. The last two questions related to what she planned to do for the following lesson, where the unknown quantity appeared on both sides of the equation. Very few students answered either of these two questions correctly, so she could see that what she had planned to do for the next lesson was, in fact, appropriate. Another notable feature of these questions is that the teacher had arranged for the correct answers to form a pattern that she could easily recognize, but the students probably would not (in this case, the first six digits of the decimal expansion of one-seventh). This made it much easier for her to evaluate the students' responses quickly.

## Exit Passes

When questions require longer responses, teachers can use the exit passes described in chapter 2. Examples of exit pass questions are:

- Why can't you have a probability greater than 1?
- What is the difference between mass and weight?
- Why are historians concerned with bias when analyzing historical sources?

Exit pass questions work best when there is a natural break in the instruction; the teacher then has time to read through the students' responses and decide what to do next. In the example in chapter 2 (page 47), the teacher discarded the exit passes, but if students write their names on the back of the cards, the teacher can use the exit passes as place settings for the next period with that class. The teacher can create either homogeneous groups, so that he or she can work with the students who are having the greatest difficulty, or heterogeneous groups, in which there is at least one student who provided a good answer. Of course, since the work hasn't been assessed, the students with the good answers don't know who they are, which leads to more open discussion.

All these techniques create student engagement while providing the teacher with evidence about the extent of each student's learning, so that the teacher is able to adjust the instruction to better meet the students' learning needs. Of course, the quality of the evidence—and, therefore, the quality of the instructional adjustments—depends on the quality of the questions asked, and so the next section of this chapter looks in more detail at what kinds of questions work best.

## Discussion Questions and Diagnostic Questions

Figure 4.7 is a *discussion* question that might be used in a middle school mathematics classroom.

Look at the following sequence:

3, 7, 11, 15, 19, . . .

Which is the best rule to describe the sequence?

A. $n + 4$

B. $3 + n$

C. $4n - 1$

D. $4n + 3$

**Figure 4.7: Example of a discussion question.**

Many students will select A; they see that each number in the sequence is four more than the number preceding it, so the rule must be "keep adding 4." Others choose B, because although rule A tells you to keep adding 4, it does not tell you where to start, whereas B indicates that the sequence starts with 3, and numbers are added to 3 to generate the remaining numbers in the sequence. Interestingly, students who choose either A or B often justify their choices by saying that their teacher told them that $n$ could be any number! The disadvantage with both A and B is that generating, say, the hundredth term requires first generating the previous ninety-nine. (Educators sometimes call these *term-to-term rules* since they allow one to generate the next term from the previous term.) Rules C and D are based on the observation that the numbers in the sequence are one fewer or three greater than the four times table. (Educators sometimes call these *position-to-term rules* since they allow one to generate each term simply by knowing its position.) Rule C allows one to find the hundredth term by multiplying 100 by 4 and subtracting 1 to get 399. Rule D works in the same way, except that one has to remember that the hundredth term is generated by setting $n$ to 99, since the first term corresponds to setting $n$ to 0, rather than 1, and then adding 3.

This question can lead to a valuable discussion in the mathematics classroom, since it allows the teacher to challenge the idea that mathematics is a right-or-wrong subject. For example, rules A and B can be thought of as correct even though they do not use the standard mathematical notation, but they are less powerful than rules C and D. Rules C and D are equivalent, but one can make an argument that C is more

elegant than D because to find the hundredth term, one sets $n$ to 100 with rule C, but one has to remember to set it to 99 with rule D.

Just because someone chooses rule C, that does not mean that he understands why rule C is superior to rule A, for example. The student may have chosen C because it looks more "mathematical." The teacher learns little just by seeing which of these alternatives students choose. She has to hear the reasons for the choices. That is why this question is a valuable *discussion* question, but it is not a good *diagnostic* question. It can lead to fruitful discussion in the class, but there is no point in asking this question unless you are going to have the discussion. If you don't have twenty minutes for the class discussion, it's probably not worth asking the question.

Consider, as an alternative, the question in figure 4.8.

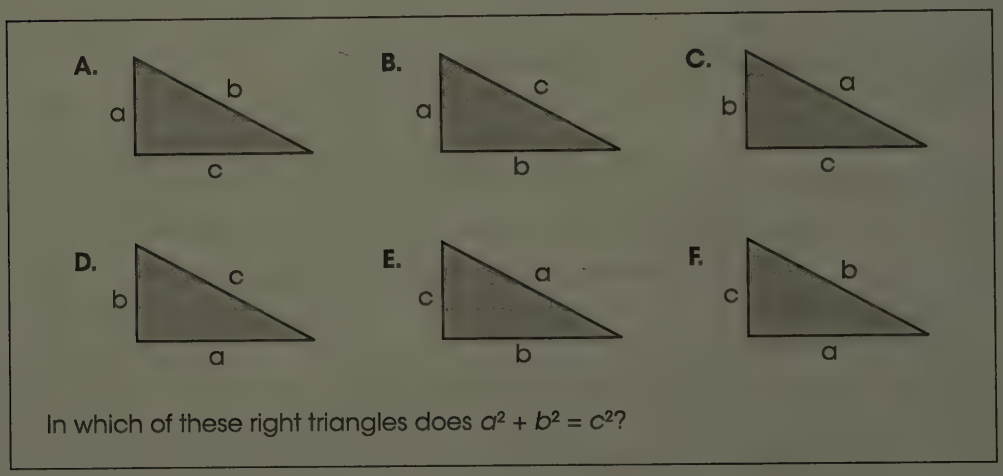



**Figure 4.8: Example of a diagnostic question.**

In some ways, this is a sneaky question, because there are two correct—and four incorrect—responses. In total, there are sixty-four possible choices of response to this question (for each of the six triangles, the student needs to decide whether the equation is true, so there are $2 \times 2 \times 2 \times 2 \times 2 \times 2$ possible combinations). Therefore, if students select B and D, and none of the others (provided they haven't copied these from other students), the teacher can be reasonably confident that they know how to apply Pythagoras's rule and can move on. Of course, if he has time, the teacher could decide to ask the students for the reasons for their choices, but the point is that, with this question, the teacher gains concrete evidence about the students' learning *without* having to have the classroom discussion.

The crucial feature of such *diagnostic* questions is based on a fundamental asymmetry in teaching: in general, it is better to assume that students do not know

something when they do than it is to assume they do know something when they don't. What makes a question useful as a diagnostic question, therefore, is that it must be very unlikely that the student gets the correct answer for the wrong reason.

In science, a teacher might ask the question in figure 4.9 to prompt discussion.

Ice cubes are added to a glass of water. What happens to the level of the water as the ice cubes melt?

A. The level of the water drops.

B. The level of the water stays the same.

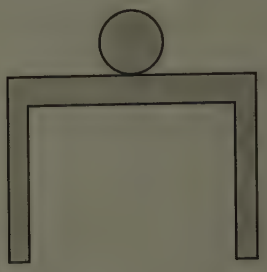
C. The level of the water rises.

D. More information is needed to be sure.

**Figure 4.9: Example of a science discussion question.**

A teacher who has been focusing on Archimedes's principle hopes that the students will choose B, but there are valid reasons for choosing alternatives. If evaporation is a significant factor, response A would be a better choice. The question does not specify that the ice cubes are floating. If so many cubes are added that they are resting on the bottom of the glass, then the water level will rise as the ice melts. Very able students may realize that B is not, in fact, correct—whatever the science teacher thinks. When ice melts, it draws heat from the surrounding water—it actually takes an appreciable amount of energy to convert ice at 0°C to water at 0°C (about 334 joules per gram, in fact). So, as the ice melts, it cools the surrounding water by more than just the effect of the ice being colder than the water. If the water is above 4°C, then as it cools, it will contract, so A would then be correct (although the amount may be too small to measure). On the other hand, if the water is below 4°C, then cooling it will cause it to expand, so C would be correct. And since all these answers depend on the assumptions made, it is fair to claim that one needs more information to be sure of one's answer, so D is correct, too. As with the previous question on mathematical sequences, the teacher learns little about the quality of his students' thinking just by seeing which responses were chosen; the teacher needs to hear the reasons for the choices, so while this may be a good *discussion* question, it is a poor *diagnostic* question.

Consider, as an alternative, the *diagnostic* question in figure 4.10 (page 112), developed by Mark Wilson and his colleagues at the University of California, Berkeley (Wilson & Draney, 2004, p. 149).



The ball sitting on the table is not moving. It is not moving because:

A. No forces are pushing or pulling on the ball
B. Gravity pulls the ball down but the table is in the way
C. The table pushes up with the same force that gravity pulls down
D. Gravity is holding it on the table
E. There is a force inside the ball preventing it from rolling off the table

*Source: Adapted from Wilson & Draney, 2004.*

**Figure 4.10: Example of a science diagnostic question.**

The first response, A, is typically selected by students with the naïve conception that if there is no movement, then there are no forces in play. The second, B, on the other hand, looks quite reasonable. Is gravity pulling the ball down? Well, we might quibble about the language, but the ball is certainly being attracted by the gravitational pull of the earth, and the table is definitely in the way. So, response B will be attractive to many students.

Response C is what the science teacher is hoping for. She has been teaching that when there is no movement, this could mean either that there are no forces acting on the object or, much more commonly, that the forces acting on the object are in equilibrium. Response D describes the situation in a similar way to response B, and response E is selected by students who, instead of seeing inertia as a property of matter, see it as a force (presumably at least in part because of the way the word *inertia* is used in everyday language).

The first and last responses, therefore, are obviously incorrect and related to well-known naïve conceptions that students have about the physical world. Response C is clearly the best answer, but what about B and D? They are correct, in an intuitive sense at least, so should the teacher accept either B or D as correct? In the science classroom, the answer is no. Responses B and D might be acceptable, naïve ways of looking at the situation, but what the teacher is trying to establish is whether her

students can think about this situation like a physicist. Physics is an unnatural way of thinking about the world—if it were natural, it wouldn't be so hard for students to learn. For example, when one ventures out in cold weather, it certainly feels as if the cold is coming in through one's clothes, but physicists have learned that it is more productive to think about the situation as one in which heat is escaping.

This question is not really asking, "What's happening here?" It is asking, "Can you think about this situation like a physicist?" And from that perspective, B and D are not acceptable. The only acceptable response is C. However, it is the plausibility of B and D that makes this such a valuable question. Students who are not secure in their understanding of forces will be tempted to opt for B or D, and so only students with a good understanding of the idea of forces in equilibrium will choose C. If a teacher asks this question, and students select C, then the teacher can be reasonably confident that they have understood the key idea about forces in equilibrium. She could, of course, ask students to explain the reasons for their choices and have a whole-class discussion, but this is not necessary for the teacher to gain valuable evidence about the students' understanding.

As far as we know, this distinction between discussion questions and diagnostic questions works in every school subject, and for content for all ages of students. A history teacher might ask the question in figure 4.11.

In which year did World War II begin?

A. 1937
B. 1938
C. 1939
D. 1940
E. 1941

**Figure 4.11: Example of a history discussion question.**

Most European historians would probably select 1939, while Americans might regard the war as having started when the United States joined the conflict in 1941. However, interesting arguments can (and have!) also been made for 1937 (the second Sino-Japanese war), 1938 (the annexation of the Sudetenland), and 1940 (the Tripartite Pact). One particularly interesting feature of this question is that it focuses attention on the issue of what the phrase *world war* actually means. In 1939, Germany was at war with Britain and France, and Japan was involved in conflicts

in Manchuria, but at that time, there was no formal alliance between Germany and Japan. However, even in 1939, the conflict was global in the sense that armed forces from British colonies such as India, Canada, Australia, and New Zealand were involved, and German colonies in Africa bordered those of Britain and France.

Again, the point is that the teacher learns little about the quality of student thinking from hearing which answer a student chooses; the teacher needs to hear reasons for the choice, and that means hearing from every student in the classroom. That's great if you have time to hear from every student, but this kind of item is useless as a quick check on student understanding.

For a good example of a diagnostic question in history, consider the question in figure 4.12, which was mentioned as a possible exit pass question in chapter 2.

Why are historians concerned with bias when analyzing sources?

A. People can never be trusted to tell the truth.
B. People deliberately leave out important details.
C. People are only able to provide meaningful information if they experience an event firsthand.
D. People interpret the same event in different ways, according to their experience.
E. People are unaware of the motivations for their actions.
F. People get confused about sequences of events.

**Figure 4.12: Example of a history diagnostic question.**

This multiple-choice question was generated by using the most interesting incorrect responses from the exit passes. Most history teachers would agree that a class in which every student selects D as the best answer is probably ready to move on. Again, a discussion might be helpful, but it is not necessary to help the teacher judge whether the students have understood the main point of this instructional sequence. It is the quality of the distractors that is crucial here—it is only because the distractors are so plausible that the teacher can reasonably conclude that the students who choose D have done so for the right reason.

A group of language arts teachers from Cherry Hill, New Jersey, who were, at the time, teaching persuasive writing to their students, generated the item in figure 4.13.

Which of these is the best thesis statement?

A. The typical TV show has nine violent incidents.
B. The essay I am going to write is about violence on TV.
C. There is a lot of violence on TV.
D. The amount of violence on TV should be reduced.
E. Some programs are more violent than others.
F. Violence is included in programs to boost ratings.
G. Violence on TV is interesting.
H. I don't like the violence on TV.

**Figure 4.13: Example of a language arts diagnostic question.**

Most language arts teachers would agree that, *within the genre of persuasive writing*, D represents the best thesis statement. If the writing were a factual reporting of a study on the amount of violence on TV, then response A might be a better thesis statement (analogous to an abstract for a research article), and F represents a perfectly reasonable opinion. Several of the other responses represent views that students hold, though, of course, the one that drives language arts teachers crazy is B—when students confuse a thesis statement with an introductory statement.

Finally, for an example from the teaching of modern foreign languages, a group of Spanish teachers from Chico Unified School District, California, developed the question in figure 4.14.

Which of the following is the correct translation for "I give the book to him"?

A. Yo lo doy el libro.
B. Yo doy le el libro.
C. Yo le doy el libro.
D. Yo doy lo el libro.
E. Yo doy el libro le.
F. Yo doy el libro lo.

**Figure 4.14: Example of a Spanish diagnostic question.**

Anyone who has taught Spanish to native English speakers knows that they have two particular difficulties in learning about male pronouns: which pronoun to choose and where it should go. This item is ingenious because in one of the responses, the

correct pronoun is in the wrong place; in another, the placement is correct, but the pronoun is incorrect; and in others, there are multiple errors. This question took the Chico teacher seventy-five minutes to generate—certainly much more time than teachers would normally take to write a single question—but this question will still be a good question in twenty years' time, because native English speakers will still be having the same difficulty with pronoun selection and placement when learning Spanish. Moreover, this question works well anywhere native English speakers are learning Spanish—it works just as well in Australia as it does in California. Good questions are both highly durable and highly portable.

Teachers can use diagnostic questions in a variety of ways. They can use them as "range-finding" questions to find out what students already know about a topic before beginning the instruction. A university teacher teaching an introductory course on botany posed the question, "What proportion of the water taken in by the roots of a corn plant is lost through transpiration?" and asked the students to hold up one, two, three, four, or five fingers according to whether they thought the answer was 10 percent, 30 percent, 50 percent, 70 percent, or 90 percent. In teaching this course over many years, she has found that this simple question gives her a good indication of students' understanding of how plants work. Students who know little about plants assume that because plants die if they cannot get enough water in through the roots, the plants would seek to minimize the amount of transpiration to minimize water loss, and thus, these students tend to select 10 percent or 30 percent. Students who understand the role played by water in transporting nutrients within the plant realize that most of the water taken in through the roots is lost through transpiration.

While using questions for range-finding at the beginning of a sequence of instruction is important, for many teachers, diagnostic questions are most useful in the middle of an instructional sequence to check whether students have understood something before moving on. The central idea here is that the teacher designs each lesson with at least one *hinge* in the instructional sequence. The hinge is a point at which the teacher checks whether the class is ready to move on by using a diagnostic question. How the lesson proceeds depends on the level of understanding shown by the students, so the direction of the lesson hinges at this point.

While the design of diagnostic questions to be used at hinge points in lessons—*hinge-point questions* or just *hinge questions* for short—is much more craft than science, work with teachers suggests that the following two principles are useful guidelines. First, it should take no longer than two minutes, and ideally less than one minute, for all students to respond to the question; the idea is that the hinge-point

question is a quick check on understanding, rather than a new piece of work in itself. Second, it must be possible for the teacher to view and interpret the responses from the class in thirty seconds (and preferably much less). If the teacher takes any longer than this to make sense of the students' responses, there is a risk of students getting off task and consequent disruptive behavior.

The main requirement of good hinge-point questions is nicely illustrated by the item in figure 4.15, used as a sample item in the National Assessment of Educational Progress (NAEP) program for fourth-grade mathematics (NAEP, 2005).

> There are two flights per day from Newtown to Oldtown. The first flight leaves Newtown each day at 9:20 and arrives in Oldtown at 10:55. The second flight from Newtown leaves at 2:15. At what time does the second flight arrive in Oldtown?

*Source: NAEP, 2005.*

**Figure 4.15: Example of a hinge-point question.**

When I ask teachers how this item might be improved, they highlight the lack of information about whether these times are a.m. or p.m., the fact that the linguistic demands may be much more taxing than the mathematical demands for some students, the fact that one has to assume that both flights take exactly the same amount of time, and so on. These are all valid criticisms, but there is a much more fundamental problem with this item that, in my experience (and I have asked this of literally thousands of teachers), not one teacher in a hundred notices: students who think there are a hundred minutes in an hour and those who know there are just sixty get the same answer.

Earlier in this chapter, I proposed that there are two good reasons to ask questions in classrooms: to cause thinking, and to provide the teacher with information that assists instructional decision making. Whether questions satisfy the first of these is relatively easy to evaluate. One requirement of questions to assist instructional decision making is that students with the right idea about whatever it is we want them to know, understand, or be able to do should give different answers than students who do not have the right idea. Or, to put it another way, ideally it would be impossible for students to give the right answer for the wrong reason. One way to improve hinge-point questions is to involve groups of teachers. One group of teachers is given the task of developing a question, which is then handed to a second group of teachers whose task it is to try to identify a way in which a student might get the correct

answer with incorrect reasoning. If they can, then the question is handed back to the first group to be refined.

In the case of the question about flights, one immediate improvement is simply to change the start time of the first flight from 9:20 to 9:05 (and presumably also to add *a.m.* or *p.m.*). With this modification, students who think there are one hundred minutes in an hour will get the answer 3:65, while those who know there are sixty minutes in an hour will answer 4:05. No question will ever be perfect, but by constantly seeking to understand the meaning behind students' responses to our questions, we can continue to refine and polish our questions and prompts.

A second requirement of questions to assist instructional decision making—much less important than the first but still useful to bear in mind when designing questions—is that the incorrect answers should be *interpretable*. That is, if students choose a particular incorrect response, the teacher knows (or at least has a pretty good guess) why they have done so. A multiple-choice question for which the incorrect responses relate to well-known naïve conceptions, such as the question on pronouns in Spanish, is one example.

While some people argue that multiple-choice questions have no place in educational assessment, there are good reasons for preferring the multiple-choice format for hinge-point questions. Multiple-choice questions are often criticized because they assess only low-order skills such as factual recall or application of a standard algorithm, although as the previous examples show, they can, if carefully designed, address higher-order skills. Another problem with multiple-choice questions used in tests is that there is no opportunity for the student to negotiate the meaning of the question with the question-setter or the machine that does the scoring.

In the classroom, these are much less important considerations, and multiple-choice questions have one great feature in their favor: the number of possible student responses is limited. When a teacher is faced with a sea of thirty whiteboards, each with a different response, it can be bewildering. Using multiple-choice questions provides a means for sorting all the students' responses ahead of time, so that precious classroom time is not spent trying to make sense of the students' answers.

When designing multiple-choice questions for classroom use, there are two key principles to make it less likely that students get the right answer for the wrong reason. The first is to use what Philip M. Sadler (1998) calls *distractor-driven multiple-choice questions*. These are questions that are written in a very different way from the traditional development method. The starting point for such a question is to begin by listing the misconceptions a student might have about a particular topic, to write the answer that a student with that misconception would give, to then write the question that would elicit that misconception, and finally to generate the correct answer.

As an example, Sadler provides the question in figure 4.16 (the percentage of eighth- through twelfth-grade students choosing each option is shown in parentheses).

The main reason for the temperature being hotter in summer than in winter is:

A. The earth's distance from the sun changes. (45 percent)
B. The sun is higher in the sky. (12 percent)
C. The distance between the northern hemisphere and the sun changes. (36 percent)
D. Ocean currents carry warm water north. (3 percent)
E. An increase occurs in greenhouse gases. (3 percent)

*Source: Sadler, 1998.*

**Figure 4.16: Example of a question designed from student misconceptions.**

The correct answer, B, was chosen by only 12 percent of students. In other words, students would have been better off guessing, because then they would have had a 20 percent chance of answering correctly. Most of the questions discussed earlier in this chapter are, in fact, distractor-driven multiple-choice questions, and were developed "backward" by starting with the misconceptions and working from there.

The second way to minimize the likelihood that a student gets a correct answer with incorrect reasoning is to have multiple correct answers, as was used in the question on Pythagoras's rule discussed earlier in this chapter. However, it is also possible to arrange for multiple correct answers to differentiate between different levels of difficulty or subtlety, as we saw in the question on figurative language in chapter 2. For example, a mathematics teacher asked a class to determine the area of a semicircle with a diameter of twenty centimeters. The students were offered five options, as seen in figure 4.17.

A. $\frac{\pi \times 20}{2}$

B. $\frac{\pi \times 20 \times 20}{2}$

C. $50\pi$

D. $\frac{\pi \times 10 \times 10}{2}$

E. $\frac{\pi}{2}\left(\frac{20}{2}\right)^2$

**Figure 4.17: Example of a question with multiple correct answers.**

Options A and B correspond to common student errors, while options C, D, and E are all correct. However, it is far more challenging for students to recognize that option E is correct than it is to determine that options C and D are correct. By having at least one correct option to a question that represents a minimum level of understanding needed for the class to move on, the teacher can build in a number of other options that keep the highest-achieving students on their toes. In particular, a high-achieving student who is told, as she leaves the classroom, that there was a correct answer that she missed is likely to be paying much closer attention in the next lesson. The power of such *multiple-correct answer questions*, especially when the distractors relate to known misconceptions, is that they allow the teacher to differentiate her instruction in an inclusive way. High-achieving students are challenged, while lower-achieving students are still able to get at least one correct answer if the instruction has been successful.

As a final variant, sometimes it makes sense to administer the question as a series of simple questions. For example, an elementary school teacher has been teaching her students about lines of symmetry in two-dimensional shapes, and toward the end of the lesson, she displays the image shown in figure 4.18.

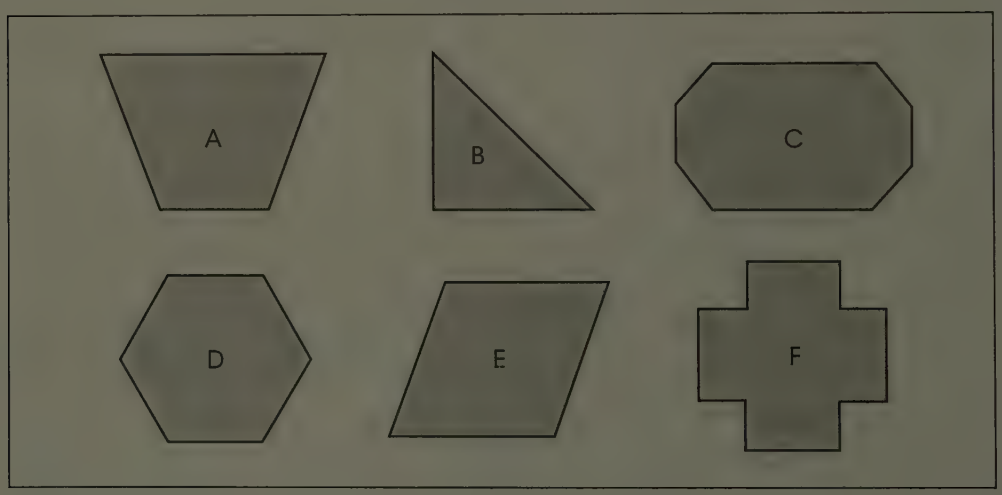



Figure 4.18: Example of a lines of symmetry question.

She calls out "A," and each student in the class has to hold up a number of fingers to show how many lines of symmetry shape A has. She then does the same for B, C, D, E, and F. She doesn't try to remember how each student responds. Instead, she focuses on just two things.

1. Are there any items that a significant proportion of the class answers incorrectly and will need to be retaught with the whole class?

2. Which two or three students would benefit from individualized instruction?

Many students indicate that they think that shape D has just three lines of symmetry and that shape E has no lines of symmetry even though it is a rhombus. She points out to the class that D has six lines of symmetry, three through the corners and three through the midpoints of the sides, and reminds the class to look for lines of symmetry that are not horizontal, vertical, or at 45°. She realizes that the incorrect answers may not necessarily indicate poor mathematical understanding. After all, in the case of shape D, holding up just three fingers could be a result of students thinking that only one hand would be necessary, and as a result of the distortion of the data projector, the image of shape E might not actually be the rhombus shape intended. Nevertheless, she understands that in teaching, as noted previously, it is better to assume students do not know something when they do than to assume that they do know something when they don't.

In this episode, the teacher administered, graded, and took remedial action regarding a whole-class quiz in a matter of minutes, without giving herself a pile of grading to do. She does not have a grade for each student to put in a gradebook, but this is a small price to pay for such an agile piece of teaching. This is, in fact, a canonical example of formative assessment: the teacher engaged every student in the class by requiring them to indicate the number of lines of symmetry for each of the six shapes, and she used the information to make on-the-fly adjustments to her instruction.

A further benefit of focusing on high-quality questions is, as mentioned earlier, their portability. Most teachers find worksheets or other teachers' lesson plans to be of limited usefulness. However, high-quality questions seem to work across different schools, districts, states, cultures, and even languages. Indeed, sharing high-quality questions may be the most significant thing we can do to improve the quality of student learning.

## Alternatives to Questions

As a final note, asking questions may not be the best way to generate good classroom discussions. Asking a class which country was most to blame for the outbreak of World War I invites students to plump for one country or another. If, instead, the teacher makes a statement, such as, "Russia was most to blame for the outbreak of World War I," students seem to respond more thoughtfully because they realize that just agreeing or dissenting is not enough—they need to provide reasons (Dillon, 1988). Similarly, asking, "Are all squares rectangles?" is likely to yield a less thoughtful discussion than framing what is, in effect, exactly the same question as a

statement: "All squares are rectangles." The quality of discussion is usually enhanced further when teachers utilize think-pair-share before students respond to teacher-led statements.

## Conclusion

Ausubel (1968) argues that the most important factor influencing learning is what the learner already knows, and that the teacher's job is to ascertain this and to teach accordingly. Students' conceptions are not random aberrations but the results of sophisticated and creative attempts to make sense of their experiences. Within a typical classroom, there is clearly not enough time for the teacher to treat each student as an individual, but with careful planning and the thoughtful application of the techniques presented in this chapter, the teacher can make the classroom a much more engaging place for students and one in which the teacher is able to make rapid and effective instructional adjustments to respond to the learning needs of all students. Once the teacher knows where learners are in their learning, she is in a position to provide feedback to the learners about what to do next—and this is the subject of the next chapter.

chapter 5

# Providing Feedback That Moves Learning Forward

It seems obvious that feedback to students about their work should help them learn, but it turns out that providing effective feedback is far more difficult than it appears. Much of the feedback that students get has little or no effect on their learning, and some kinds of feedback are actually counterproductive. This chapter reviews the research on feedback, specifically discussing the quality of feedback and how learners react to feedback, and investigates how memory works and why that knowledge is important when dispensing feedback. It then posits the idea of feedback as a recipe for future action and provides several practical techniques teachers can implement to give their students feedback that moves learning forward.

## The Quality of Feedback

The power of feedback to improve classroom learning was vividly demonstrated in a study by Maria Elawar and Lyn Corno (1985). A group of eighteen sixth-grade teachers in three schools in Venezuela received seven hours of training on how to provide constructive written feedback on the mathematics homework produced by their students. The feedback included specific comments on errors, suggestions to the students about how to improve, and at least one positive remark. A second group of teachers received the same training as the first group, but gave constructive feedback to half their classes and just scores to the other half. A third group of teachers received no training and graded homework as normal (that is, by giving only scores). The students receiving the constructive feedback learned twice as fast as the

control-group students—in other words, they learned in one week what the other students took two weeks to learn. Furthermore, in the classes given constructive feedback, the improvement in learning occurred across the achievement range, attitudes toward mathematics were more positive, and the achievement gap between male and female students was reduced (Elawar & Corno, 1985).

Ruth Butler (1988) investigated the effectiveness of different kinds of feedback in twelve sixth-grade classrooms in four schools in Israel. On the first day, all students were given a booklet containing two tasks. The first task gave students a picture with five concentric circles, each of which contained three or four letters from the Hebrew alphabet. Students were asked to construct as many words as they could starting with one of the letters in the center, and then taking a letter from one or more successive bands. The second task on the first day asked students to complete two examples from a standard test of divergent thinking where students had to think of different uses of familiar objects, based on the Minnesota Tests of Creative Thinking (Torrance & Templeton, 1963).

Two days later, the work was returned to the students. In one classroom in each of the schools, students received a comment on how their work could be improved, such as, "You thought of quite a few correct words. Maybe you could think of more short words/more long words/even longer words." In another classroom in each of the four schools, students were given a grade by scoring all the work, and then grading the students "on a curve" from 40 to 99 (normally distributed). In the third classroom in each school, each student got a grade and a comment, side by side. Then the students were asked to attempt some similar tasks and told that they would get the same sort of feedback as they had received for the first lesson's work. Again, the work was collected and scored.

Butler then analyzed the performance of the highest- and lowest-performing students in the twelve classes by selecting, on the basis of their previous grades in language and mathematics, the highest- and lowest-performing 25 percent of students.

Those given only grades made no progress from the first lesson to the second—their work was no better. When asked whether they wanted to continue doing similar work, those who had received high grades indicated that they did, but those who had received low grades did not. The students given only comments scored, on average, thirty percentage points higher on the work done in the second lesson than that done in the first (although, of course, they did not know this because they had not been given grades), and all these students indicated that they wanted to carry on doing similar work. The obvious question is, What happened to the students given both scores and comments?

When I ask teachers what they think happened in this study, many assume that the students given both scores and comments progressed at least as much as those given comments. After all, if the comments alone produced a thirty percentage point improvement in scores, then giving both scores and comments should have been even more informative.

Other teachers suggest that there was some trade-off between the two kinds of feedback. For example, some teachers say that they expect an improvement in scores because of the effects of the comments, but there may have been some polarization of attitudes as a result of the scores; in other words, students who got high scores were motivated and wanted to carry on, while low scorers were demotivated and wished to move on to something else. A second possible trade-off works the other way, with students making no progress due to the scores' effect, but with attitudes kept positive by the encouragement provided by the comments.

Most teachers, therefore, are surprised to learn that the effect of giving both scores and comments was the same as the effect of giving scores alone. Far from producing the best effects of both kinds of feedback, giving grades alongside the comments completely washed out the beneficial effects of the comments; students who got high grades didn't need to read the comments, and students who got low scores didn't want to. And yet giving both a grade (or score) and some kind of comment is probably the most prevalent form of feedback to students in North America (and indeed in most other countries). This study (and others like it, to follow) shows that if teachers are providing careful diagnostic comments and then putting a score or a grade on the work, they are likely to be wasting their time (see Lipnevich & Smith, 2008, for a review of more recent studies, and a similar study with college students). They might as well just give a score or a grade—the students won't learn anything as a result, but the teacher will save a great deal of time.

Butler (1987) found some clues as to why feedback can have these effects. This time, two hundred fifth- and sixth-grade students in eight classes spent a period working on a variety of divergent thinking tasks. Again, the researchers collected student work and gave students one of four kinds of feedback on this work at the beginning of the second period two days later.

1. In two classes, the students were given comments.
2. In two classes, the students were given grades.
3. In two classes, the students were given written praise.
4. In two classes, the students were given no feedback at all.

The quality of the work done in the second period was compared to that done in the first. The quality of the work of the students who had been given comments had improved substantially compared to their work in the first period, but those given grades and praise had made no more progress than those given absolutely no feedback on their work.

At the end of the second period, the students were given a questionnaire. The questionnaire sought to establish what factors influenced the students' decisions about whether to expend effort in the classroom and what factors the students believed determined whether they were successful or not. Specifically, the questionnaire was designed to elicit whether the students attributed their expenditure of effort and their success to ego-related factors or to task-related factors, as shown in table 5.1.

**Table 5.1: Ego- and Task-Related Attributions**

| Attribution of | Ego | Task |
|---|---|---|
| Expenditure of effort | To do better than others<br>To avoid doing worse than others | Interest<br>To improve performance |
| Success | Ability<br>Performance of others | Interest<br>Effort<br>Experience of previous learning |

*Source: Butler, 1987.*

Students who were given comments had high levels of task-involvement, but their levels of ego-involvement were the same as those given no feedback. However, while those given written praise and those given grades had comparable levels of task-involvement to the control group, their levels of ego-involvement were substantially higher. As noted, the provision of grades and written praise had no effect on achievement; their only effect was to increase the sense of ego-involvement. This, as anyone involved in guidance and counseling work in schools knows, is bad news. To change a student's behavior, it is generally far more effective to focus on criticizing the behavior rather than the student (in effect, giving task-involving rather than ego-involving feedback).

These findings are also consistent with research carried out in the 1970s that showed clearly that praise was not necessarily a good thing—in fact, the best teachers appear to praise slightly less than average (Good & Grouws, 1975). It is the quality rather than the quantity of praise that is important, and in particular, teacher praise is far more effective if it is infrequent, credible, contingent, specific, and genuine

(Brophy, 1981). It is also essential that teachers relate praise to factors within an individual's control, so praising a gifted student simply for being gifted is likely to lead to negative consequences in the long term (Dweck, 2006, 2015).

The timing of feedback is also crucial. If it is given too early, before students have had a chance to work on a problem, then they will learn less. In a review of forty research reports on the effects of feedback in *test-like events*—such as questions embedded in programmed learning materials, review tests at the end of a block of teaching, and so on—Robert Bangert-Drowns, Chen-Lin Kulik, James Kulik, and MaryTeresa Morgan (1991) found that what mattered was the degree of *mindfulness* (or thoughtfulness) in the students that the feedback generated. When students could peek ahead and look at answers to questions before they had tried to answer them, they learned significantly less than when they had to attempt to answer the questions before getting feedback.

Malcolm Simmons and Peter Cope's (1993) study provides a direct demonstration of the effect of mindfulness. Pairs of students between nine and eleven years old worked on angle and rotation problems. Some students worked on the problems on a computer, using the programming language Logo, and some worked on the problems using pencil and paper. The students working in Logo were able to use a *trial-and-improvement* strategy that enabled them to get a solution with little mental effort. For those using pencil and paper, working out the effect of a single rotation was much more time-consuming, giving these students an incentive to think carefully, and this greater mindfulness led to more learning.

The key idea in all this—that what matters is the mindfulness with which students engage in the feedback—means that sometimes less is more. A study of sixty-four third graders required the students to engage in arithmetical reasoning tasks, and varied the kind of support they received (Day & Cordón, 1993). Half of the students were given a scaffolded response when they got stuck—in other words, they were given only the minimum amount of support to get them unstuck and to make progress. The other half of the students were given a complete solution to the problem on which they were stuck and then given a new problem to work on. Students given the scaffolded response learned more and retained their learning longer than those given a full solution (Day & Cordón, 1993). In a sense, this is hardly surprising, since those given the complete solutions had the opportunity for learning taken away from them. As well as saving the teachers time, developing skills of intervening as little as possible—just enough to get the students going—promotes better learning.

Jon Saphier (2005) provides a good example of this kind of feedback.

> Teacher: What part don't you understand?
>
> Student: I just don't get it.
>
> Teacher: Well, the first part is just like the last problem you did. Then we add one more variable. See if you can find out what it is, and I'll come back in a few minutes. (p. 92)

Most teachers have had the experience of giving a student a new task only for the student to ask for help immediately. When the teacher asks, "What can't you do?" a common reply is, "I can't do any of it." In such circumstances, the student's reaction may stem from anxiety about the unfamiliar nature of the task, and it is often possible to support the student by saying something like, "Copy out that table, and I'll be back in five minutes to help you fill it in." This is usually all the support the student needs. Copying out the table forces the student to look in detail at how the table is laid out, and this busywork can provide time for the student to make sense of the task herself.

Another good example of scaffolded feedback comes from Ian Smith (2008). An art teacher had sketched a face and, in collaboration with the class, determined seven criteria for a successful portrait, including the idea that the eyes should be halfway down the face and that the distance between the eyes should be roughly the same as the width of one eye, and so on. She was then able to give feedback on the fulfillment of each of the seven criteria to a student in the format shown in figure 5.1.

| ✓ | ✗ | ✗ | ? | ✗ | ✓ | ✓ |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

*Source: Smith, 2008.*

**Figure 5.1: Feedback grid for a face-drawing task.**

This communicated clearly to the student which criteria needed attention, took up very little of the teacher's time, and still left plenty of work for the student to do.

An obvious question that arises from these studies is, Does it make a difference whether a teacher gives feedback orally or in writing? One of the few studies to address this was that of eighty Canadian students in three groups learning to write the major scales in their music classes (Boulet, Simard, & De Melo, 1990). One group was given written feedback, a list of weaknesses, and a work plan; the second group received oral feedback on the nature of their errors, plus a chance to work on improvement in class; and the third group was given no feedback. At the beginning of the study, there were no differences between the three groups in terms of their previous achievements in music, scores on a test of musical aptitude, and scores on

a measure of their ability to learn. All groups fell short of the mastery level of 80 percent set for this task, but the students in the second group, who had been given oral feedback, scored significantly higher than the students in the other two groups (whose scores were not significantly different from each other). Because the treatment of group 2 differed in three ways from that of group 1, it is not possible to determine what made the difference. However, from their observations, the researchers indicate that whether the teacher gives feedback orally or in writing is much less important than the fact that group 2 was given time, in class, to use the feedback to improve their work.

The preceding studies were all conducted on school-age students, so an obvious question is whether the same conclusions would apply to college students. To answer this question, Jeffrey Nyquist (2003) looked for studies on the effects of feedback for college students, and specifically looked for studies that met the following three requirements.

1. The study had to involve an experimental manipulation of a characteristic relevant to feedback.
2. Participants had to be college-aged learners.
3. The study had to involve measures of academic performance, reported in sufficient detail for a standardized effect size to be calculated.

He identified a total of 86 relevant studies, which between them involved 12,920 individuals, and reported a total of 185 main results. The average effect of feedback was 0.4 standard deviations, equivalent to moving an average student up to the 66th percentile. To investigate whether different kinds of feedback had different effects, he allocated the studies to one of five categories, according to the kind of feedback provided.

1. **Weaker feedback only:** Studies in which students were given only the knowledge of their own score or grade (often called *knowledge of results*, or just KR)
2. **Feedback only:** Studies in which students were given their own score or grade, together with either clear goals to work toward or feedback on the correct answers to the questions they attempted (*knowledge of correct results*, or KCR)
3. **Weak formative assessment:** Studies in which students were given information about the correct results, together with some form of explanation (KCR+e)

4. **Moderate formative assessment:** Studies in which students were given information about the correct results, together with some explanation and suggestions for how to improve
5. **Strong formative assessment:** Studies in which students were given information about the correct results, together with some explanation plus a specific activity to undertake

The results are shown in table 5.2 (these are corrected values, and thus differ slightly from the values published in Nyquist, 2003).

Table 5.2: Effects of Feedback on College Students' Achievement

| Category | Number of Studies | Impact on Average Student |
|---|---|---|
| Weaker feedback only | 31 | 56th percentile |
| Feedback only | 48 | 64th percentile |
| Weak formative assessment | 49 | 60th percentile |
| Moderate formative assessment | 41 | 65th percentile |
| Strong formative assessment | 16 | 71st percentile |

*Source: Adapted from Nyquist, 2003.*

While the pattern isn't perfect, it is clear that the more faithfully the principles of formative assessment are implemented, the greater the impact on student achievement. In particular, the effect of strong formative assessment (effect size 0.56) is four times as great as feedback that only provided students with information about whether their work was correct or not (effect size 0.14). As far as we can tell, the effects of feedback are similar for all learners, from preK to college.

While the values in table 5.2 paint a convincing picture, it is important to remember that the values here are averages, and there was, in fact, considerable divergence in the size of the effects of feedback on achievement. Specifically, 24 of the 185 results were negative—feedback actually lowered performance—a finding that has been confirmed by other researchers.

Avraham Kluger and Angelo DeNisi (1996) looked at every study they could find that had ever been done on the effects of feedback in schools, colleges, and workplaces over a ninety-year period (from 1905 to 1995). They defined a feedback intervention as "actions taken by (an) external agent(s) to provide information regarding some aspect(s) of one's task performance" (Kluger & DeNisi, 1996, p. 255) and found over 3,000 research studies (2,500 journal articles and 500 technical reports) that looked at the impact of such feedback interventions on performance.

Of course, these studies varied in their quality, and to be sure that poor-quality studies were not being included, Kluger and DeNisi (1996) established a number of criteria for inclusion in their review. First, there had to be two groups of participants for whom the only difference (as far as the researchers could judge) was whether they had received feedback or not. Second, a study had to have had at least ten participants (some, particularly in the medical field, had only a single participant). Third, the study had to include some kind of measurement of performance with sufficient details of the measurements of impact provided to be able to calculate the size of the impact of feedback on performance.

Astonishingly, only 131 of the original 3,000 studies—approximately 4 percent—satisfied these criteria of scientific quality. This figure was so low that Kluger and DeNisi (1996) reviewed the excluded studies to make sure that worthwhile studies were not unduly rejected. They concluded that those studies rejected could not shed useful light on the effects of feedback. In total, the 131 accepted studies reported a total of 607 experimental results involving 23,663 observations on 12,652 individuals. Perhaps even more surprisingly, in 231 of the 607 experimental results provided in the 131 high-quality studies, providing feedback actually lowered performance.

This is one of the most counterintuitive results in all of psychology. In each of the studies, feedback was intended to improve performance, but in almost two out of every five carefully conducted studies, the participants would have done better if the feedback had simply not been given.

With such a large number of studies, there are obviously many things that could affect the effectiveness of feedback, but there are two main reasons why feedback can fail to have its desired effects. The first is that what really matters with feedback is how students react to it, and the second is that much feedback fails to take into account how our minds (and particularly our memories) work.

## How Learners React to Feedback

To try to understand why feedback could have such counterproductive—and unexpected—effects, Kluger and DeNisi (1996) looked in detail at the studies to determine when feedback does and does not improve performance. They pointed out that when the feedback draws attention to a gap between one's current performance and the goal, what happens depends on whether the current performance is higher or lower than the goal.

When the feedback tells an individual that he has already surpassed the goal, one of four things can happen. Obviously one hopes that the individual would seek to change the goal to one that is more demanding, but it might also be taken as a signal

for the individual to ease off and exert less effort. Alternatively, when success comes too easily, the individual may decide that the goal itself is worthless and abandon it entirely, or reject the feedback as being irrelevant.

When, as is more common, the feedback indicates that current performance falls short of the goal, there are again four responses. The recipient may change the goal—for example, when a student decides to settle for a B even though an A might be within her grasp but is felt to be too much work or too risky in terms of one's self-image to attempt. A second response is to abandon the goal altogether, as seen in the *learned helplessness* (Dweck, 1975) displayed by students who, in the face of repeated indications that they are falling short, decide that something is forever beyond them—for example, when a student says, "I'm no good at mathematics." The third type of response is to reject the feedback. This is quite common in workplace settings when, for example, someone who feels he has contributed well beyond what was expected gets a neutral evaluation from a supervisor. The effect of such rejection of feedback is to lower commitment (and therefore, possibly, to reduce performance). The fourth kind of response—and presumably the one intended by the person giving feedback—is to change one's behavior so as to increase performance to match the goal. These responses to feedback are summarized in table 5.3.

**Table 5.3: Possible Responses to Feedback**

| Response Type | Feedback Indicates Performance Exceeds Goal | Feedback Indicates Performance Falls Short of Goal |
|---|---|---|
| **Change behavior** | Exert less effort | **Increase effort** |
| **Change goal** | **Increase aspiration** | Reduce aspiration |
| **Abandon goal** | Decide goal is too easy | Decide goal is too hard |
| **Reject feedback** | Ignore feedback | Ignore feedback |

Only the two kinds of responses highlighted in bold are likely to improve performance. The other six, at best, do nothing, and at worst, lower performance, sometimes to a considerable degree.

The research Kluger and DeNisi (1996) reviewed also shows that it is very difficult, if not impossible, to predict which of these responses will occur. It will depend on the individual receiving the feedback, the kind of task on which feedback is given, and the recipient's perceptions of the person giving the feedback.

The use of praise can increase motivation, but then it often becomes necessary to use praise all the time to maintain the motivation. Moreover, in this situation, it is very difficult to maintain praise as genuine and sincere. In contrast, the use

of feedback improves performance when it is focused on what needs to be done to improve, and particularly when it gives specific details about *how* to improve.

In concluding their review of the effects of feedback, Kluger and DeNisi (1996) point out that the biggest problem is the failure of most of the studies they reviewed to consider long-term effects. Even when the effect of feedback is to increase motivation, this may be a short-term improvement at the expense of a long-term worsening of performance. They suggest that, instead, research on feedback should focus less on the actual impact on performance and more on the kinds of responses that are triggered in the individual as a result of the feedback.

In a series of research studies spanning over thirty years, Carol Dweck of Stanford University and her colleagues have investigated, among other things, how students make sense of their successes and failures in school by asking students questions such as, "When you get an A, why is that?" or "If you got an F, why might that be?" Dweck and her colleagues found that there were three strong themes running through the students' responses (Dweck, 2000).

The first was whether the success or failure was due to factors relating to the individual or due to outside factors (in other words, how the attribution was *personalized*). For example, "I got an A because I did a good piece of work" would be an *internal* attribution because the cause is seen as being within the individual. "I got an F because Mr. Smith hates me" would be an *external* attribution, because the cause is seen as being outside the individual. It is quite common to find individuals attributing their successes internally and their failures externally. There is an old Chinese adage that captures this tendency beautifully: "Success has a thousand fathers, but failure is an orphan."

The second theme was whether success was seen as being due to factors that were likely to be long-lasting or whether they were more likely to be transient (in other words, the permanence or stability of the factors). For example, when attributing success, students are likely to attribute "being smart" as *stable* (that is, long-lasting); if the student got an A this time because she is smart, she is likely to get an A next time because she will still be smart. If, on the other hand, a student says, "I got an A because I worked really hard on this assignment," then this is attributing success to an *unstable* (that is, transient) factor. Next time, whether the student gets an A will depend on how much effort he puts into that next assignment.

The third was the *specificity* of the attribution: whether success or failure is seen as being due to factors that affect performance in all areas or just the area in question. For example, some students overgeneralize their successes. They believe that because they are told they are good at one thing, they are good at everything. Students can

also overgeneralize failures and believe that because they fail at one thing, they are a failure at everything. (These are both *global* attributions.) Table 5.4 gives some examples of attributions of success and failure (see also Dweck, 2006).

**Table 5.4: Dimensions of Attributions of Success and Failure**

| Attribution | Success | Failure |
|---|---|---|
| **Personalization** | Internal: "I got a good grade because it was a good piece of work."<br>External: "I got a good grade because the teacher likes me." | Internal: "I got a low grade because it wasn't a very good piece of work."<br>External: "I got a low grade because the teacher doesn't like me." |
| **Stability** | Stable: "I got a good grade because I'm good at that subject."<br>Unstable: "I got a good grade because I was lucky in the questions that came up." | Stable: "I got a bad grade because I'm no good at that subject."<br>Unstable: "I got a bad grade because I hadn't reviewed the material before the test." |
| **Specificity** | Specific: "I'm good at that, but that's the only thing I'm good at."<br>Global: "I'm good at that, so I'll be good at everything." | Specific: "I'm no good at that, but I'm good at everything else."<br>Global: "I'm useless at everything." |

Dweck (2000) and others find that there is a slight tendency for boys to be more likely to attribute their successes to stable causes (such as ability) and their failures to unstable causes (such as lack of effort and bad luck). This would certainly explain the high degree of confidence with which many boys approach examinations for which they are rather unprepared. More controversially, the same research suggests that there is a slight tendency for girls to attribute their successes to unstable causes (such as effort) and their failures to stable causes (such as lack of ability), leading to the learned helplessness mentioned previously.

It is important to remember that these are not sharp differences between boys and girls—the overlap in how males and females attribute successes is far, far greater than the differences between males and females—but these differences may in part explain why female students are now outperforming male students all over the world (OECD, 2016). The best learners consistently attribute both success and failure to internal, unstable causes. They believe "It's up to me" (internal) and "I can do something about it" (unstable). The tendency is for girls to do this for their successes, although not for their failures, while the tendency for boys is to attribute neither successes nor failures to internal, unstable causes (successes tend to be attributed to internal, stable causes, while failures tend to be attributed to external, unstable causes). Regardless of the current tendencies, learning in classrooms will be considerably enhanced if students embrace this idea of "It's up to me, and I can do something

about it." When I talk to students about these issues, I often use examples from sports, such as Michael Jordan, Tom Brady, and Mike Piazza.

Although it is not true that Michael Jordan was cut from his high school varsity basketball team, he did experience a significant reversal. His high school, Laney High School, ran a junior varsity team in addition to the main varsity team. Although the expectation was that most tenth graders would play on the junior varsity team, some exceptionally gifted athletes in the tenth grade did play on the varsity team. Jordan and his friend Leroy Smith attended a basketball camp in the summer after ninth grade and were encouraged by the coach of the varsity team, Pop Herring, to try out for the varsity team in the tenth grade. In the end, Smith made it, and Jordan did not. The assistant head coach at the time, Fred Lynch, said, "Leroy was not a better basketball player than Mike, he just had size. We didn't have a lot of tall kids, and Leroy was 6'6", 6'7" . . . and Pop Herring thought we had plenty of guards but needed size" (Pickeral, 2009).

Although technically Jordan was not cut from the varsity team, the effect on him was galvanizing, as he recalled in his acceptance speech when he was inducted into the National Basketball Association's hall of fame on September 11, 2009:

> Leroy Smith was the guy that when I got cut, he made the team . . . he started the whole process with me, because when he made the team and I didn't, I wanted to prove—not just to Leroy Smith, not just to myself—but to the coach who actually picked Leroy over me—I wanted to make sure you understood, 'You made a mistake, dude.' (Jordan, 2009)

In his junior and senior years, he did play for his high school's varsity team, and played well, amassing over 1,400 points, though not enough to stand out nationally. At the end of his junior year, he was not on the list of the top three hundred high school prospects being considered by colleges (Bondy, 1999).

When Tom Brady was drafted into the National Football League in 2000, he wasn't drafted until the sixth round (he was actually the 199th player selected), and was the seventh quarterback chosen that year (Massey & Thaler, 2005).

Perhaps most remarkably, when Mike Piazza was drafted by the Los Angeles Dodgers in 1988, he was drafted 1,390th. In other words, the twenty-six major league baseball teams in existence at that time selected 1,389 players ahead of him (and only 43 players after him!). He was the 63rd player drafted by the Dodgers that year, and he wasn't even drafted as a catcher—at the time, he was a first baseman (Epstein, 2013). Realizing that he probably had a better chance of a major league career as a catcher, he began playing catcher in the Winter Leagues, and played twenty-one games for the Anaheim Angels in 1992 (in which he batted a modest .232). In

the following year, however, he was the National League's Rookie of the Year. Over the next fourteen years, he averaged over .300 and is now widely regarded as the best offensive catcher of all time, with career totals of 427 home runs and 1,335 runs batted in.

Each of these three individuals received feedback that they weren't good enough, but each decided in the face of that feedback to improve rather than give up and do something else. The determination to do better was crucial in each of these cases.

Of course, whether a student sees feedback as relating to something that is permanent or transient depends on the student's attitude. One student may see feedback as sending the message, "You're not smart enough," while another student may see the same feedback as saying, "You're not smart enough—yet." What matters is whether students see their future potential as limited by the current performance or not.

Students who believe in *fixed* ability will see any piece of work as a chance either to reaffirm their ability or to be shown up. If they are confident in their ability to achieve what the teacher asks of them, then they will attempt the task. However, if their confidence in their ability to carry out their task is low, then they may well avoid the challenge (especially if they think others will succeed), and this can be seen in classrooms every day. Many students decide that, taking all things into account, they would rather be thought lazy than dumb, and refuse to engage with the task. This is a direct consequence of the belief that ability is fixed.

In contrast, those who see ability as incremental—having what Carol Dweck now calls a *growth mindset*—see all challenges as chances to learn, to get smarter. Therefore, these students will try harder in the face of failure (Dweck, 2006). These views of ability are generally not global—a student can believe that ability in mathematics is fixed and that ability in athletics is incremental, in that the more one trains at, say, triple jump, the more one's ability at that athletic endeavor increases. Therefore, what we as teachers need to do is ensure that the feedback we give students supports a view of ability as incremental rather than fixed—making sure they understand that by working, they're getting smarter. This feedback must also take into account an understanding of how memory (and, consequently, learning) works—something that will be discussed in the next section.

## How Memory Works

Most people see human memory as a kind of recording device. Things that are experienced are recorded in some way, and, over time, unless they are constantly refreshed, these recordings decay. If we cannot recall facts or events, we tend to assume that the memories of those facts or events have decayed so much that they

have disappeared. However, it turns out that things are a little more complex, and interesting, than that.

Many people cannot recall the license plate number of their very first car, and assume that the number has been forgotten. But if they were shown a list of ten license plate numbers, most people would easily be able to recognize the number, which would not be possible if the number had really been forgotten. The memory is still there. What is hard is retrieving it. The same insight provides an explanation of the *tip of the tongue phenomenon*—the fact that people cannot recall something at a particular time, but they know that they know it, and are often able to recall it a short time later, especially if they stop trying to recall it. Other things can be easily recalled at a given point in time, but are never effectively learned. For example, most people can remember the number of their hotel room during the stay, but weeks later, cannot recall it. More important, they would not be able to pick out the number from a list of numbers. While the number is easily recalled during the stay, it is not effectively learned (in the sense of changing long-term memory).

In an extensive research program spanning almost forty years, Robert Bjork and his colleagues researched how our memories actually work, which is well summarized in Soderstrom and Bjork (2015). This research shows that human memory works very differently from physical or digital storage mechanisms.

For example, with physical or digital storage, retrieval does not change the stored information, nor any other information. In contrast, scientists have discovered that when humans retrieve things from memory, the retrieved material becomes more memorable, and—more surprising—things that are not retrieved become less easy to retrieve in the future.

Robert Bjork suggests that there are two aspects of memory that determine how easy things are to recall, which he calls *storage strength* and *retrieval strength* (Bjork & Bjork, 1992). Storage strength is an indication of how well something has been learned, while retrieval strength is how easy something is to recall at a given moment in time. Storage strength is a one-way street—it can increase, but it cannot decrease (except perhaps through injury to the brain). On the other hand, retrieval strength goes up and down independently of storage strength. Not using a particular memory causes its retrieval strength to fall, and recalling something causes retrieval strength to rise. However, retrieval strength is limited, so increasing the retrieval strength of some items in memory decreases the retrieval strength of other things.

This may seem rather arcane and irrelevant, but there are two key findings from the research of Robert Bjork and others that have huge implications for learning in general, and feedback in particular. The first is that storage strength increases every

time an item is restudied or retrieved from memory, but successfully retrieving something from memory increases storage strength more than restudying the same item. The second is that the harder it is to retrieve something from memory, the greater the positive impact on storage strength. In other words, successfully retrieving something from memory generates more learning when retrieval strength is *low* than when it is *high*—when there are, in Robert Bjork's (1994) memorable phrase, "desirable difficulties" in learning (p. 193).

These ideas help explain many of the research findings on feedback. Feedback that provides students with information about the correct answer to something that students answer incorrectly is the equivalent of restudying the same material. This can help learning, because every time students recall something, their storage strength increases, but feedback is likely to be *more* effective if it helps students retrieve things themselves. To put it simply, hints are likely to be more effective than answers (see Finn & Metcalfe, 2010).

Another finding from the feedback research is the rather strange fact that feedback is often more effective when it is delayed. For many years, researchers assumed that feedback acted as a kind of reinforcement of learning, so immediate feedback should be more effective than delayed feedback. However, in many studies, it has been found that delaying feedback makes it more effective—what is sometimes called the *delay-retention effect* (Kulhavy, 1977). Robert Bjork's *new theory of disuse*, as he calls it, explains why (Bjork & Bjork, 1992). Delaying feedback means that, when it is given, feedback is being provided when retrieval strength is lower than it would otherwise have been, and thus the feedback has a greater impact on storage strength (namely, on long-term learning).

The problem in all this is that many of these ideas go against the grain. In a study of students in the upper-level engineering course on continuous time signals and systems at the University of Texas at El Paso, students were divided at random into two groups (Mullet, Butler, Verdin, von Borries, & Marsh, 2014). One group received feedback on assignments as soon as they were completed, while the others received delayed feedback (either after one to two weeks or after three to four weeks). The students getting immediate feedback averaged 84 percent on the end-of-course exams, while those getting delayed feedback averaged 93 percent, and there was no difference in scores between those with shorter and longer feedback delays. In a follow-up experiment, the same students got immediate feedback on some assignments and delayed feedback on other assignments. The students averaged 65 percent on exams that tested the content on which they had received immediate feedback,

and 81 percent on the content on which they had received delayed feedback (and again, there was no difference in scores for longer versus shorter delays).

However, in the second experiment, where the same students were given immediate feedback on some assignments and delayed feedback on other assignments, 90 percent of the students said they greatly preferred immediate feedback, and 79 percent thought that they had benefited a lot more from immediate feedback (Mullet et al., 2014).

In fact, it is not just students who think immediate feedback is better; many teachers think so too, and this seems to be because they fail to distinguish between the level of *performance* in the learning task and the amount of *learning* that takes place (Bjork, 1994). When we give students lots of support (for example, in the form of feedback) to complete a particular task, then the level of the students' performance on that task increases. However, the amount of learning that takes place (namely, the increase in storage strength) decreases because the student is not having to work so hard. Conversely, when students have to struggle in the learning task, the quality of their performance on this task reduces, but the amount of learning that takes place increases. In general, there is an inverse relationship between the quality of performance on a task and the amount of learning that occurs as a result of completing that task.

What this means is that teachers have to strike a careful balance. Students like getting feedback immediately, but when feedback is provided as students are working, the feedback may scaffold students' thinking so strongly that they do not develop the ability to think their own way through a problem or a task. Moreover, providing feedback immediately has a smaller impact on student learning than feedback that is delayed somewhat. There will be times that providing feedback immediately is necessary to maintain students' motivation, but the important point here is that, as far as possible, the feedback should leave the thinking with the student. Supporting students as they work through *desirable difficulties* in their work is likely to have a greater impact on long-term learning than feedback that just tells students what to do.

## Feedback as a Recipe for Future Action

All this suggests that providing effective feedback is very difficult. Get it wrong, and students give up, reject the feedback, or choose an easier goal. Even when the students engage with the feedback, there is still a danger that the focus will shift to short-term rather than long-term goals. That is why I have become more and more interested in the practices of high-achieving sports coaches.

In many high schools, those involved in academic programs treat the school as a talent refinery. Their job is to deliver the curriculum to the students. Some students will get it and thrive, and others will not. Although most states require certain courses to be passed to qualify for a high school diploma, beyond that, if students cannot pass a course (such as algebra 2 or trigonometry), then that's OK—it's obviously not their subject. In other words, the school functions rather like an oil refinery—its job is to sort the students into different layers. Those involved in athletics programs cannot afford to do this. The football coach does not have the luxury of saying, "We don't have a good enough quarterback, so we're not going to play football this year." Instead, the football coach sees her or his job as making the best quarterback the team has as good as he can be, and doing the same at every other position. In other words, athletics coaches tend to see schools not as talent refineries but as talent incubators or talent factories. They see their job as not just identifying talent but also nurturing it, and even producing it, often getting out of athletes more than the athletes themselves believed they could achieve.

Coaches do this through the provision of *feedback that moves learning forward.* In 1998, when Paul Black and I published *Inside the Black Box: Raising Standards Through Classroom Assessment*, we recommended that feedback during learning be in the form of comments rather than grades, and many teachers took this to heart. Unfortunately, in many cases, the feedback was not particularly helpful. Typically, the feedback would focus on what was deficient about the work submitted (which the students were not able to resubmit), rather than on what to do to improve their future learning. In such situations, feedback is rather like the scene in the rearview mirror rather than through the windshield. Or as Douglas Reeves (2008) once memorably observed, it's like the difference between having a medical and a postmortem.

Feedback functions formatively *only if the learner uses the information fed back to him or her to improve performance.* If educators intend the information fed back to the learner to be helpful but the learner cannot use it to improve his or her performance, it is not formative. I remember talking to a middle school student who was looking at the feedback his teacher had given him on a science assignment. The teacher had written, "You need to be more systematic in planning your scientific inquiries." I asked the student what that meant to him, and he said, "I don't know. If I knew how to be more systematic, I would have been more systematic the first time." This kind of feedback is *accurate*—it is describing what needs to happen—but it is not *helpful* because the learner does not know how to use the feedback to improve. It is rather like telling an unsuccessful comedian to be funnier—accurate, but not particularly helpful, advice.

Again, the differences in such practices between academic and athletic programs are illuminating. Imagine a young fast-pitch softballer is struggling—her earned run average (ERA) is 10 (if you know nothing about fast-pitch softball, that's not very good). If her coach were like many high school teachers, she might say, "Here's my advice: you need to get your ERA down." Accurate but not helpful. She knows she needs to get her ERA down, but she needs to know *how.*

The coach might look at her pitching and realize that she is having a lot of runs hit off her because she is trying, unsuccessfully, to pitch a rising fastball. This is a fastball thrown with so much rotation on the ball that as it reaches the plate, it rises sharply and is quite unhittable. Of course, if it doesn't rise, it's just a fastball over the middle of the plate, which is why she is getting hit for a lot of runs. So, the coach says to the pitcher, "I know what's going wrong. It's your rising fastball. It's not rising." Again, accurate, but still not helpful.

However, if the coach tells the pitcher that she is not dropping her pitching shoulder enough to deliver the pitch from below the knee, then this gives the athlete something to work with. The secret of effective feedback is that saying what's wrong isn't enough; to be effective, feedback must provide a *recipe for future action.*

This is inherent in the origin of the term *feedback*—a term borrowed from engineering. Although the term has been around for well over a hundred years (Wiliam, in press b), mathematician and philosopher Norbert Wiener (1948) brought it into widespread use. The important feature of feedback as it is used in engineering is that it forms part of a *feedback loop.* Using a thermostat to regulate a room's temperature is a classic example of a feedback loop.

The system has the following four key elements.

1. A means of setting the desired state (temperature setting)
2. A means of establishing the current state (the thermometer)
3. A means of comparing the current state with the desired state (the thermostat)
4. A means of bringing the current state in line with the desired state (the furnace or the cooling system)

For engineers, information about the discrepancy between the current state and the desired state is useless unless there is also a mechanism within the feedback loop to bring the current state closer to the desired state. In education, we use the term *feedback* for any information given to students about their current achievements. Just telling the students that their current performance falls short of where they need to

be isn't feedback that an engineer would recognize. If there is nothing connecting the thermostat to the furnace, then there is no feedback loop and, therefore, to an engineer, no feedback.

To be effective as a recipe for future action, educators must design feedback so as to progress learning. In other words, the feedback must embody a model of progression, and here, again, is where much coaching in athletics programs is well designed. It is not enough to clarify the current state and the goal state. The coach must design a series of activities that will move athletes from their current state to the goal state. Often coaches will take a complex activity, such as the double play in softball or baseball, and break it down into a series of components, each of which athletes need to practice until they reach fluency, and then the coaches will assemble the components together. Not only does the coach have a clear notion of quality (the well-executed double play), he or she also understands the *anatomy* of quality: he or she is able to see the high-quality performance as being composed of a series of elements that can be broken down into a developmental sequence for the athlete.

This skill of being able to break down a long learning journey—from where the student is right now to where he or she needs to be—into a series of small steps takes years for even the most capable coaches to develop. The best coaches say they are always learning how to be better at coaching. However, even when a teacher or a coach is able to pinpoint which aspects of a student's performance need to be improved, it turns out that many well-intentioned attempts to improve that performance fail, because they fail to take into account how human memory works.

## Practical Techniques

If I had to reduce all the research on feedback into one simple overarching idea, at least for academic subjects in school, it would be this: *feedback should cause thinking.* All the practical techniques discussed in this section work because, in one way or another, they get the students to think, rather than react emotionally to the feedback they are given. We will now discuss the following practical techniques: minus, equals, plus; feedback for future action; three questions; techniques for utilizing feedback; and grading practices that support learning.

### Minus, Equals, Plus

When I talk to teachers about the first of the two studies by Ruth Butler (1988) discussed previously—the one in which students were given comments, scores, or both (page 124)—I ask them, "When the students were given both comments and scores,

which did they look at first?" Everyone realizes, of course, that it was the score. What is more interesting is what the students looked at next: somebody else's score.

As soon as students compare themselves with someone else, their mental energy becomes focused on protecting their own sense of well-being rather than learning anything new. In one school I worked with, the principal insisted that there be a score or a grade on each student's work every two weeks. The language arts teachers gave students the traditional letter grades at the end of each marking period, but during the marking period, they used a grading scheme that they called "minus, equals, plus." As well as comments, each student received a symbol of –, =, or +, depending on whether the work submitted was not as good as, about the same as, or better than his or her last work on the topic. The students who used to get A grades typically hated this system, because they had to keep working harder to get a "+". It is particularly interesting when high-achieving students in these classes compare their grades with other students' grades and find that they have received an "=" or even a "–" when students with lower absolute achievement get a "+." The important feature of such a grading scheme is that it feeds back to learners about things that are within their control, such as whether they are improving, rather than things over which they have less influence, such as how they compare with other students in the class.

## Feedback for Future Action

To be effective, feedback needs to direct attention to what's next, rather than focusing on how well or poorly the student did on the work, and this rarely happens in the typical classroom. In some situations, feedback is seen as punishment, instead of as the formative tool it can be. Pam Hayes, a middle school mathematics teacher, once told me about a conversation she had with a fifth grader in which the student said, "When you get a lot of feedback on your work, it means it wasn't very good." When asked to explain this rather surprising comment, the girl pointed out that successful work is usually just given a (high) grade and a comment like "Good job," whereas less successful work is returned to the student with lots of annotation from the teacher. To this girl, the more "feedback" you got, the worse your work must have been. In many classrooms, teachers require students to do corrections for homework, leaving high achievers with nothing to do. Used in this way, feedback really is punishment.

If, however, we embrace the idea of feedback as a recipe for future action, then it is easy to see how to make feedback work constructively: don't provide students with feedback unless you allow time, in class, to work on using the feedback to improve their work. Then feedback is not an evaluation of how well or how poorly one's work was done, but rather a matter of "what's next?" in which students have control over the future outcome.

## Three Questions

One technique for structuring this that is particularly effective when responding to a piece of student writing is *three questions*. As the teacher reads each student's work, when she sees something on which she would like the student to reflect, she places a numbered circle at that point in the text. Underneath the student's work, the teacher writes three questions, each one relating to the relevant numbered circles, and leaves space for the student's response. Then, the student takes the first ten or fifteen minutes of the next lesson to respond to the three questions the teacher posed. The important feature of this technique is that no matter how bad or good the student's work is, everyone has the same amount of work to do. Feedback is no longer a post-mortem, but now more like a medical exam.

## Techniques for Utilizing Feedback

This idea that feedback is about "what's next?" also addresses another shortcoming of much current practice. I often ask teachers whether they believe that their students spend as much time utilizing the feedback they are given as it has taken the teacher to provide it. Typically, less than 1 percent of teachers believe this to be the case, and this needs to change.

The first fundamental principle of effective, utilizable classroom feedback is that *feedback should be more work for the recipient than the donor.* Giving students comments, rather than grades or scores, is obviously useful, but most teachers still find it difficult to get the students to read the comments. Charlotte Kerrigan, a high school English language arts teacher, had been giving only comments to her students for a while but was still unhappy with the amount of attention the students were giving to her comments, so she made a small but extremely powerful change in the way she provided feedback. Her tenth-grade class had just completed essays on a Shakespeare play they had been studying. Kerrigan collected the essays, and instead of writing her comments in the students' notebooks, she did so on strips of paper. Each group of four students received their four essays and the four strips of paper, and the group had to decide which comment belonged with which essay. Because students had no idea, at least at the outset, whether the comment related to their own work or that of one of their peers, they read the comments for what they said, rather than taking them personally. Furthermore, seeing what comments had been given to the work of others helped them understand the features of more successful work.

A second principle of effective feedback is that *feedback should be focused.* We generally give our students large amounts of what we call feedback (although an engineer

probably wouldn't), but it is usually of moderate quality, and we generally don't require students to do much with it. In giving feedback, less is often more.

I learned this the hard way when I first became a teacher educator, working with preservice teachers and observing them during their periods of practical teaching. I would sit at the back of their classrooms and assiduously note all the errors they had made in their teaching—I would frequently generate as many as three pages of comments on all the errors the student teacher had made in a forty-five-minute lesson. At the end of the period, I would bestow this wonderful feedback on the hapless student. I was frustrated that my wonderful feedback did not seem to be having any effect. After a while, I realized that the problem was that I was giving far too much feedback; I needed to give less, but more focused, feedback. Rather than handing over a whole list of errors to be corrected, I started saying things like, "Over the next two weeks, I want you to work on these two things: first, before you give any important instructions to the class, make them put their pens down, and second, make sure that you don't talk over the students—if they start talking to each other when you are talking, stop talking." I gave less feedback and had more impact by being more focused.

A third principle is that *feedback should relate to the learning goals that the teacher has shared with the students.* If the teacher provides a scoring rubric, then it is important that the feedback relates to that rubric. If there are learning intentions and success criteria for the work, then the feedback should loop back to those. This sounds obvious, but I have lost count of the number of times I have seen teachers provide students with rubrics or success criteria and yet fail to use these in framing their feedback to the students.

Mathematics teachers may be thinking that ideas such as comment-only grading may work well in language arts and social studies, but not so well in mathematics. After all, if a student has solved twenty equations, and the teacher places a check mark next to fifteen of them and a cross next to the other five, the student can work out his score of fifteen out of twenty, even if the teacher does not put a score of 75 percent on the work. As noted previously, however, what is important is not the form that the feedback takes but the effect it has on students.

Putting a check or a cross next to each of the solutions leaves nothing for the student to do, except maybe correct those that are incorrect. An alternative would be to say to the student, "Five of these are wrong. You find them; you fix them."

This technique is particularly well suited to mathematics, as it is often easier to check whether a solution is correct (for example, by substituting for a solved variable back into the original equation). However, it can also be readily adapted to other

subjects. When reviewing a final draft of a piece of writing, a language arts teacher may place a dot in the margin of each line where there is something that needs attention. For weaker students, she may replace the dot with a *g* for an error in grammar, an *s* for a spelling error, a *p* for punctuation, and so on, thus differentiating the feedback. The important point is that the feedback is focused, is more work for the recipient than the donor, and causes thinking rather than an emotional reaction.

These techniques can make feedback more effective as a way of moving students' learning forward, but this will be undermined unless we also change the most significant kind of feedback in most American schools and colleges, and that is the thorny issue of grading.

## Grading Practices That Support Learning

From the research discussed previously, it should be clear that the grading practices prevalent in most U.S. middle schools and high schools are actually lowering student achievement (Butler, 1987, 1988). Worse, typical grading practices don't even do the one thing they are supposed to do, which is tell us what students know. Consider:

> Imagine, for a moment, a school that has an eight-week marking period, with students receiving a grade each week. Lesley starts out with four *A*s but ends up with four *C*s. Overall, of course, she gets a *B*. Chris, on the other hand, starts out with four *C*s but ends up with four *A*s. He gets a *B* too.
>
> But who has learned more? In terms of overall achievement, Chris, with his four final *A*s, seems to have mastered the content for the marking period and really deserves an *A*. Conversely, Lesley, with her four final *C*s, seems far from mastering the content, but she gets a *B* because of her good start. The fact is that our current grading practices don't do the one thing they are meant to do, which is to provide an accurate indication of student achievement. (Clymer & Wiliam, 2006/2007, p. 36)

We can't get rid of grades—the whole American system of college selection relies on them—but what we can do is design smarter grading systems that provide accurate information about student achievement while supporting student learning. We will now discuss three techniques for smarter grading systems: (1) grading after learning, (2) grading by learning needs, and (3) allowing grades to decrease.

### *Grading After Learning*

The key to providing accurate grading while supporting student learning is a principle outlined by Alfie Kohn (1994): "Never grade students while they are still learning" (p. 41). As soon as students get a grade, the learning stops. We may not like it, but

the research reviewed here shows that this is a relatively stable feature of how human minds work. So, we have to deal with it and design assessment systems accordingly.

If grades stop learning, students should be given them as infrequently as possible. In high school, there may be an argument for one per marking period but certainly no more. In middle schools, there may be a case for grades once a year, but in elementary schools, the use of grades appears to be entirely unjustified and, indeed, unnecessary.

Many administrators realize this but continue to mandate grades because they believe that parents want them, and surveys of parents often show support for grades, but this is hardly an informed choice. Parents often support grades only because they want to know how their children are doing in school, and they assume that grades are the only way to find out. However, parents have no idea what their children's grades mean, because, as Paul Dressel (1957) remarked over half a century ago:

> A grade can be regarded only as an inadequate report of an inaccurate judgment by a biased and variable judge of the extent to which a student has attained an undefined level of mastery of an unknown proportion of an indefinite material. (p. 6)

Once we have them hooked, students want grades, too, but rather than allowing them to become codependents in an unhealthy relationship, it would surely be better not to get students addicted to grades in the first place.

### *Grading by Learning Needs*

We need classroom assessment systems that are designed primarily to support learning and deal in data that are recorded at a level that is useful for teachers, students, and parents in determining where students are in their learning. Such fine-scale evidence can always be aggregated for summative reporting. It is not possible to go the other way: from aggregate reports of achievement to learning needs.

For example, a swimming teacher observing students in a swimming pool had drawn a grid listing the students' names in the first column. The other four columns were headed as follows: arms, legs, breathing, and timing. As she observed each student, she entered a 0, 1, or 2 in each column, according to the level of competence displayed by the student in that aspect of swimming. She could, if she wanted to, add up the scores and give each student a total out of eight, but knowing that someone has scored seven out of eight is useless for planning instruction. On the other hand, knowing that a student's arms, legs, and breathing are fine but the timing requires attention is useful to the teacher (and would also be useful to another teacher if one had to take over teaching this group).

Middle school science specialist Jacki Clymer and I describe how these principles could be implemented in a science classroom (Clymer & Wiliam, 2006/2007). For each marking period, the teacher identifies key learning outcomes. For example, for the first marking period, he or she may adopt a total of ten areas of interest (some of these might cover multiple learning outcomes).

1. The appropriate use of laboratory equipment
2. Metric unit conversion and labeling
3. Calculation of density
4. Application of density (floating, sinking, layering, thermal expansion)
5. Density as a characteristic property
6. The phases of matter (at a molecular level)
7. Gas laws
8. Communication (graphing)
9. Communication (lab reports)
10. Inquiry skills

For each of the ten areas of interest, the teacher identifies sources of evidence. For example, the evidence for the appropriate use of laboratory equipment comes from observations, from a homework assignment on safety, and from the laboratory reports completed by the students. Based on the evidence collected, the teacher scores each area for each student as either 0 (no evidence of mastery), 1 (some evidence of mastery), or 2 (strong evidence of mastery), and keeps these data in a spreadsheet. The teacher uses the "conditional formatting" feature in the spreadsheet to highlight a cell in green if there is a 2 in the cell, yellow if there is a 1, and red if there is a 0. The result is an instant summary of the achievement of the class, as shown in figure 5.2 (page 149).

Since there are ten areas for this marking period, each student's total is divided by twenty to yield a percentage, which the teacher can then use to provide an indication of what the grade would be if the marking period ended at that point. However, at any time before the end of the marking period, students can provide further evidence of their competence. So, for example, if Emma Bettany wants to know what she needs to do to get an A, the teacher can quickly tell her that all she needs to do is provide evidence that she has mastered one of the three areas for which the teacher has, to date, seen only evidence of partial mastery.

| | | | Can Use Equipment | Can Convert Metric Units | Can Calculate Density | Can Apply Density | Can Apply Density as a Property | Can Apply Phases of Matter | Understands Gas Laws | Can Draw Graphs | Can Write Reports | Displays Inquiry Skills | Percent | Grade |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Period | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | | |
| Last name | First name | Outcome | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | | |
| Allen | James | | 2 | 2 | 2 | 1 | 2 | 2 | 2 | 1 | 1 | 0 | 75 percent | C |
| Ayears | Liam | | 2 | 2 | 2 | 2 | 2 | 0 | 1 | 1 | 0 | 1 | 65 percent | D |
| Baldwin | Lee | | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 0 | 75 percent | C |
| Bettany | Emma | | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 85 percent | B |
| Birch | Leah | | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 1 | 0 | 0 | 65 percent | D |
| Burns | Robert | | 2 | 2 | 1 | 1 | 2 | 0 | 1 | 1 | 0 | 0 | 50 percent | D |
| Cobern | David | | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 0 | 1 | 80 percent | B |
| Creasey | Simon | | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 0 | 0 | 65 percent | D |
| Darby | Hannah | | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 1 | 80 percent | B |
| Eastwood | Luke | | 1 | 2 | 1 | 2 | 2 | 1 | 2 | 1 | 0 | 0 | 60 percent | D |
| Ferguson | Mark | | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 0 | 1 | 85 percent | B |
| Forbes | Sarah | | 1 | 0 | 1 | 0 | 2 | 1 | 2 | 1 | 0 | 0 | 40 percent | D |
| Goodger | Mark | | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 95 percent | A |
| Hall | Mark | | 1 | 2 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 1 | 85 percent | B |
| Howells | Georgie | | 2 | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 0 | 0 | 70 percent | C |
| Hudson | Kirsty | | 2 | 2 | 1 | 2 | 2 | 1 | 2 | 1 | 0 | 0 | 65 percent | D |
| Hurley | Victoria | | 1 | 2 | 0 | 0 | 2 | 1 | 2 | 0 | 0 | 2 | 50 percent | D |
| Langan | Jennifer | | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 90 percent | A |
| Larkin | Andrew | | 1 | 1 | 0 | 2 | 2 | 1 | 2 | 2 | 0 | 0 | 55 percent | D |

**Figure 5.2: Example grading scheme for grading by learning needs.**

continued ↓

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Leach | Jonathan | | 2 | 2 | 1 | 2 | 2 | 2 | 1 | 2 | 0 | 1 | 75 percent | C |
| Lowings | Charlotte | | 2 | 1 | 0 | 2 | 2 | 2 | 2 | 1 | 2 | 1 | 75 percent | C |
| Mcglashan | Scott | | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 0 | 0 | 0 | 50 percent | D |
| Parr | Amy | | 2 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 0 | 75 percent | C |
| Ringham | Grace | | 0 | 1 | 0 | 2 | 2 | 1 | 1 | 1 | 0 | 0 | 40 percent | D |
| Rosamond | Lee | | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 0 | 0 | 70 percent | C |
| Rose | Peter | | 1 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 80 percent | B |
| Ryder | Thomas | | 2 | 1 | 2 | 1 | 2 | 1 | 2 | 2 | 0 | 1 | 70 percent | C |
| Skeats | William | | 1 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 90 percent | A |
| Walton | Emma | | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 100 percent | A |
| Average facility | | | 76 percent | 86 percent | 71 percent | 84 percent | 98 percent | 78 percent | 88 percent | 69 percent | 29 percent | 31 percent | | |

*Source: Adapted from Clymer & Wiliam, 2006/2007.*

At the end of the marking period, the students take a test, which confirms the evidence collected up to that point. When the performance on the test indicates a different level of mastery from that indicated on the basis of the classwork, the teacher treats this as a matter needing further investigation (rather than, for example, just averaging the scores). Typically, the teacher will interview the student to probe the extent of the student's understanding of the matter in question.

Perhaps the most profound impact of such a grading system is that it pushes both teacher and students into thinking about longer-term learning. If a student shows mastery of something at the beginning of the marking period but then fails to do so later, his grade can go down.

This is exactly what we found in a study of the implementation of this system with an eighth-grade class. Students became more engaged in monitoring their own learning; frequently asked for clarification, both from the teacher and from their peers; and regarded the teacher more as a coach than a judge. Their achievement went up, too (Clymer & Wiliam, 2006/2007).

### *Allowing for Grades to Decrease*

A particularly important feature of this grading system is that it avoids the ratchet effect present in most grading systems—the idea that a student's grade for a particular piece of work can never go down. When students are allowed multiple resubmissions and can improve work without penalty, this can have negative consequences. For example, in many high schools, students know they can turn in very poor-quality work, get feedback on it, and improve it for a passing grade. The problem with such systems is that there is no incentive for students to make the first submission as good as it can be, because it can always be remediated in a second submission. Districts that allow students to resubmit work multiple times need to consider creating systems that avoid such problems. Deanna Holen, a physical education teacher in Chico, California, designed a grading rubric that was constructed in such a way that no matter how poor the original submission, a resubmission could bring it up to a passing grade. However, her system also ensured that the better the first submission was, the higher the final grade would be, as shown in figure 5.3 (page 152; D. Holen, personal communication, March 31, 2007).

| | | Grade on first submission | | | | |
|---|---|---|---|---|---|---|
| | | A | B | C | D | F |
| Grade on second submission | A | A | A | B | C | C |
| | B | B | B | C | C | D |
| | C | C | C | C | D | D |
| | D | D | D | D | D | F |
| | F | F | F | F | F | F |

Figure 5.3: Example grading scheme to encourage both improvement and good first attempts.

Another way to provide similar incentives is to allocate 50 percent of the available points to the first submission and 50 percent to the improvement shown in the work as a result of responding to the feedback.

Some teachers question whether we should ever be satisfied with any work that is not of a high standard. Joe Rubin, a science teacher in San Francisco, gives only one grade: an A. When he grades work, it either comes back with an A or is returned with a note saying it is not yet ready to be evaluated, together with some suggestions as to how it might be improved. Of course, the students who need multiple attempts to get an A may well run out of time, and thus complete fewer assignments, but the message this teacher is sending to all students is that only the best is good enough. More important, he sends the message that everyone can do work of excellent quality, although some need more support and guidance than others. In sum, he emphasizes quality over quantity.

## Conclusion

The word *feedback* was first used in engineering to describe a situation in which information about the current state of a system was used to change the future state of the system, but this has been forgotten, and any information about how students performed in the past is routinely regarded as useful. It is not. In this chapter, we have seen that in almost two out of every five carefully designed scientific studies, information given to people about their performance lowers their subsequent performance. We have also seen that when we give students feedback, there are eight things that can happen, and six of them are bad (table 5.2, page 130).

Some ways to give effective feedback have been described in this chapter, but every teacher will be able to come up with many more, provided that the key lessons from the research on feedback are heeded. If we are to harness the power of feedback to

increase student learning, then we need to ensure that feedback causes a cognitive rather than an emotional reaction—in other words, *feedback should cause thinking* by creating desirable difficulties. Feedback should be focused; it should relate to the learning goals that have been shared with the students; and it should be more work for the recipient than the donor. Indeed, the whole purpose of feedback should be to increase the extent to which students are owners of their own learning, which is the focus of the next two chapters.

chapter 6

# Activating Learners as Instructional Resources for One Another

Even though there is a substantial body of research that demonstrates the extraordinary power of collaborative and cooperative learning to produce tangible and substantial increases in students' learning, educators rarely deploy these practices effectively in classrooms. This chapter explores the role that learners can play in improving the learning of their peers and discusses how teachers can effectively enable students to work together to become meaningful, valuable resources for each other's learning. It concludes with a number of specific classroom techniques that educators can use to put these principles into practice.

At the outset, it is important to note that while some researchers talk about *cooperative learning* and others talk about *collaborative learning*, there is little consensus about the meanings of the terms. For some people, the terms are relatively interchangeable, while for others, there are clear differences between the two. In K–12 settings, the term *cooperative learning* is more common (for example, Johnson & Johnson, 2009; Slavin, Hurley, & Chamberlain, 2003). In higher education, one common way of distinguishing the two is to use the term *cooperative learning* for situations where students work together to achieve a shared learning goal the teacher has established, and to use the term *collaborative learning* when the students themselves decide what the goal should be (see, for example, Springer, Stanne, & Donovan, 1999).

However, despite the preceding distinction, in this book I will use the term *cooperative learning* to refer to both cooperative and collaborative learning. As is clear

from chapter 3, one of the key requirements of effective formative assessment is that learners are clear about what it is they will be learning. Sometimes the teacher will determine the goals, sometimes the students themselves will, and sometimes the goals will be the result of a negotiation between the students and the teacher. As far as formative assessment is concerned, while teachers and students do need to be clear about the goals of the learning, any of these methods of goal setting may be used, as they do not directly impact the *practice* of formative assessment. Thus, any distinction between *cooperative* and *collaborative learning* in terms of who sets the goals is irrelevant to the purpose of this book.

Another area of potential confusion, especially in higher education, is how peers can work together to meaningfully improve each others' work. The focus of peer assessment is often on whether students can give each other accurate grades or not. Indeed, this is what is meant by the vast majority of research on peer assessment in higher education (see, for example, Falchikov & Goldfinch, 2000; Jones & Alcock, 2012). If teachers want to involve peers in assessment for summative purposes, then this is an important issue. However, the focus in this chapter is on helping students at all levels *improve*, rather than evaluate, the work of their peers.

## Effective Cooperative Learning

Having reviewed the available research evidence, Robert Slavin, Eric Hurley, and Anne Chamberlain (2003) conclude that "research on cooperative learning is one of the greatest success stories in the history of educational research" (p. 177). Exactly why cooperative learning has such a profound effect is still a matter of some debate, although there appear to be four main factors.

1. **Motivation:** Students help their peers learn because, in well-structured cooperative learning settings, it is in their own interests to do so, and so effort increases.
2. **Social cohesion:** Students help their peers because they care about the group, again leading to increased effort.
3. **Personalization:** Students learn more because their more able peers can engage with the particular difficulties a student is having.
4. **Cognitive elaboration:** Students who provide help in group settings have to think through the ideas more clearly.

All these factors have a role to play, but some appear to be more powerful than others. In particular, focusing on social cohesion, without attending to the other factors, appears to have little impact on student learning. A review of ninety-nine studies

finds that cooperative learning in which group rewards depended on the *aggregate* of the learning of *individual* members produced four times the impact on learning than was found when rewards were based on a single group product (Slavin, 1995).

Personalization and cognitive elaboration are important too. A review of seventeen studies of peer interaction on the learning of mathematics in elementary, middle, and high schools finds that help in the form of answers or procedural information has no benefit for those giving help and produces a significant drop in the achievement of those getting help. On the other hand, when the help takes the form of elaborated explanations (namely, explaining why a particular answer is correct, rather than just giving the correct answer to a peer), both those giving and getting help benefit, and this benefit is especially great for those giving help, producing at least a 50 percent increase in the rate of learning (Webb, 1991).

One particularly surprising finding is that the effect of peer tutoring can be almost as strong as one-on-one instruction from a teacher. John Schacter's (2000) study of 109 students in fourth-, fifth-, and sixth-grade classrooms finds that students working in student-led groups learn almost as much as students getting one-on-one tutorial instruction from a teacher, and those in student-led groups actually learn more than those in teacher-led groups.

In fact, under certain circumstances, peer tutoring can be more effective than one-on-one tutorial instruction from a teacher. I once asked two fifth-grade girls whether they preferred to get help from the teacher or from each other. One of the girls said, "I much prefer to get help from my friends because teachers have got this weird kind of language, if you know what I mean." This didn't surprise me, as most teachers have had the experience of giving what they think is a perfectly lucid explanation of something to a student only for the student to respond with, "Huh?" More galling, often a peer will contribute something that seems completely unintelligible, and the student says, "Oh right, I get it now." The fact that students have a kind of shorthand communication is nothing new. What was surprising was what the girl said next. She said, "And if I haven't understood it after the second explanation from the teacher, I pretend I have." I asked her why she did this, and she said it was because she was aware of how busy the teacher was and didn't want to take up too much of her time. Her friend admitted that she did this, too. I then interviewed some boys who also said that they pretended that they understood something when, in fact, they didn't, typically not out of concern for the teacher's time but because they did not want to appear foolish in front of the teacher.

Because of these conversations, I started looking at the expressions on the faces of students when the teacher was engaged with them in one-on-one tutorial instruction, and I realized that they couldn't wait for the ordeal to end. Hardly ever would

a student interrupt a teacher for clarification or to ask the teacher to go over something a second time. And yet, when working with peers, a student would ask the peer to slow down or to go over something again and again until it was understood. Peer tutoring can, under the right circumstances, generate more effective learning than would be possible with one adult for every student, because of the change in power relationships.

Effective cooperative learning requires the presence of two elements. First, there must be group goals, so that students are working *as* a group, not merely working *in* a group. In other words, students must be working to achieve a single common goal, rather than different goals. For example, students working together to produce a single story would be working *as* a group. Students exchanging ideas and then writing their own individual stories would be students working *in* a group.

Second, there must be individual accountability, so that individual students cannot ride the coattails of others' work. In other words, one person failing to do what he or she is meant to do *must* harm the group's chances of success. This is common in sports, where one fielder missing a catch harms the whole team's chances of winning. It is also common in the orchestra and band, where one student playing an incorrect note or missing a cue affects the entire group's performance. However, as we shall see, it seems to be more difficult to ensure individual accountability in academic subjects.

Some studies find that the benefits of cooperative learning environments are greater for lower achievers (for example, Boaler, 2002); others find benefits for high achievers (for example, Stevens & Slavin, 1995). Overall, it appears that cooperative learning is equally effective for students at all achievement levels, as long as the two key features—group goals and individual accountability—are present (Slavin et al., 2003). Having said that, cooperative learning interventions seem to be particularly beneficial for students of color at the undergraduate level (Springer et al., 1999), and the value of collaborative learning environments for African American students in particular has been emphasized by Wade Boykin and his colleagues (Boykin, Coleman, Lilja, & Tyler, 2004; Boykin, Lilja, & Tyler, 2004).

Although the two requirements—group goals and individual accountability—seem straightforward, they cut across some widely held assumptions about fairness for groups and individuals. For example, most teachers accept as fair the idea of group rewards for behavior. A technique that some schools use to promote good behavior in classrooms is *secret student*. The idea is that each day, the teacher selects one member of the homeroom at random to be the secret student, but the identity of the secret student is revealed only to the teachers teaching that particular student that day. Each teacher then tells the homeroom teacher whether the behavior of the secret student was satisfactory, and if the proportion of periods in which the behavior

of the secret student exceeds a prespecified threshold (for example, 80 percent, all but one, or all but two), then the class is credited with a good day and the identity of the secret student is revealed to the class. If, however, the behavior of the secret student is not good enough, then the identity of the student is *not* revealed. The class receives some group reward that depends on the number of good days that are recorded. While one might have some concerns about the operation of the secret student scheme in practice, the use of peer pressure to promote good behavior seems to be broadly acceptable.

However, the practice is notably less acceptable when applied to academic work. For example, imagine a situation where, at the end of the week, a teacher quizzes his or her class on the material, and each group receives the lowest-scoring group member's score (of course, the teacher doesn't reveal the lowest-scoring member's name). In other words, the achievement at the *trailing edge* characterizes the group.

Many teachers find this type of assessment completely unacceptable. They point out that the score they award to the group, being that of the lowest-scoring member, does not represent the group's level of achievement, which is an entirely fair point. It would certainly be invalid to infer anything about the individual group members' achievement from this score, and it would be quite wrong to enter this score in a gradebook as an indication of the individual group members' achievement. But if, instead, the teacher and students take the score to indicate how well the students worked *as a group*, then the awarded score seems more reasonable.

The fact that many teachers view group rewards for good behavior as entirely legitimate but trailing-edge scoring as unacceptable reveals how difficult it is to secure the benefits of cooperative learning in classrooms. In fact, what teachers describe as cooperative learning in their classrooms rarely has the features that would make it effective. One survey of eighty-five elementary school teachers in two districts finds that although 93 percent of the teachers said they employed collaborative learning, follow-up interviews with twenty-one of the teachers showed that only five teachers implemented collaborative learning in such a way as to create both group goals and individual accountability (Antil, Jenkins, Wayne, & Vadasy, 1998). Furthermore, only one of the twenty-one teachers implemented collaborative learning that satisfied the more complex criteria Elizabeth Cohen (1994) proposed—namely, open-ended tasks that emphasize higher-order thinking, group tasks that require input from other members, multiple tasks related to a central intellectual theme, and roles assigned to different group members.

In the final section of this chapter, I present a number of practical techniques for activating students as learning resources for one another that draw on the research

on cooperative and collaborative learning. In particular, the techniques provide practical ways for teachers to create group goals and individual accountability in their classrooms.

## Practical Techniques

The remainder of this chapter presents a number of techniques that teachers have found useful in getting started with the process of activating students as learning resources for one another. Inevitably, a number of these techniques involve students assessing each other's work, and this raises a number of ethical concerns. Perhaps the most important is whether students should be involved in summative assessment, and in my view, the answer is a firm no. I think it is quite wrong for one student to be placed in the position of evaluating the achievement of another student for the purpose of reporting to parents or others. The purpose of peer assessment should be simply, and purely, to help the individual being assessed improve his work. This section will consider the following techniques: C3B4ME, peer improvement of homework, homework help board, two stars and a wish, end-of-topic questions, error classification, student reporter, preflight checklist, I-you-we checklist, reporter at random, group-based test preparation, and "If you've learned it, help someone who hasn't."

### C3B4ME

In this technique, students must seek assistance from at least three other students before seeking it from the teacher, hence the description *see three before me* (C3B4ME). Some teachers support this with a poster declaring, "There is more than one teacher in this room." This technique can work in conjunction with the *no hands up* technique. When you randomly select a student to answer a question, before he or she responds, have the student brainstorm his or her answer with three other random students—asking if they agree or would like to add to the answer.

### Peer Improvement of Homework

One middle school mathematics teacher tried to reduce the amount of time she spent checking homework by keeping the students unaware of whether—or how—their homework was going to be checked and graded. Sometimes she gave the students a rubric, and they had to grade their own work. Sometimes she told students to swap their homework with a neighbor and grade each other's, and sometimes she asked one group of four students to look at the work of another group.

When she began to implement this approach regularly, three surprising things happened. First, more students did the homework. If students had not completed the homework, then they were not allowed to participate in the group evaluation

activity, and even though these students were, in effect, being allowed to do in class what they should have done at home, they regarded being excluded from the evaluation as sufficient punishment to ensure that next time they handed in something to be evaluated. Second, their work was neater. Students appeared to care more about communicating clearly with each other than about communicating to the teacher. Third, and perhaps not surprisingly, the quality of the homework was also higher.

## Homework Help Board

Another approach for dealing with issues arising from homework is the homework help board. At the beginning of the day (in elementary school) or at the beginning of the lesson (in middle school and high school), students indicate on the homework help board any questions that they had about the homework. Students who think they can help with these questions are then encouraged to seek out the question submitters and provide assistance.

## Two Stars and a Wish

*Two stars and a wish* is a very simple technique for getting started with peer assessment. When any student gives feedback on another student's work, he or she has to provide two things he thinks were good about the work (the two stars) and a suggestion for improvement (the wish). Students can write the comments on sticky notes so that if the recipient doesn't find the feedback helpful, he or she can easily remove it.

To improve the quality of the feedback, the teacher collects the sticky notes once the students have responded to the feedback and, using a document camera, displays each note to the class (anonymously). The teacher then asks the class to vote on whether they think that the comment would be useful to them. The students in the class then discusses the salient features of the best-rated feedback, and the teacher displays the resulting criteria on a poster in the classroom for future reference.

## End-of-Topic Questions

It is quite common for a teacher to reach the end of an exposition, a chapter, or a unit and ask the class, "Any questions?" Of course, few students are willing to raise their hands to ask a question because they do not want to look foolish in front of the rest of the class. To overcome this, the teacher can say, "In your groups, decide if you have any questions." Just the chance to talk through the matter with their peers can make students more confident about asking questions in front of the class—after all, if no one in the group has a solution, then it's clearly not a dumb question. Some teachers go even further and insist that each group comes up with at least one question. The teacher collects the questions, sorts them quickly, and deals with all the questions on the same issue at the same time. Gathering all the questions

before beginning to answer the questions is particularly useful because it allows the teacher to make connections between different issues that the students may not realize are connected. Some teachers give the students sticky notes for their questions, but because they are adhesive, it is more difficult to sort the questions rapidly. For that reason, many teachers have found that a sheet of paper, cut vertically into five strips (approximately 2 inches by 8½ inches), provides the best compromise.

Getting students to write their questions down also helps them develop literacy skills, and teachers have reported that their students become better at asking questions. One high school mathematics teacher reported that she noticed a real change in the precision with which her students asked questions. She said that students used to say things like, "I can't do quadratics," and when the teacher asked what it was about quadratics they couldn't do, the students would respond by saying things like, "I can't do any of it." However, after some months of getting students to ask questions on strips of paper, she found that the students were more precise and focused, saying things like, "I can't do quadratics when there's a minus in front of the $x$-squared." This allowed the teacher to clarify her teaching and assist students with relearning specific topics.

## Error Classification

This technique is useful when you can classify errors in a relatively straightforward way. For example, a Spanish teacher may collect final drafts of a piece of writing in Spanish, use a pencil to underline errors in the text, and return the work to the students. The students then have to classify the errors they made (for example, errors in tense, gender, pronouns, or possessives). After determining their errors and the areas in which they have made few errors, students identify a buddy with complementary strengths to help them correct their work.

## Student Reporter

Other teachers have experimented with end-of-lesson reviews. One method that appears to be particularly effective is to form the class into groups five minutes before the end of the lesson and ask each group to produce a list of things it has learned during the lesson. Each group then reports one thing it has learned to the class. (The number of items in the list needs to be the same as the number of groups so that each group is guaranteed to have at least one thing on its list that has not been mentioned by the other groups.)

An alternative to this is to have a student reporter lead the end-of-lesson discussion. At the beginning of the lesson (or at the end of the previous one), a student is appointed as a reporter for the lesson. The teacher then conducts the lesson as normal

but finishes ten minutes before the scheduled end of the lesson. The student reporter summarizes the lesson's main points and tries to answer any remaining questions that students in the class may have (one teacher calls the selected student the *captain* so the summary of the lesson is the *captain's log*, as in *Star Trek*!). If she or he can't answer the questions, then the reporter asks members of the class to help out.

Teachers who have tried this out have found that, although initially reluctant, students are quickly lining up to play the role of reporter; they see it as an opportunity to contribute to the learning of the whole class. One interesting approach is to ask the reporter, as he is observing the lesson, to make up questions to ask the class at the end. The teacher then uses the best of these questions for a "three-fourths of the way through a unit" test.

## Preflight Checklist

This technique is particularly valuable when there are a number of clear requirements that submitted work has to satisfy. For example, science teachers routinely require laboratory reports to be structured in a standard way—for example, the report follows the sequence of question, hypothesis, method, results, and conclusion; the report has a title that is underlined; the pages have margins; diagrams and graphs are drawn and labeled in pencil; and so on. Before a student can submit an assignment, it must be signed off on by a buddy, who checks that all the required features are present. When the teacher grades the assignment, if any of the items on the preflight checklist are not up to standard, it is the buddy rather than the student submitting the assignment who is taken to task, thus creating a measure of accountability for the buddy to take the task seriously.

## I-You-We Checklist

At the end of a group activity, each student records something about his or her own contributions, something about other individuals' contributions, and an evaluation of the whole group's quality of work (hence, *I-you-we*). When students know that each member of the group will be assessing both their own contributions and those of others, it creates an incentive for students to be more honest and accurate about their own contributions. Moreover, by asking each member of the group to reflect on how the group worked together, students are more likely to be able to work more effectively together in the future (Johnson, Johnson, & Smith, 1998).

## Reporter at Random

Many teachers advocate assigning students roles, such as chair, timekeeper, facilitator, scribe, and so on, when they are working collaboratively. This can be a good

idea, but in general, it is a bad idea to assign the role of reporter at the outset. As soon as students know that someone else will have to report back, they can ease off, as the need for individual accountability weakens. Because each student knows that he or she may be asked to report back, he or she will be more likely to pay attention, concentrate, and contribute. More important, students who are particularly keen for the group to do well will ensure that others are paying attention, because if the person reporting does not give a good account of the group's work, everyone in the group will be associated with that report.

## Group-Based Test Preparation

When students are preparing for a test, one way to make reviewing more engaging is to organize them into groups of five or six and assign each student one aspect of the work to review. Each student is given a card on which details of a task are written, with suggestions for how it could be carried out. For instance, a teacher I observed was preparing a seventh-grade class for a test on the work they were just completing on plants. Each group was given six cards with the main topics they had covered (pollination, photosynthesis, starch test, seed dispersal, germination, and flowers) and a brief description of the task. For example, the card for pollination read as follows:

> You have been asked to explain to the rest of the people in your group exactly what happens when an insect-pollinated plant reproduces. You should be able to describe where the pollen seed goes from and to, and how pollination results in the formation of a seed.

The following day, each member presented to the group. The rest of the group then responded to the presentation using colored cups (see chapter 7) to traffic-light their responses: green means "better than I could have done"; yellow means "about the same as I would have done"; and red means "not as good as I would have done." After the voting, the group decided what needed to be added to the explanation to make it a good one.

## "If You've Learned It, Help Someone Who Hasn't"

One objection to cooperative learning is that it holds back the able students. However, as discussed previously, if the students engaged in peer tutoring are providing elaborated explanations rather than just answers, then there is compelling evidence that both those who give and those who receive help will benefit.

Since cooperative learning cuts across the accepted demarcation of roles in classrooms, some teachers have found that high-achieving students have resisted peer tutoring because they claim that, as well as being held back, they are being asked to

do the teacher's job for her. This appears to be particularly marked in mathematics, where there can be a great gulf between the highest achievers and the lowest achievers, particularly in terms of the speed with which assignments are completed. One tactic that is useful in countering this attitude is to make it clear that to be a good mathematician, getting the right answer isn't enough. One also has to communicate one's findings to other mathematicians. Being asked to explain how an answer was reached develops skills of mathematical communication that the high achiever needs to develop, even if she just wants to be a pure mathematician.

The attitude toward group work also depends on cultural norms. In the United States, there is the saying, "The squeaky wheel gets the grease." In Japan, there is a different saying: "*Deru kui wa utareru*," which, roughly translated, means, "The protruding nail gets hammered into place." Many observers of mathematics teaching in Japan and the United States (for example, Stigler & Hiebert, 1999) have pointed out that Japanese teachers seem to worry far more about keeping the class together; if one student has already understood the topic, then her task is to help other students who haven't.

As noted, the students who learn most in group work are those who give help and those who receive it, provided the help is in the form of elaborated explanations rather than just answers. If there is a large range of ability within a group, students in the middle can become disengaged spectators, watching the high achievers teach the low achievers. They don't need the help that the weakest students need, but they also aren't forced to articulate their thinking as the strongest students do when they teach others. Sometimes, therefore, teachers may choose to organize students into half-ability groups—weak achievers with middle, and middle with strong—to engage as many students as possible in an activity that helps learning.

Another concern that teachers often have about this kind of approach is that it's not fair. One student gets everything right, and another makes many errors, but then the teacher or a peer shows the student what to do, and they are both at the same point. The fact that the first student got there more quickly than the second should somehow be recognized by giving him a higher score. But think about what happens when I take my car to the DMV for its annual emissions check. My car, because it's been serviced recently, passes the first time. My neighbor's car fails, but at least he's told what's wrong. They don't just say, "Bring the car back when it's better." When he takes the car back with the problem fixed, he gets the same certificate I received. I don't get a bigger one, edged in gold, because I got it right the first time. The certificate simply shows that the car's emissions levels are satisfactory. In the same way, the primary purpose of assessment should not be to sort and rank and grade students.

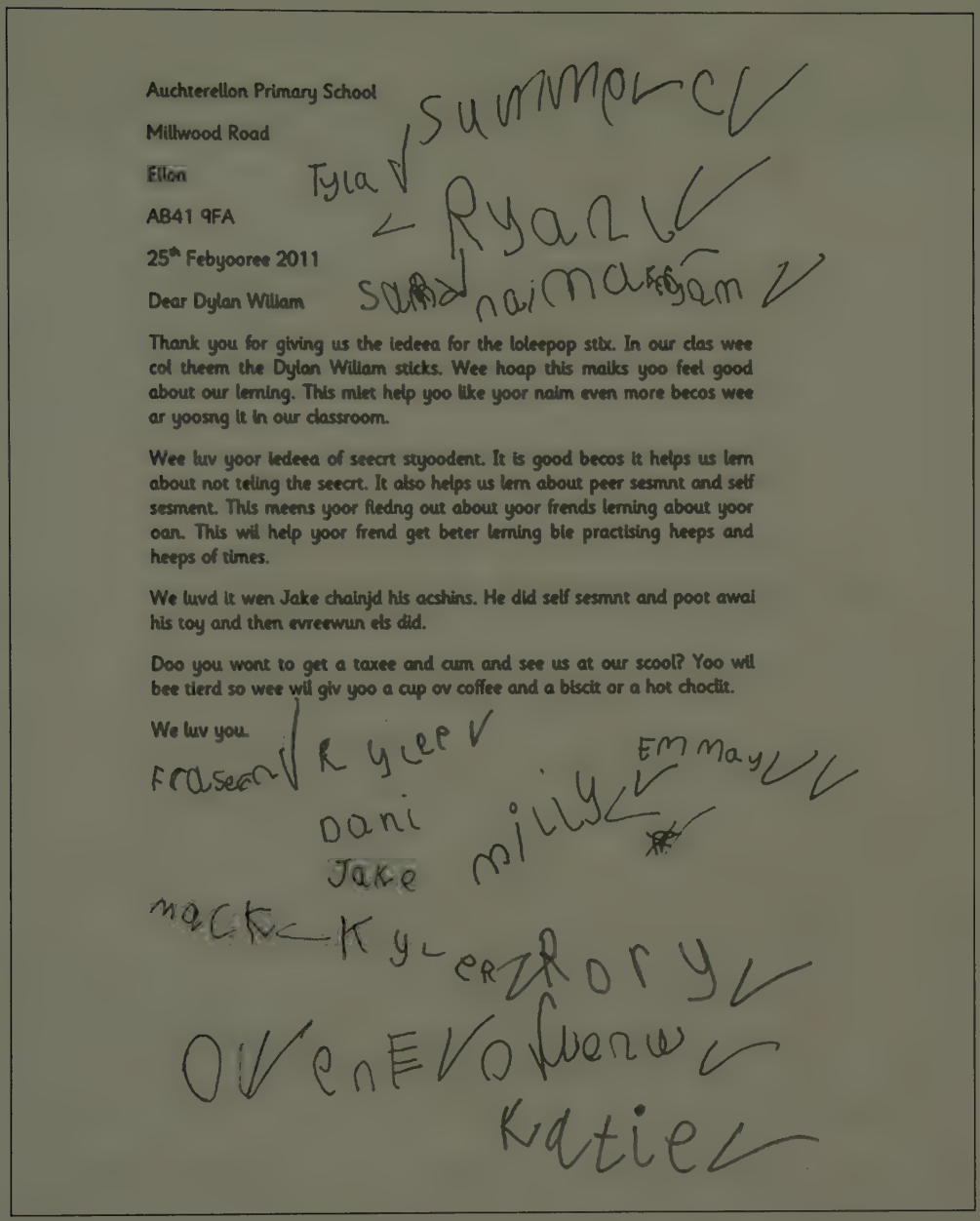
Rather, the main purpose of assessment should be to provide information to the teacher to make real-time adjustments to his instruction in order to better meet the learning needs of his students. As we saw in chapter 5, we need to be careful not to create incentives for students to turn in low-quality work.

As a final incentive, I include in figures 6.1 and 6.2 a couple of letters I received from a school in Scotland, after I had worked with their teachers on some of the ideas of formative assessment discussed in this book. In Scotland, students who achieve their fourth birthday before the end of February in any given year attend school from the middle of the following August, so that the youngest children in the class (called primary 1) will be four years and five months old when they start. The school in question has two primary 1 classes, and their teachers sent me these letters that their students had written to me. It is clear that the teachers not only took many of the ideas to heart but also shared with their students where they had gotten the ideas from, and why they were using them. Again, they were assisting students by helping them see the value—and, indeed, the enjoyment—in becoming instructional resources for each other's learning.

## Conclusion

In this chapter, we have seen that activating students as learning resources for one another produces tangible and substantial increases in students' learning. Every teacher I have ever met has acknowledged that you never really understand something until you try to teach it to someone else. And yet, despite this knowledge, we often fail to harness the power of peer tutoring and other forms of collaborative learning in our classrooms.

This chapter has presented a number of classroom techniques that can be used with students of almost any age and that can readily be incorporated into practice. Many of these techniques focus specifically on peer assessment, which can be especially powerful if teachers gear it toward improvement rather than evaluation—students tend to be much more direct with each other than any teacher would dare to be. However, it is important to realize that peer assessment also benefits the individual who gives help. Making students provide feedback to each other forces them to internalize the learning intentions and success criteria but in the context of someone else's work, which is much less emotionally charged. Activating students as learning resources for one another can, therefore, be seen as a stepping-stone to students becoming owners of their own learning—the subject of the next chapter.



Auchterellon Primary School

Millwood Road

Ellon

AB41 9FA

25th Febyooree 2011

Dear Dylan William

Thank you for giving us the iedeea for the loleepop stix. In our clas wee col theem the Dylan William sticks. Wee hoap this maiks yoo feel good about our lerning. This miet help yoo like yoor naim even more becos wee ar yoosng it in our classroom.

Wee luv yoor iedeea of seecrt styoodent. It is good becos it helps us lern about not teling the seecrt. It also helps us lern about peer sesmnt and self sesment. This meens yoor fiedng out about yoor frends lerning about yoor oan. This wil help yoor frend get beter lerning bie practising heeps and heeps of times.

We luvd it wen Jake chainjd his acshins. He did self sesmnt and poot awai his toy and then evreewun els did.

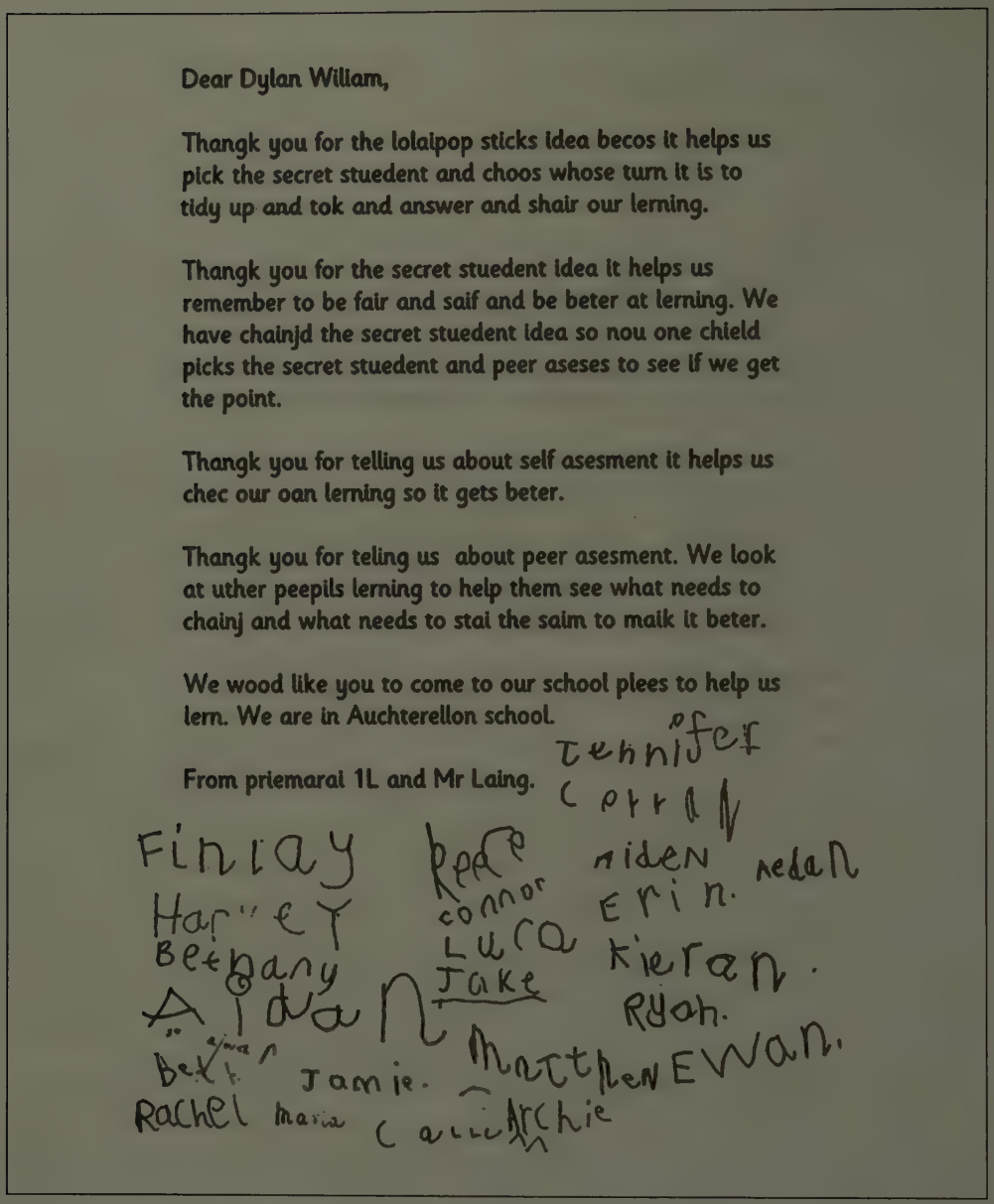
Doo you wont to get a taxee and cum and see us at our scool? Yoo wil bee tierd so wee wil giv yoo a cup ov coffee and a biscit or a hot choclit.

We luv you.

Tyla Summer Ryan Sara naimatoom Fraser Kyler Dani Jake milly Emma Mack Kyler Rory Owen Evo Katie

*Source: Reprinted with permission from Auchterellon Primary School, 2011.*

Figure 6.1: First response letter reflecting the value of activating learners as instructional resources for one another.



Dear Dylan William,

Thangk you for the lolalpop sticks idea becos it helps us pick the secret stuedent and choos whose turn it is to tidy up and tok and answer and shair our lerning.

Thangk you for the secret stuedent idea it helps us remember to be fair and saif and be beter at lerning. We have chainjd the secret stuedent idea so nou one chield picks the secret stuedent and peer aseses to see if we get the point.

Thangk you for telling us about self asesment it helps us chec our oan lerning so it gets beter.

Thangk you for teling us about peer asesment. We look at uther peepils lerning to help them see what needs to chainj and what needs to stai the saim to maik it beter.

We wood like you to come to our school plees to help us lern. We are in Auchterellon school.

From priemarai 1L and Mr Laing.



*Source: Reprinted with permission from Auchterellon Primary School, 2011.*

Figure 6.2: Second response letter reflecting the value of activating learners as instructional resources for one another.

chapter 7

# Activating Learners as Owners of Their Own Learning

In the introduction to his book *Guitar*, Dan Morgan (1965) writes, "No one can teach you to play the guitar" (p. 1). I found this rather puzzling, since the subtitle of the book is *The Book That Teaches You Everything You Need to Know About Playing the Guitar*. However, Morgan (1965) clarifies by adding, "But they can help you learn" (p. 1). This is pretty obvious, really. Whether learning to play a musical instrument, a sport, or a whole range of other human endeavors, we intuitively grasp that teachers do not create learning; only learners create learning. And yet our classrooms seem to be based on the opposite principle—that if teachers try really hard, they can do the learning for the learners. Accountability regimes that mandate sanctions for teachers, for schools, and for districts, but not for students, only exacerbate this.

This chapter reviews the research evidence on the impact of getting students more involved in their learning and shows that activating students as owners of their own learning can produce extraordinary improvements in their achievement. It discusses the advantages and disadvantages of student self-assessment and self-testing and then considers self-regulated learning in the context of metacognition and motivation. As with the previous four chapters, it concludes with a number of practical techniques for classroom implementation.

## Student Self-Assessment

To many, the phrase *student self-assessment* conjures up images of students giving themselves grades and diplomas, and the reaction from teachers is often predictable, including phrases like "lunatics running the asylum" or "foxes guarding the henhouse." There is, in fact, evidence that students can assess themselves quite accurately for summative purposes (see, for example, Darrow, Johnson, Miller, & Williamson, 2002), but only when the stakes are low. Whether or not students can assess themselves accurately for summative purposes is completely irrelevant to the topic of this chapter, which is whether or not students can develop sufficient insights into their own learning to improve it.

The answer is yes. The potential of student self-assessment for raising achievement was vividly demonstrated in a study of twenty-five elementary school teachers in Portugal (Fontana & Fernandes, 1994). Over a twenty-week period, the teachers met for two hours each week, during which they were trained in the use of a structured approach to student self-assessment that involved both a prescriptive component and an exploratory component.

The prescriptive component took the form of a series of hierarchically organized activities, from which the teacher selected based on diagnostic assessments of the students. For the exploratory component, each day at a set time, students organized and carried out individual plans of work, choosing tasks from a range offered by the teacher. The students had to evaluate their performance against their plans once each week. The progression within the exploratory component had two strands: over the twenty weeks, the tasks and areas in which the students worked took on the students' own ideas more and more; and the criteria that the students used to assess themselves became more objective and precise.

In the first two weeks, students chose from a set of carefully structured tasks and were then asked to assess their own performance. For the next four weeks, students constructed their own mathematical problems following the patterns of those used in weeks 1 and 2, and evaluated them as before, but this time, the students were required to identify any problems they had had and whether they had sought appropriate help from the teacher. Over the subsequent four weeks, students were given additional sets of learning objectives and again had to devise problems, but were not given examples by the teacher. Finally, in the last ten weeks, students were allowed to set their own learning objectives, to construct relevant mathematical problems, to select appropriate apparatus, and to identify suitable self-assessments.

In order to evaluate the impact of the self-assessment activities on the students' progress, the achievement of the 354 students taught by the twenty-five teachers in

the study group was compared with that of 313 students taught by twenty teachers who had not been involved in the study in any way but were matched in terms of age, qualifications, and experience, and were using the same curriculum scheme for the same amount of time. To further ensure comparability, the twenty control teachers were provided with the same amount of in-service professional development, but it was not focused on student self-assessment. A standardized mathematics test was administered to all 667 students at the beginning of the twenty-week study and again at the end. The scores of those in the control-group teachers' classes went up by 7.8 points, while the scores of those in the classes of teachers employing self-assessment rose by 15 points. In other words, through the development of their self-assessment skills, students managed to learn in twenty weeks what would otherwise have taken thirty-eight weeks to learn. Using self-assessment in these twenty-five classrooms had almost doubled the rate at which students were learning.

This, of course, is just one study, but there is now increasing evidence that getting students to assess themselves is one of the most straightforward and inexpensive ways to increase student achievement (see, for example, chapter 9 of Butler, Schnellert, & Perry, 2017). One particularly effective way of doing this is through self-testing.

## Self-Testing

Most people think that American students are over-tested, but in fact, there is now considerable evidence that we actually need more testing in American schools, albeit of a very different kind to that which happens in most schools currently.

In 2013, a group of five American psychologists of education investigated findings from psychological science that students could use to improve their own learning (Dunlosky, Rawson, Marsh, Nathan, & Willingham, 2013). They restricted their investigation to techniques that students could use on their own to improve their learning, without the need for technological support or any guidance from a teacher. They included some techniques where the research evidence seemed reasonably strong, and others because they were techniques students reported that they used regularly, such as rereading and highlighting. In the end, they selected ten techniques for further investigation.

1. **Elaborative interrogation:** Generating an explanation for why an explicitly stated fact or concept is true
2. **Self-explanation:** Explaining how new information is related to known information, or explaining steps taken during problem solving

3. **Summarization:** Writing summaries (of various lengths) of to-be-learned texts
4. **Highlighting or underlining:** Marking potentially important portions of to-be-learned materials while reading
5. **Keyword mnemonic:** Using keywords and mental imagery to associate verbal materials
6. **Imagery for text:** Attempting to form mental images of text materials while reading or listening
7. **Rereading:** Restudying text material again after an initial reading
8. **Practice testing:** Self-testing or taking practice tests over to-be-learned material
9. **Distributed practice:** Implementing a schedule of practice that spreads out study activities over time
10. **Interleaved practice:** Engaging in practice that mixes different kinds of problems, or study that mixes different kinds of material, within a single study session

After extensive review of the research evidence on each of these techniques (their review cited over four hundred references), they concluded that for five of the techniques—summarization, highlighting, the keyword mnemonic, imagery use for text learning, and rereading—there was only limited evidence of impact on student achievement. They concluded that summarization and the use of imagery did work for some students on some tasks, but little was understood about when summarization worked and when it didn't, so recommending its use for all students would not be a good idea. The idea of generating keywords worked in some contexts but was hard to implement in others, so again it could not be recommended as a general strategy. Rereading and highlighting were two techniques that students mentioned often, but while these were sometimes effective, they were often not, so again could not be recommended for general use.

Three techniques received *moderate* ratings, on the basis that the techniques worked a lot of the time, but not every time: (1) elaborative interrogation, (2) self-explanation, and (3) interleaved practice. For these three techniques, while there was some evidence that they worked, there wasn't much evidence that they were effective in educational (as opposed to laboratory) settings. This, of course, did not mean that these techniques would not work in educational settings; merely, there was not much evidence on this point either way.

Two techniques—(1) practice testing and (2) distributed practice—received *high* ratings because (a) they were effective with learners of different ages and abilities, (b) they were shown to boost students' performance across many kinds of tasks, and (c) there was plenty of evidence that they worked in educational contexts. Given the discussion of Robert Bjork's work in chapter 5, this should not come as a surprise. Practice testing gives students practice in retrieving things from memory (*retrieval strength*), and therefore increases long-term memory. Moreover, when practice is distributed—when students test themselves some time after they study—then the retrieval strength of the items being recalled will be lower, so successful retrieval will have a greater impact on storage strength.

There is also a second benefit of practice testing, which comes when students find out if they were correct or not, and this is known as the *hypercorrection* effect. For many years, it was thought that if a student was really confident that an answer he or she had given to a question was correct, then it would be difficult for him or her to accept that an erroneous answer was actually incorrect. In fact, it seems that the opposite is true. High-confidence errors are, in fact, much easier to correct than low-confidence errors. This has been shown by experiments in which students are asked to complete a test and, for each answer they give, they are asked to rate their confidence in their answer on a scale of one to five. Then they are told, for each answer, whether it is correct or not, and for incorrect answers, they are given the correct answer. If they are tested again on the same content some weeks later, the questions they answered incorrectly in the first test are more likely to be answered correctly on the second test if the confidence in the answer given in the first test was high (Metcalfe, Butterfield, Habeck, & Stern, 2012). We don't know exactly why this is. It could be because students who were confident they were right are more embarrassed by getting things wrong, and, therefore, remember that they got them wrong for a longer period of time, but whatever the reason, the effect seems to be reasonably robust. Thus, if students complete a practice test and get immediate feedback on their answers, students will get the benefit of the hypercorrection effect for those questions where they were confident they were correct.

The problem in all this, of course, is that students do not like tests, and students will likely greet with dismay any attempt to increase the amount of testing. The solution to this dilemma is realizing that students do not gain any additional benefit from practice testing when a score is recorded in a teacher's gradebook. The benefit of testing comes from the retrieval practice (especially when retrieval strength is low) and from the hypercorrection effect, where students were confident that incorrect answers were actually correct. In other words, the best person to mark a test is the person who just took it. We need more testing but less grading. Students should be

encouraged to test themselves using practice tests and check their own answers, but they should not be required to tell anyone how well they did (unless they want to, of course).

The big idea here is that students should take more responsibility for their own learning, so that over time, they are able to manage their own learning as well as, or even better, than any teacher could. In other words, in the psychological jargon, they become self-regulating learners.

## Self-Regulated Learning

Self-regulated learning enables the learner to coordinate cognitive resources, emotions, and actions in the service of his or her learning goals (Boekaerts, 2006). Some (for example, Winne, 1996) emphasize the cognitive aspects of this process—does the learner have the necessary knowledge, skills, strategies, and so on to reach the goal? Others (for example, Corno, 2001) point out that many students possess the necessary skills but do not use them in classrooms, which suggests that the problem is not a lack of skill but rather a lack of motivation or volition. As mentioned in chapter 4, still others, such as Alan Schoenfeld (1985), point out that sometimes it is their beliefs that get in the way—if students believe that all problems can be solved in five minutes or less, they may well give up on problems that they could solve but where it takes more than five minutes to get a solution.

Since the 1970s, there has been a great deal of research in these areas, and two in particular—metacognition and motivation—have been intensively studied (see, for example, Flavell, 1976; Wigfield, Eccles, & Rodriguez, 1998). This book summarizes the research in these areas in the next two sections before weaving the two threads together.

### Metacognition

John Flavell (1976), widely credited with inventing the term, defines *metacognition* as follows:

> "Metacognition" refers to one's knowledge concerning one's own cognitive processes and products or anything related to them, e.g., the learning-relevant properties of information and data. For example I am engaging in metacognition (metamemory, metalearning, metaattention, metalanguage, or whatever) if I notice that I am having more trouble learning A than B; if it strikes me that I should double-check C before accepting it as a fact; if it occurs to me that I had better scrutinise each and every alternative in any multiple-choice type task situation before

> deciding which is the best one; if I sense that I had better make a note of D because I may forget it; if I think to ask someone about E to see if I have it right. In any kind of cognitive transaction with the human or non-human environment, a variety of information processing activities may go on. Metacognition refers, among other things, to the active monitoring and consequent regulation and orchestration of these processes in relation to the cognitive objects or data on which they bear, usually in the service of some concrete goal or objective. (p. 232)

Metacognition, therefore, includes knowing what one knows (*metacognitive knowledge*), what one can do (*metacognitive skills*), and what one knows about one's own cognitive abilities (*metacognitive experience*). The research shows clearly that "the most effective learners are self-regulating" (Butler & Winne, 1995, p. 245) and, more important, that training students in metacognition raises their performance (for example, Lodico, Ghatala, Levin, Pressley, & Bell, 1983) and allows them to generalize what they have learned to novel situations (Hacker, Dunlosky, & Graesser, 1998).

In recent years, the IMPROVE teaching approach developed by Zemira Mevarech and her colleagues for teaching mathematics has proven to be particularly effective. The name IMPROVE is an acronym for the seven stages of the teaching approach.

1. Introducing new concepts
2. Metacognitive questioning
3. Practicing
4. Review and reducing difficulties
5. Obtaining mastery
6. Verification
7. Enrichment

In the *metacognitive questioning* phase, students take turns in asking and answering three kinds of questions: (1) comprehension questions, such as "What kind of problem is this?"; (2) strategic questions, such as "What kinds of methods are likely to be useful in solving this problem?"; and (3) connection questions, which ask students to relate the current problem to previous problems they have solved (Mevarech & Kramarski, 2014).

While the evidence shows clearly that students who use metacognitive skills have higher achievement, such skills are useful only if students are motivated to use them.

## Motivation

Most people are familiar with the distinction between intrinsic and extrinsic motivation: whether the motivation for doing something comes from the fact that it is inherently interesting or enjoyable or comes because it will lead to some other valued outcome (Ryan & Deci, 2000). If individuals undertake only those things that are inherently interesting or enjoyable, then they are unlikely to learn to read, write, or play a musical instrument. We are generally motivated to learn these things because we value the consequence, whether it is avoiding punishment for not doing homework; reaching some external goal we have set for ourselves, such as learning to drive; or learning how to play a favorite song on the guitar.

Much of the literature about motivation in schools treats motivation rather like a substance in students' brains: some students have a lot of it, and others don't. When students fail to learn, we blame their lack of motivation. At the other extreme, there are those who believe that it is the teacher's job to motivate the students. If the students don't learn, it is because the teacher was not a sufficiently good motivator, so the cause of the failure to learn is the teacher.

Lately, however, the traditional model—that motivation is a cause of academic achievement—has come into question, because when we look at data on students' achievement and their motivation in school, the idea that high levels of motivation result in higher levels of achievement does not seem to fit the evidence very well. In fact, the evidence, such as it is, suggests that causation is running in the opposite direction. Motivation is not a cause but a *consequence* of achievement (Garon-Carrier et al., 2016).

This way of thinking is particularly marked in the work of Mihaly Csikszentmihalyi, a psychologist at the University of Chicago. In his book *Flow: The Psychology of Optimal Experience*, Csikszentmihalyi (1990) describes various situations in which individuals become completely absorbed in the activities in which they engage:

> A dancer describes how it feels when a performance is going well: "Your concentration is very complete. Your mind isn't wandering, you are not thinking of something else; you are totally involved in what you are doing. Your energy is flowing very smoothly. You feel relaxed, comfortable and energetic."
>
> A rock climber describes how it feels when he is scaling a mountain: "You are so involved in what you are doing [that] you aren't thinking of yourself as separate from the immediate activity. You don't see yourself as separate from what you are doing."

> A mother who enjoys the time spent with her small daughter: "Her reading is the one thing she's really into, and we read together. She reads to me and I read to her, and that's a time when I sort of lose touch with the rest of the world, I'm totally absorbed in what I'm doing."
>
> A chess player tells of playing in a tournament: " . . . the concentration is like breathing—you never think of it. The roof could fall in and, if it missed you, you would be unaware of it." (pp. 53–54)

Csikszentmihalyi describes this sense of being completely absorbed in an activity as *flow*. Flow can arise because of one's intrinsic interest in a task, as with the mother reading to her daughter, but can also arise through a match between one's capability and the task's challenge level. Boredom occurs when the level of challenge is low and the level of capability is high. General anxiety results when the level of challenge is high and the level of capability is low. Apathy results when both are low. However, flow results when both capability and challenge are high.

This way of thinking about motivation is radical because it does not locate the problem in the teacher or the learner, but rather in the match between challenge and capability. In the traditional view of motivation, if the student is not motivated, it is the fault either of the teacher or of the student. But if we see motivation not as a cause but as an *outcome*—an emergent property of getting the match between challenge and capability right—then if the student isn't motivated, that's just a signal that the teacher and the learner need to try something different.

However, it will not be enough that an activity is absorbing if the cost of engaging in the task is seen by the student as being too high, whether this is in terms of the opportunity cost that attempting a task might take or negative consequences such as the risk to one's self-image if unsuccessful (Eccles et al., 1983). The goals that students actually pursue in classrooms will depend on complex trade-offs between cost and benefit.

We know that students are more motivated to reach goals that are specific, are within reach, and offer some degree of challenge (Bandura, 1986; Schunk, 1991), but when the goals seem out of reach, students may give up on increasing competence and instead avoid harm, by either focusing on lower-level goals they know they can reach or avoiding failing altogether by disengaging from the task, as we saw in chapter 5. It might be assumed from this that competition is unhelpful, but focusing on increasing competence within teams to compete against other teams has been found to increase student achievement in mathematics, provided the competition was focused on relative improvement among the groups (Linnenbrink, 2005).

It is also worth noting that while students' motivation and their belief in their ability to carry their plans through to successful completion—what psychologist Albert Bandura (1997) termed *self-efficacy*—tend to decline as students go through school, what the teacher does can make a real difference. A study of 1,571 students in 84 mathematics classrooms from fifth to twelfth grades finds that students provided with positive constructive feedback are more likely to focus on learning rather than performance (Deevers, 2006).

## Integration of Motivational and Metacognitive Perspectives

This discussion may appear to have brought us a long distance from classroom formative assessment, but fulfilling the potential of formative assessment requires that we recognize that assessment is a two-edged sword. Assessment can improve instruction, but it can also impact the learner's willingness, desire, and capacity to learn (Harlen & Deakin Crick, 2002). Although we don't yet know everything about the most effective learning environments, the existing research on metacognition and motivation provides clear and strong evidence that activating students as owners of their own learning is an essential component.

At the outset, it is important to acknowledge that most students want to do well at school. Very few students want to fail academically, if only to keep their parents from nagging too much. However, although students want to do well in an abstract sense, the extent to which they engage in learning activities depends on a number of factors.

When educators invite students to participate in a learning activity, students use a number of sources of information to decide what they are going to do. Consider the following six.

1. **Current perceptions of the task:** Does this look like a task I can successfully complete?
2. **Previous experiences on similar tasks:** Have I undertaken similar tasks before, and was I successful or not? Did I enjoy them?
3. **Beliefs about the subject or task:** Is this a subject or a task that I think I am good at, or is it a subject that I know I find difficult?
4. **Beliefs about ability and the role of effort in the subject:** Is this a subject that I think I need to be really smart to be successful in, or is it a subject where I can be successful if I try really hard?
5. **Interest in the subject (personal versus situational):** Is this a subject or a topic that I am interested in (personal interest)? Has the task

been presented in a way that makes me interested in undertaking it (situational interest)?

6. **Costs and benefits:** How much effort will I have to put into this task to be successful? Will it be worth it? How will I feel if I try my best and do not succeed? Will I be embarrassed or would it be OK to try and still fail? Is it likely that others will succeed?

Although weighing all these different factors is obviously a complex process, the students ultimately make a very simple choice. They direct their energies along one of two pathways (*dual pathways*): either toward preserving well-being or toward growth (Boekaerts, 1993). To preserve well-being, students might put little effort into the task, so that they can claim that if they are unsuccessful, it is because they weren't trying, because the task was "boring." They can copy work from other students or from the internet, or they can "play the fool." All these responses are ways of preserving a sense of well-being. The alternative is to try one's best on the task, with all the risks that this entails, in order to increase one's capability—in other words, to grow.

The important point about this dual-processing self-regulation model of what students do in classrooms, developed by Monique Boekaerts (1993), a Belgian psychologist of education, is that it is dynamic, and a student's decision can change rapidly. If a teacher says to a class, "This task is very difficult, and most people take at least three attempts to get it right," then this has the effect of changing the cost-benefit tradeoff. The benefits of success are higher (successfully completing a hard task is good), and the costs of failure are lower (since failure is expected, at least in the first few attempts). A student who was disinclined to engage in the task may now choose to do so because the balance of costs and benefits has changed. The teacher may present the task in such a way as to make it more interesting to the student—for example, by presenting it as a mystery or a puzzle to be solved. The teacher may undertake growth mindset activities with students to change their views about the role of ability and talent (see, for example, Blackwell, Trzesniewski, & Dweck, 2007). The student may find ways to lower the threat to self-image, thus allowing a shift of energy and attention back to the growth pathway.

Boekaerts's model suggests that the most important thing is creating learning environments that encourage students to activate the growth rather than the well-being pathway. We cannot possibly anticipate all the factors that a student may consider in deciding whether to pursue growth rather than well-being, but there are a few things that can be done to tip the scales in the right direction.

Consider the following five.

1. Share learning goals with students so that they are able to monitor their own progress toward them.
2. Promote the belief that ability is incremental rather than fixed. When students think they can't get smarter, they are likely to devote their energy to avoiding failure.
3. Make it more difficult for students to compare their achievement with others.
4. Provide feedback that contains a recipe for future action rather than a review of past failures (a medical exam rather than a postmortem).
5. Use every opportunity to transfer executive control of the learning from the teacher to the students to support their development as autonomous learners.

And if you figure out a way to do all that, please let me know. The fact that we know what needs to be done is not the same as doing it. Continuously developing one's teaching is extraordinarily difficult. The good news is that you don't need to start from scratch. Rather, you can build on the achievements of other teachers who have already developed techniques, such as those in the next section, to encourage the growth pathway or mindset in your students.

## Practical Techniques

There is no doubt that activating students as owners of their own learning produces substantial increases in learning, but it is not a quick fix. Many teachers find that students' first attempts at self-assessment are neither insightful nor useful. One teacher I worked with said, "Self-assessment is like making pancakes; you usually end up throwing the first one away." It is also a difficult course to steer. A homeroom teacher was addressing some behavior problems that had arisen in a middle school class's mathematics lessons. The homeroom teacher decided to try to help the mathematics teacher by getting the students to do a self-assessment of their own performance and asked each student to complete the sentence, "I would learn better in mathematics lessons if . . ." One girl thought about this for quite a while, and in the end, she wrote, "I would learn better in mathematics lessons if I had a better mathematics teacher." I have been in her mathematics classroom, and she has a point. However, when the homeroom teacher saw what the student had written, she told the student that this was not acceptable, so the student went back to her seat and

wrote, "I would learn better in mathematics lessons if I brought the right equipment to school, paid attention in class, and remembered to do my homework." Originally, the student thought that her homeroom teacher really wanted her to engage in working on what would improve her learning of mathematics, but the homeroom teacher hadn't provided sufficient guidance about what was and was not up for negotiation. More important, the self-assessment activity had not been framed sufficiently well to ensure that the students focused on things that were within their control. In such cases, the students end up playing a game of "guess what's in the teacher's head," and there is little, or no, worthwhile learning. As we will see, self-assessment can be uncomfortable for both student and teacher, but the benefits are great, and once teachers get used to involving the students in their own learning, it is almost impossible to go back.

Many of the techniques described in chapter 6 can be adapted for self-assessment, but the following are some techniques that are specifically designed to encourage students to reflect on their own learning. This section will discuss the following techniques: traffic lights, red or green disks, colored cups, learning portfolios, and learning logs.

## Traffic Lights

Many teachers use the *traffic lights* technique to activate students as owners of their own learning. At the beginning of the lesson, the teacher shares the learning intentions and any associated success criteria with the students, and at the end of the period, the students have to assess the extent to which they have achieved the intended learning by placing a colored circle against the learning intentions that they wrote in their notebooks at the start of the lesson. Green indicates confidence that the intended learning has been achieved. Yellow indicates either ambivalence about the extent to which the intended learning has been achieved or that the objectives have been partially met. Red indicates that the student believes that he or she has not learned what was intended. The problem is that this is a self-report, and as noted earlier, self-reports are not particularly reliable. One teacher addressed this by saying to the class, "Reds over here with me; greens help the yellows; yellows make sure the greens understand this as well as they think they do." By changing the meaning of green from "I feel confident" to "I am now ready to teach this to someone else," the teacher made it much less likely that students would signal green just because they wanted to look good.

Traffic lights can also be used for test preparation. One teacher had been looking at how her students prepared for tests and was concerned to see that many of her students seemed to employ as their primary strategy flipping through the pages of their

notebooks, presumably hoping that some of it would stick. She was particularly surprised to see that many students spent as much time reviewing the things that they were confident about as they did reviewing things they had not learned. To improve the quality of preparation, the teacher encouraged students to place a colored dot in the top corner of each page to signal the students' confidence with the material. When preparing for the test, students could then skip over the material they had flagged as green and concentrate on those areas in which they were less confident. While such an approach depends on the accuracy of the students' judgments, which are, after all, self-reports, the fact that the only audience for this self-assessment is the student herself increases the likelihood that she will be honest.

## Red or Green Disks

A high school algebra teacher had been using traffic lights for a while and started thinking about how she could get more real-time information. She gave each student in the class a CD-sized disk, green on one side and red on the other. At the beginning of the period, the green side faced up, but, as the lesson progressed, if students wanted to signal that they thought the teacher was going too fast, they flipped the disk over to red. She found that students who hadn't asked a single question that year in class were willing to flip the disk over to red to show that they were getting confused.

A few weeks later, something rather remarkable occurred. Using an overhead projector, the teacher was showing a class of students who were taking algebra 1 (for the *third* time) how to solve equations. She became so invested in what she was doing that she forgot to check for any red disks. A student at the back of the class had flipped her disk over to red a few minutes earlier and was getting more and more frustrated that the teacher had not checked for red disks. She looked at her two friends seated on either side, and seeing that they too had flipped their disks over to red, she grabbed her neighbors' two disks and waved all three disks in the air to flag the teacher to slow down. When I first heard this story, I thought this must have been an extraordinarily confident student, but over time, as more and more teachers have used these techniques, more stories of this kind have emerged. When teachers open the channels of communication with the students, the students will use them.

Sometimes the honesty of students can be disarming—even scary. Dave Tuffin, a science teacher I once observed, was teaching a lesson to a seventh-grade class on a hot summer afternoon. He described it as "one of those lessons where you know it's not working but you can't stop yourself"—an experience that will be familiar to every teacher. He was trying to get more and more animated in order to wake up the class, but in reality, he was in a hole, and still digging. Then one student

raised his hand and said, "Sir, this isn't working, is it?" Afterward Dave described his feelings, saying that a year earlier, he would have been angry with the student for embarrassing him in that way. But for months, he had been trying to get the students to take more responsibility for their own learning, and so he said, "You're right. What shall we do?" Not one student suggested having a recess or giving up on the science. Instead, the whole class had a mature conversation about how to improve their learning of the lesson. Perhaps the most important thing about this story is the reaction of the teacher. We saw in chapter 5 how important it is to get the egos of the students out of the learning situation, but this story shows how important it is to get the teacher's ego out of the classroom, too. The student was not trying to embarrass Dave. He was throwing him a lifeline, and Dave had the grace to realize this and grab it.

## Colored Cups

Another teacher I observed had tried the red or green disks method, but found that the disks were difficult to see clearly from the front of her classroom due to the fluorescent lighting. So she went to a party store and bought colored cups, in red, yellow, and green. In her classroom, each student is given one of each of the colored cups, and the lesson begins with the green cup showing. If the student wants to signal that the teacher is going too fast, then the student shows the yellow cup, and if a student wants to ask a question, then the red cup is displayed. Why would any student show red? Because in this classroom, the rule is that as soon as one student shows red, the teacher uses the Popsicle sticks to pick another student at random, and the selected student has to go to the front of the classroom to answer the question being posed by the student who showed red. The teacher describes this as converting her classroom into one gigantic game of "chicken."

This technique neatly encapsulates two key components of effective formative assessment—engagement and responsiveness. If a student is showing yellow or green, he can be called on to explain the work to someone else, which requires the students to be monitoring their own learning and, therefore, engaged. And the flow of information from the students about the pace of instruction helps the teacher make adjustments to be more responsive to the students' learning needs.

## Learning Portfolios

Many schools encourage students to keep portfolios of their work, but too often, these are maintained in the same way as an artist's portfolio—to display the latest and best work. In her portfolio, an artist might have something in gouache, something in acrylic, something in watercolor, something in oil, and something in

charcoal. If the artist produces a new work in, say, gouache, that she considers to be better than what is already in the portfolio, then she will replace the old gouache piece with the new one. Focusing on the latest and best in this way creates what might be termed a *performance portfolio*, which supports summative assessment well but tends to obscure the learning journey.

For an incremental view of ability, a learning portfolio is far more useful. When students do better work, the teacher adds to the portfolio rather than replacing earlier work, to allow students to review their learning journeys. Looking back at earlier examples of their persuasive writing, for instance, can help students see what has developed, which has two immediate benefits. First, seeing what has improved and thus identifying a trajectory of development means the student is likely to be able to see how further improvement might be possible. Second, focusing on improvement means the student is more likely to see ability as incremental rather than fixed, which, as we saw in chapter 5, is a key characteristic of the most effective learners.

Students can start developing such learning portfolios at a very young age. One kindergarten teacher I observed gets her students to paint a self-portrait each month and places these portraits in the students' learning portfolios. In January, the students review their four self-portraits from September, October, November, and December, and are asked to reflect on what they think has gotten better in their painting. In discussion with her teacher, one girl decided that the most salient feature was "my arms don't come out of my head anymore." This student may not be the best painter in the class, but she knows that with continued effort, she will improve.

## Learning Logs

One technique that teachers have found useful as a way of getting students to reflect on their learning is to ask students to complete a learning log at the end of a lesson. Of course, there is nothing new in this, but some teachers find a slight variant on the usual method useful. Instead of providing responses to one or two self-assessment questions from the teacher, students are invited to respond to no more than three of the following prompts.

- Today I learned . . .
- I was surprised by . . .
- The most useful thing I will take from this lesson is . . .
- I was interested in . . .
- What I liked most about this lesson was . . .

- One thing I'm not sure about is . . .
- The main thing I want to find out more about is . . .
- After this session, I feel . . .
- I might have gotten more from this lesson if . . .

Getting students to choose which three of these statements they respond to seems to encourage a more thoughtful approach to the process of reflecting on their learning.

## Conclusion

Teachers have a crucial role to play in designing the situations in which learning takes place, but only learners create learning. Therefore, it is not surprising that the better learners are able to manage their learning, the better they learn. All students can improve how they manage learning processes and become owners of their own learning. However, this is not an easy process. Reflecting critically on one's own learning is emotionally charged, which is why developing such skills takes time, especially with students who are accustomed to failure.

This chapter has provided research evidence along with a number of practical techniques that teachers can use to increase the engagement of their students and their own responsiveness to their students' needs. In the epilogue, the main themes of this book are reviewed, concluding with a few words of advice for teachers in taking the ideas presented in this book into their own classrooms.

# Epilogue

Every generation believes it is witnessing unprecedented change—in society, in the world of work, and in the demands that are made of people. However, there are good reasons to believe that this time, it really is different. As Princeton economist Alan Blinder (2009) points out, as many as fifty million American jobs are at risk of being offshored by 2035. Some of these are low-skill jobs, but many are not. Although China has overtaken America as the world's largest manufacturer, the average American worker is three to four times as productive as the average Chinese worker (United Nations Statistics Division, 2010). This is not because they work with more expensive equipment, but mostly because of intangibles such as ways of organizing work, the intellectual property being used, and so on (Kling & Schulz, 2009).

America became the world's manufacturer in the second half of the 20th century because, between 1910 and 1940, it made extraordinary investments in mass public education for students up to the age of eighteen at a time when most other equally developed economies thought that education beyond the age of fifteen would be wasted on those not going on to higher education (Goldin & Katz, 2008). This investment created both a skilled workforce and a set of consumers who were well enough educated to be able to adopt the new technology (Bhidé, 2008).

The important feature of the investment that was made in educating America's youngsters between 1910 and 1940 was that it did not try to predict the future, and it was successful precisely for that reason. The education provided was general education, not vocational training targeted at specific jobs. It is more, and better, *general* education for more of the population that will guarantee America's future prosperity.

If we look at previous attempts to improve reading and mathematics, we would probably get very depressed. The data from the National Assessment of Educational Progress (Rampey, Dion, & Donahue, 2009) show that while there have been some

improvements in the performance of nine- and thirteen-year-olds, the reading and mathematics levels of seventeen-year-olds are the same as they were in the early 1970s. However, as we saw in chapter 1, the failure to improve education arises primarily from a failure to understand the importance of teacher quality.

During the 20th century, most policies for educational improvement assumed that all teachers were more or less the same. Now we understand that the quality of the teacher is the single most important determinant of how much students learn, and in the classrooms of the most effective teachers, students from disadvantaged backgrounds learn as much as those from advantaged backgrounds. The impact of teacher quality is profound.

A study from Harvard University shows that the impact of having an outstanding teacher in kindergarten can be detected in the average hourly rates of pay of those students thirty years later (Chetty et al., 2010). The same study shows that the economic value generated by an outstanding kindergarten teacher is around $320,000 every year.

There is now widespread agreement about the importance of teacher quality—indeed, it appears that differences in the quality of schools are almost entirely accounted for by differences in the quality of the teachers at the schools. To raise student achievement, we have to improve teacher quality.

Given the failure of most attempts to improve teacher performance, it is hardly surprising that many economists of education have suggested that the only way to improve teacher quality is through deselecting the least effective teachers and replacing them with better ones (Hanushek, 2010). However, such a change will take thirty years to work through the system, by which time it will be too late. Therefore, while we should, of course, seek to ensure that the best people are attracted to, and remain in, teaching, we must also seek to improve the performance of existing teachers, through professional development focused on what the research shows as making the biggest difference to students.

In chapter 2, we saw that the research evidence available indicates that when formative assessment practices are integrated into the minute-to-minute and day-by-day classroom activities of teachers, substantial increases in student achievement—of the order of a 50 to 70 percent increase in the speed of learning—are possible, even when outcomes are measured with externally mandated standardized tests. Moreover, these changes are not expensive to produce; classroom formative assessment is approximately twenty times as cost-effective in raising achievement as class-size reduction. The currently available evidence suggests that there is nothing else remotely affordable that is likely to have such a large effect.

Chapters 3 through 7 each reviewed one of the five key strategies of classroom formative assessment.

1. Clarifying, sharing, and understanding learning intentions and success criteria
2. Eliciting evidence of learning
3. Providing feedback that moves learning forward
4. Activating learners as instructional resources for one another
5. Activating learners as owners of their own learning

For each of these strategies, I presented evidence of its impact on student learning and provided a number of practical classroom techniques for implementing the strategy. Each of the techniques requires little in the way of technology and can be easily adapted to use in the teaching of any subject, for any age range.

Some of the techniques are relatively new, but most are not. They have been used by the best teachers for tens, if not hundreds, of years. It is not the techniques that are new; rather, it is the framework provided by classroom formative assessment that shows how they fit together, and it is the evidence that shows the impact of such techniques on student achievement.

The problem with being provided with so many techniques is that, as psychologist Barry Schwartz (2003) points out in *The Paradox of Choice: Why More Is Less*, too much choice can be paralyzing—and dangerous. When teachers try to change more than two or three things about their teaching at the same time, their teaching typically deteriorates, and they go back to doing what they were doing before. I advise you to choose one or two of the techniques in this book and try them out in your classroom. If the techniques appear to be effective, the goal should be to practice them until they become second nature. If they are not effective, you can modify them, or try another technique. No technique is likely to work for all teachers, but all teachers will, I believe, find something here that will work for them. For more guidance about how to set up teacher self-help groups on implementing classroom formative assessment, see Wiliam and Leahy (2015).

In her book *A Return to Love*, Marianne Williamson (1992) writes, "Our deepest fear is not that we are inadequate. Our deepest fear is that we are powerful beyond measure" (p. 190). We now know that the teacher is the most powerful influence on how much a student learns, and that teachers can continue to make significant improvements in their practice throughout their entire careers. If all teachers accept the need to improve practice, not because they are not good enough but because they

can be even better, and focus on the things that make the biggest difference to their students, we *will* be able to prepare our students to thrive in the impossibly complex, unpredictable world of the 21st century.

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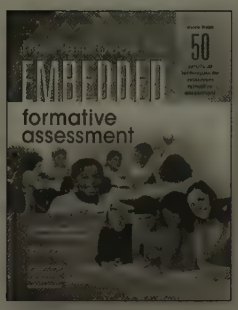
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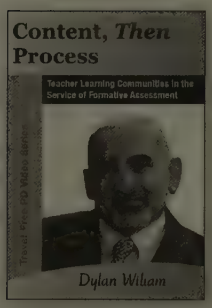
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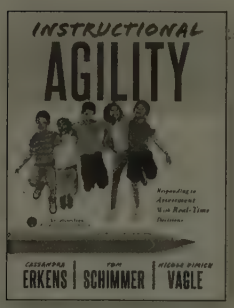
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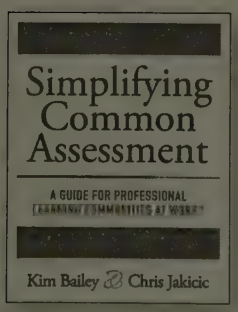

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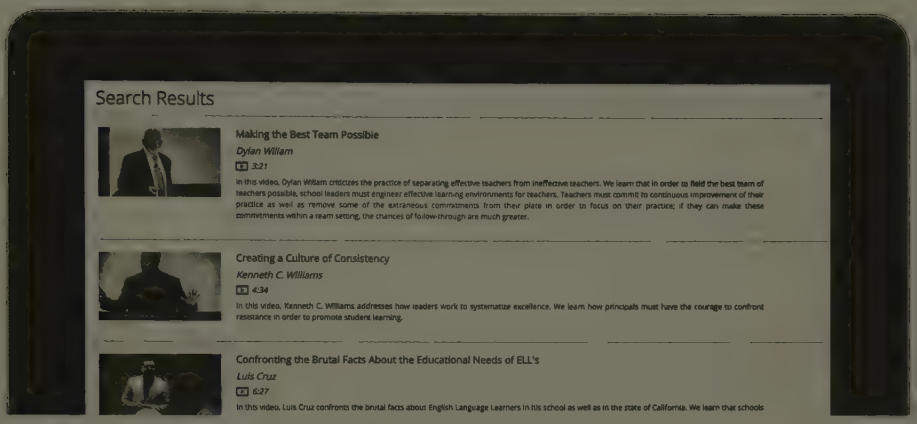

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